
Optimal Rates for Feasible Payoff Set Estimation in Games

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Abstract

We study a setting in which two players play a (possibly approximate) Nash equilibrium of a bi-matrix game, while a learner observes only their actions and has no knowledge of the equilibrium or the underlying game. A natural question is whether the learner can rationalize the observed behavior by inferring the players' payoff functions. Rather than producing a single payoff estimate, inverse game theory aims to identify the entire set of payoffs consistent with observed behavior, enabling downstream use in, *e.g.*, counterfactual analysis and mechanism design across applications like auctions, pricing, and security games. We focus on the problem of estimating the set of feasible payoffs with high probability and up to precision ϵ on the Hausdorff metric. We provide the first minimax-optimal rates for both exact and approximate equilibrium play, in zero-sum as well as general-sum games. Our results provide learning-theoretic foundations for set-valued payoff inference in multi-agent environments.

1. Introduction

The central question in *inverse game theory* is the following (Waugh et al., 2011; Kuleshov and Schrijvers, 2015): given observed equilibrium behavior, what utility functions could have generated it? We study the problem of recovering the *entire* set of possible payoffs compatible with the players' observed behavior, given action samples generated by their strategies. We are interested in the minimax sample complexity of approximately identifying such a set. Even when equilibrium behavior is observed exactly (let alone through samples), the problem of inferring an agent's utilities from behavior is inherently underdetermined, as multiple payoff functions can rationalize the same observed play (Ng and Russell, 2000; Waugh et al., 2011). This issue is often re-

ferred to as *unidentifiability*. For example, the zero payoff matrix is compatible with any equilibrium. This ambiguity is usually addressed by imposing additional structure, such as restricting attention to parametrized low-dimensional payoff families (Waugh et al., 2011; Kuleshov and Schrijvers, 2015), or by choosing a priori a selection principle such as maximum entropy (Vorobeychik et al., 2007; Waugh et al., 2011), or bounded-rationality models (such as quantal response) (Yu et al., 2022).

Rather than imposing additional modeling assumptions or fixing an *a priori* selection principle, we focus on identifying the entire set of payoff functions consistent with the observed strategies, *i.e.*, the feasible payoff set. The idea of this approach is to defer the selection of a particular payoff to a later stage, thereby making the entire process modular and more adaptable to the requirements of the application at hand. For example, given the entire set of possible equilibria, one may select equilibria according to a maximum-entropy criterion (Waugh et al., 2011), add additional structural constraints to the set (*e.g.*, if we know that the game is symmetric or has a potential-game structure), or choose a robust representative such as the incenter solution (Cui et al., 2025). This set-valued perspective is standard in structural econometrics and inverse game theory (see, among others, Brown and Matzkin (1996); Manski (2003); Tamer (2010); Waugh et al. (2011)). However, the corresponding learning problem remains poorly understood from a theoretical standpoint, and even in the simplest multi-agent settings, we lack a sharp characterization of the *statistical* difficulty of such set-valued identification.

In this work, we address this gap by providing minimax-optimal sample complexity guarantees for the problem of recovering the feasible payoff set from the observed behavior of two players in an unknown bimatrix game playing a Nash equilibrium. In particular, we consider two players repeatedly interacting in an unknown n -dimensional normal-form bimatrix game with payoff matrices (A, B) . The players play a (possibly approximate) Nash equilibrium (x, y) of the game, and the learner observes i.i.d. action pairs (X_t, Y_t) sampled from the product distribution $x \times y$. The learner's goal is to infer the *feasible payoff set*: the set of all payoff matrices (A', B') for which the strategies (x, y) constitute a Nash equilibrium (or an α -Nash equilibrium) of (A', B') . The goal of the paper is to answer the following

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question: *How many equilibrium samples are required to estimate the feasible payoff set up to Hausdorff error ϵ with probability at least $1 - \delta$?* Our main contribution is the first minimax-optimal characterization of this question. Our results cover both *general-sum* and *zero-sum* games, and both approximate and exact Nash equilibria.

Technical challenges An important technical ingredient in our analysis is the correspondence (a set-valued function) that maps equilibrium strategies to the feasible payoff sets that rationalize them. Regarding the lower bounds, we first show that, when observing exact equilibria (*i.e.*, $\alpha = 0$), the strong discontinuity of such correspondence makes the problem unlearnable. Therefore, we relax the learning requirement and provide minimax-optimal rates when the strategy profiles have a minimum probability of playing each action that is bounded away from 0. Conversely, when $\alpha > 0$, we can show that such discontinuities are somewhat mitigated. Interestingly, our results reveal that small values of α can substantially degrade the rates through an unavoidable dependence on α^{-1} . Informally, the cause is that when (x, y) play some actions with probability $\propto \alpha$, small changes in the strategy profiles lead to large changes in the payoff sets. For the upper bounds, the major technical obstacle in obtaining tight rates is establishing this α^{-1} dependence in the approximate-equilibrium case. To this end, focusing on a term that depends only on the x -player, we first show that the Hausdorff error scales with $\mathcal{O}(\alpha^{-1}(\hat{x} - x)^\top Ay)$, where $\hat{x}$ is the empirical mean of x after m samples and A is any feasible payoff for (x, y) (an analogous argument holds for the y -player). Hence, to get a tight rate, we need to show that the empirical value of the game $\hat{x}^\top Ay$ converges to the true value $x^\top Ay$ as $\mathcal{O}(\sqrt{\alpha(n + \log(1/\delta))/m})$, where m is the number of samples (while standard concentration of $\hat{x}$ to x would only yield $\mathcal{O}(\sqrt{n + \log(1/\delta)/m})$). We argue that such rates are achievable by studying a fractional knapsack problem that arises in controlling the error.

1.1. Related Works

Early empirical work on payoff learning treats utilities as regression targets when realized utilities are observed for sampled strategy profiles, enabling supervised payoff approximation (Vorobeychik et al., 2007). Closer in spirit to our setting, Kuleshov and Schrijvers (2015) introduced inverse game theory in the context of succinctly-representable games and studied the computational problem of finding utilities that rationalize an observed equilibrium (primarily via feasibility of linear/convex programs under structural restrictions). Waugh et al. (2011) study an inverse equilibrium problem (for correlated equilibria) in a parametrized setting, and recover a polytope of compatible payoff functions used to compute the maximum-entropy equilibrium. Recently, Goktas et al. (2024) studied how to recover a sin-

gle payoff-parameter/Nash-equilibrium pair consistent with observed behavior, relying on some form of oracle access to payoffs during the inverse optimization. This is fundamentally different from our setting, where we aim to recover the feasible set of payoff matrices from i.i.d. equilibrium action samples alone. A related line of works focus on learning adversary models in Stackelberg games, with the main goal of enabling behavioral modeling of attackers in security games (Haghtalab et al., 2016; Sinha et al., 2016; Wu et al., 2022). This problem differs from ours in the fact that the learner can probe the adversary (*i.e.*, the player with unknown payoffs) by observing their responses to ad hoc strategies. In the context of sponsored search auctions, Nekipelov et al. (2015) studies how to estimate players' valuations from observed contextual data, by assuming each player places bids following a no-regret algorithm. Finally, Ling et al. (2018) attempts to fit parametric games end-to-end by replacing Nash equilibrium with differentiable fixed points, such as quantal response equilibrium, enabling gradient-based estimation of game parameters.

Related to our work is the single-agent version of our problem, usually studied in inverse reinforcement learning (IRL). In IRL, the objective is to infer rewards for which a demonstrated policy is optimal (Abbeel and Ng, 2004). It is well known that even in single-agent settings, the problem is fundamentally *ill-posed*, as multiple reward functions may be compatible with the observed behavior (Ng and Russell, 2000). Analogously to the game theory setting, there have been several attempts to resolve this ambiguity by focusing on specific selection criteria such as maximum margin, maximum entropy, and minimum Hessian eigenvalue (Ratliff et al., 2006; Metelli et al., 2017; Zeng et al., 2022). In this work, we take inspiration from an alternative recent line of research that attempts to overcome the ambiguity issue by estimating the *feasible reward set*, *i.e.*, the entire set of reward functions compatible with the observed data (Metelli et al., 2021; Lindner et al., 2022; Metelli et al., 2023; Poiani et al., 2024). Remarkably, this approach captures the inherent ambiguity of the problem, supports robust prediction and counterfactual reasoning, and avoids committing to arbitrary equilibrium-selection or regularization choices.

There are many works that extend the single-agent IRL problem to the multi-agent IRL problem (MAIRL), including, *e.g.*, (Natarajan et al., 2010; Yu et al., 2019; Fu et al., 2021; Reddy et al., 2012; Zhang et al., 2019). As in inverse game theory, most MAIRL approaches resolve the inherent ambiguity through a priori selection principles or additional modeling assumptions. There are recent exceptions that study the learning problem of returning the entire set of feasible rewards (Freihaut and Ramponi, 2024; Tang et al., 2024). However, unlike what happens in game-theoretic applications such as mechanism design, auctions, and pricing, the primary goal of (MA)IRL is to serve as a preprocess-

ing step within a larger pipeline: the learned model and payoff are subsequently used to train multi-agent policies, which are then evaluated in the true environment. As noted by [Freihaut and Ramponi \(2024\)](#), in strategic settings this pipeline lacks guarantees without further restrictions (such as uniqueness of equilibria or structural assumptions that enforce it, e.g., entropy-regularized Markov games), since the new learned equilibria for the approximated payoffs might perform arbitrarily bad in the original setting.

Finally, a closely related area is *revealed preference* analysis, which asks whether observed price/purchase observations can be rationalized by utility maximization and constructs utility representations consistent with the available observations ([Samuelson, 1948](#); [Afriat, 1967](#); [Varian, 1982](#)). There is a related line of works studying the sample complexity of learning to predict optimal bundles of items for a buyer with an unknown demand function from some specific class (e.g., linear, separable piecewise-linear concave, and Leontief) from observed choices ([Beigman and Vohra, 2006](#); [Zadimoghaddam and Roth, 2012](#); [Balcan et al., 2014](#)).

2. Preliminaries

Mathematical notation For $n \in \mathbb{N}$, we denote by Δ_n the n -dimensional simplex. Furthermore, $[n]$ denotes the set $\{1, \dots, n\}$. We also denote by e_i the i -th standard basis vector of $\mathbb{R}^n$. For any $x \in \Delta_n$, we denote by $\text{supp}(x) = \{i \in [n] : x_i > 0\}$. Let $\mathcal{X}_1, \mathcal{X}_2$ be two non-empty subsets of a metric space $(\mathcal{X}, d)$. Then, the Hausdorff distance ([Rockafellar and Wets, 2017](#)) $H_d(\mathcal{X}_1, \mathcal{X}_2)$ is:

$$\max \left\{ \sup_{x_1 \in \mathcal{X}_1} \inf_{x_2 \in \mathcal{X}_2} d(x_1, x_2), \sup_{x_2 \in \mathcal{X}_2} \inf_{x_1 \in \mathcal{X}_1} d(x_2, x_1) \right\}.$$

We note that H_d is defined based on the metric d . In this work, we will be primarily interested in the case where d is the ℓ_∞ -norm between (subsets of) Euclidean spaces. From now on, we will drop the dependency on d .

General-Sum Games and Nash-Equilibrium A general-sum game (GSG) is specified by two matrices $A, B \in [-1, 1]^{n \times n}$. For mixed strategies $x, y \in \Delta_n$, the expected losses incurred by the two players are $x^\top A y$ and $x^\top B y$, respectively. Each player seeks to choose a mixed strategy that minimizes their own expected loss. Given $\alpha \geq 0$, we say that a pair of strategies x, y is an α -Nash equilibrium for $\mathcal{G}$ if the following conditions are met: $x^\top A y \leq e_i^\top A y + \alpha$ for all $i \in [n]$, and $x^\top B y \leq x^\top B e_j + \alpha$ for all $j \in [n]$. For any x, y we denote by $\mathcal{G}_\alpha(x, y)$ the set of all matrices for which x, y is an α -Nash equilibrium, that is, the set of matrices A, B that verify the following constraints for all indices $i, j \in [n]^2$:

$$x^\top A y \leq e_i^\top A y + \alpha \quad \text{and} \quad x^\top B y \leq x^\top B e_j + \alpha.$$

Formally, we can define the following sets:

$$\begin{aligned} \mathcal{G}_\alpha^x(x, y) &= \{A \in [-1, 1]^{n \times n} : x^\top A y \leq e_i^\top A y + \alpha \quad \forall i\} \\ \mathcal{G}_\alpha^y(x, y) &= \{B \in [-1, 1]^{n \times n} : x^\top B y \leq x^\top B e_j + \alpha \quad \forall j\}, \end{aligned}$$

so that we can write $\mathcal{G}_\alpha(x, y) = \mathcal{G}_\alpha^x(x, y) \times \mathcal{G}_\alpha^y(x, y)$.

Zero-Sum Games and Nash-Equilibrium A zero-sum game (ZSG), instead, is specified by a single matrix $A \in [-1, 1]^{n \times n}$. In this case, $x, y \in \Delta_n \times \Delta_n$ is an α -Nash equilibrium if $x^\top A e_j - \alpha \leq x^\top A y \leq e_i^\top A y + \alpha$ holds for every $i, j \in [n]^2$. For any pair of mixed strategies x, y , we denote by $\mathcal{Z}_\alpha(x, y)$ the set of all matrices for which x, y is an α -Nash equilibrium, that is:

$$\begin{aligned} \mathcal{Z}_\alpha(x, y) &= \{A \in [-1, 1]^{n \times n} : x^\top A e_j - \alpha \leq x^\top A y \\ &\quad \text{and } x^\top A y \leq e_i^\top A y + \alpha, \quad \forall i, j \in [n]^2\}. \end{aligned}$$

Note that $\mathcal{Z}_\alpha(x, y) \subseteq \mathcal{G}_\alpha(x, y)$. Indeed, it is sufficient to slice $\mathcal{G}_\alpha(x, y)$ on the hyperplane where $A = -B$.

Remark 2.1. When α is not specified, we refer to the exact equilibrium case. There, we will simply write e.g., $\mathcal{G}(x, y)$ and $\mathcal{Z}(x, y)$ instead of $\mathcal{G}_\alpha(x, y)$ and $\mathcal{Z}_\alpha(x, y)$.

Learning Framework We consider a setting in which two players follow a pair of mixed strategies $x, y \in \Delta_n \times \Delta_n$ that constitute an α -Nash equilibrium of an unknown general-sum game. The objective of the agent is to recover the feasible payoff set $\mathcal{G}_\alpha(x, y)$ of matrices for which (x, y) is an α -Nash equilibrium. In particular, the interaction framework is as follows: during each round $t \in \mathbb{N}$, the learner only observes $X_t, Y_t \sim x, y$. The learning algorithm takes in input a precision level $\epsilon > 0$ together with a maximum risk parameter $\delta \in (0, 1)$ and the parameter $\alpha \in [0, 1]$ and is composed of (i) a stopping rule τ_δ that controls the end of the data acquisition phase, and (ii) a recommendation rule, which we express as a function f that takes in input a pair of matrices A, B and returns 1 if the agent believes that (A, B) belongs to $\mathcal{G}_\alpha(x, y)$ and 0 otherwise. We note that f uniquely describes a set $\hat{\mathcal{G}}_\alpha = \{(A, B) : f((A, B)) = 1\}$, which denotes the guess of the agent for all the pair of matrices belonging to $\mathcal{G}_\alpha(x, y)$. Hence, in the following, we will refer to $\hat{\mathcal{G}}_\alpha$ as the recommendation rule (similarly, $\hat{\mathcal{Z}}_\alpha$ is the guess of the agent for the set $\mathcal{Z}_\alpha(x, y)$).

Among the possible learning algorithms, we are interested in those that are (ϵ, δ) -correct w.r.t. the Hausdorff distance. More precisely, the algorithm has to guarantee that:

$$\mathbb{P}(H(\mathcal{G}_\alpha(x, y), \hat{\mathcal{G}}_\alpha) \geq \epsilon) \leq \delta$$

in the general-sum case, with an analogous requirement for $\mathcal{Z}_\alpha$ in the zero-sum setting.

We make a remark that distinguishes the two learning objectives. At first, one might think that the problem of learning

$\mathcal{Z}_\alpha(x, y)$ could be easily solved once we have a set $\hat{\mathcal{G}}_\alpha$ that approximates $\mathcal{G}_\alpha(x, y)$. Nonetheless, we did not find any trivial way to slice a generic set $\hat{\mathcal{G}}_\alpha$ so that it approximates $\mathcal{Z}_\alpha(x, y)$. On one hand, one might be tempted to obtain $\hat{\mathcal{Z}}_\alpha$ by slicing $\hat{\mathcal{G}}_\alpha$ on $A = -B$. However, this only ensures one direction of the Hausdorff distance.¹ Similarly, one could think of slicing $\hat{\mathcal{G}}_\alpha$ by taking all the pair of matrices (A, B) which are almost zero-sum, e.g., $\|A - (-B)\|_\infty \leq \epsilon$. However, this set might be too large and could fail to satisfy the other direction of the Hausdorff distance. For these reasons, we study zero-sum and general-sum games as separate problems, providing minimax rates for both these settings.

3. Sample Complexity Lower Bound

In this section, we study the sample complexity of any (ϵ, δ) -correct algorithm that learns, with high probability, the set of payoffs induced by an unknown strategy profile (x, y) . In Section 3.1 we discuss the case of $\alpha = 0$, and in Section 3.2 we discuss the case when $\alpha > 0$.

3.1. On Exact Equilibria

We begin by highlighting a fundamental challenge in learning the feasible sets $\mathcal{G}(x, y)$ and $\mathcal{Z}(x, y)$ from data generated by an exact equilibrium. To build intuition for the problem, we first consider a simple general-sum example. Let $n = 2$ and consider two strategy profiles (x^0, y^0) and (x^1, y^1) such that $y^0 = y^1 = (1, 0)$, $x^0 = (0, 1)$, and $x^1 = (\pi, 1 - \pi)$ for some π .² Then, it is easy to see that a matrix A belongs to $\mathcal{G}^x(x^0, y^0)$ if and only if $a_{21} \leq a_{11}$, while A belongs to $\mathcal{G}^x(x^1, y^1)$ if and only if $a_{21} = a_{11}$. Hence, as $\pi \rightarrow 0$, we have a pair of instances that are statistically indistinguishable (i.e., the KL divergence between the two instances approaches 0 as $\pi \rightarrow 0$) while the Hausdorff between the payoff sets is constant and much larger than ϵ . This behavior is illustrated in Figure 1 (left): for example, the payoff matrix with $a_{21} = -1$ and $a_{11} = 1$ (marked with P^* in the figure) belongs to $\mathcal{G}(x^0, y^0)$ but lies at a constant distance from every point on the line $a_{21} = a_{11}$. Consequently, when x is unknown and must be learned from samples, one cannot distinguish between (x^0, y^0) or (x^1, y^1) , and therefore cannot output an ϵ -correct estimate of the feasible set. In general, we have the following result.

Theorem 3.1. *Let $\alpha = 0$ and $\epsilon < 1$. Then, there exists a problem instance $(x, y) \in \Delta_n^2$ such that, for any $\pi \in (0, 1)$ and (ϵ, δ) -correct algorithm, it holds that:*

$$\mathbb{E}_{x, y}[\tau_\delta] \geq \Omega\left(\frac{\log(\frac{1}{\delta})}{\log(\frac{1}{1-\pi})}\right). \quad (1)$$

This holds both for General and Zero-Sum Games.

These results show that there exists an instance in which

¹Note that, in principle, the resulting set could even be empty.

²We use exactly this construction to prove Theorem 3.1.

learning $\mathcal{G}(x, y)$ and/or $\mathcal{Z}(x, y)$ is essentially impossible when $\alpha = 0$. Indeed, taking the limit of Equation (1) for $\pi \rightarrow 0$ leads to an infinite complexity lower bound. The cause of this phenomenon is intrinsic in the definition of an exact Nash equilibrium. Indeed, the condition for $A \in \mathcal{G}^x(x, y)$,³ i.e., $x^\top A y \leq e_i^\top A y \forall i \in [n]$, can equivalently be rewritten as follows:⁴

$$x^\top A y = e_i^\top A y \quad \forall i : x_i > 0 \quad (2a)$$

$$x^\top A y \leq e_i^\top A y \quad \forall i : x_i = 0. \quad (2b)$$

This explains the strong discontinuity of $\mathcal{G}^x(\cdot, y)$ when the strategy profile of the x -player changes support, as we observed in the example.

To mitigate this issue, a natural assumption is to restrict attention to problem instances in which each action is played with probability bounded away from zero. This is exactly what was done in the IRL problem (Metelli et al., 2023), where the authors have found a similar phenomenon for the inverse reinforcement learning problem. Specifically, for any x, y , let $\pi_{\min}(x, y) \in (0, \frac{1}{n}]$ as:

$$\pi_{\min}(x, y) = \min \left\{ \min_{i: x_i > 0} x_i, \min_{j: y_j > 0} y_j \right\}.$$

Then, we assume that the agent is dealing with a set of instances such that $\pi_{\min}(x, y) \geq \pi_{\min}$ for some $\pi_{\min} > 0$.⁵ In this case, we derive the following lower bound.

Theorem 3.2. *Let $\alpha = 0$, $\pi_{\min} > 0$, and $\epsilon < 1/384$. Then, there exists a problem instance (x, y) such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}_{x, y}[\tau_\delta] \geq \Omega \left(\frac{\log(\frac{1}{\delta})}{\log(\frac{1}{1-\pi_{\min}})} + \frac{n + \log(\frac{1}{\delta})}{\epsilon^2} \right). \quad (3)$$

This result holds both for General and Zero-Sum Games.

We can interpret the first term as the worst-case complexity of learning the support of (x, y) . The second term, instead, arises from the fact that, even after the support is identified, the strategies (x, y) must still be estimated accurately in order to return an (ϵ, δ) -correct set in the Hausdorff metric.

³Similar reasoning holds for $\mathcal{G}^y(x, y)$ and $\mathcal{Z}(x, y)$ as well.

⁴This is a well-known result, e.g., Gintis (2000, Section 3.6). For completeness, we report a formal statement in Lemma D.8.

⁵We observe that in some settings assuming that all actions are played with positive probability could be too demanding. Another possibility, would be to relax the learning requirement, e.g., to retrieve the set of matrices for which (x, y) is an α -Nash equilibrium with $\alpha > 0$. As we discuss later in this section, this problem is learnable. Furthermore, since (x, y) is an exact equilibrium, the true (unknown) payoffs also belong to $\mathcal{G}_\alpha(x, y)$ or $\mathcal{Z}_\alpha(x, y)$. In this way, one can still guarantee to recover a set that contains an approximation of the true payoffs.

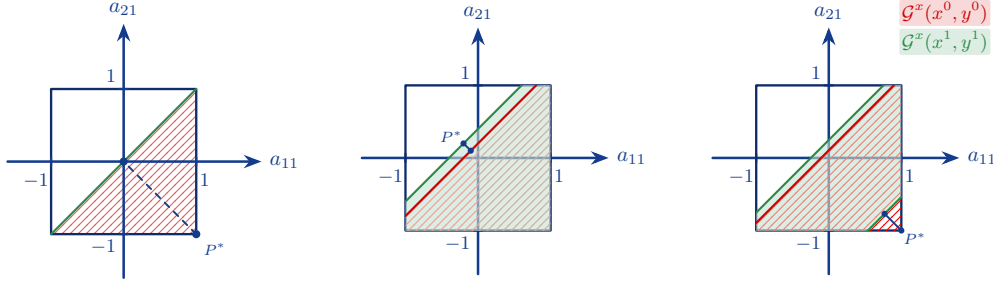


Figure 1. Visual representations of payoff sets $\mathcal{G}_\alpha^x(\cdot, \cdot)$ for several pairs of instances (x^0, y^0) (striped red region) and (x^1, y^1) (solid green region). In each figure, a point P^* is used to illustrate the Hausdorff distance between the two areas. The y vector is $(1, 0)$ in all problems. (Left): we set $\alpha = 0$, $x^0 = (0, 1)$, $x^1 = (\pi, 1 - \pi)$ for any $\pi \in (0, 1)$; (Center): $\alpha = 0.2$, $x^0 = (0, 1)$, and $x^1 = (\pi, 1 - \pi)$ for $\pi = 0.05$; (Right): $\alpha = 0.1$, $\epsilon = 0.1$, $x^0 = (\frac{\alpha}{2}, 1 - \frac{\alpha}{2})$ and $x^1 = (\frac{\alpha(1+3\epsilon)}{2}, 1 - \frac{\alpha(1+3\epsilon)}{2})$.

3.2. On Approximate Equilibria

In this section we provide a lower bound for the case in which players are playing an α -approximate Nash equilibrium. A natural question is whether the difficulty of learning the support of the equilibrium strategies again leads to an infinite sample-complexity bound. We argue that, in this setting, the answer is *no*. The presence of α in the definitions of $\mathcal{G}_\alpha(x, y)$ and $\mathcal{Z}_\alpha(x, y)$ prevents these sets from having a representation analogous to Equation (2). Intuitively, this removes the strong discontinuity exhibited by $\mathcal{G}(x, y)$ and $\mathcal{Z}(x, y)$ when (x, y) vary across strategy profiles with different supports. Informally, the parameter α acts as a tolerance that absorbs the error contributed by actions that are played with small probability. To have an intuitive picture of this phenomenon, it is instructive to consider the simple instance that we analyzed for the $\alpha = 0$ setting. We recall that $y = y^0 = y^1 = (1, 0)$, $x^0 = (0, 1)$, and $x^1 = (\pi, 1 - \pi)$ for some π . Let $\Delta_A = a_{21} - a_{11}$. Then, we have that

$$\begin{aligned} A \in \mathcal{G}_\alpha^x(x^0, y^0) &\Leftrightarrow -2 \leq \Delta_A \leq \alpha; \\ A \in \mathcal{G}_\alpha^x(x^1, y^1) &\Leftrightarrow \Delta_A \in [\max\{-2, -\frac{\alpha}{\pi}\}, \min\{2, \frac{\alpha}{1-\pi}\}]. \end{aligned}$$

As $\pi \rightarrow 0$ (or, equivalently, as $\pi \ll \alpha$), the two sets collapse. Thus, α is absorbing the error of actions that are played rarely. These sets are depicted in Figure 1 (center).

While this discussion is reassuring, an important question remains: what is the *worst-case* dependence of the lower bound on the parameter α ? One can anticipate that, as $\alpha \rightarrow 0$, the sample complexity must diverge, since actions played with vanishing probability increasingly influence the geometry of the feasible sets. To address this question, we establish the following result.

Theorem 3.3. *Let $\alpha \in (0, 1/4)$ and $\epsilon \leq 1/128$. Then, there exists a problem instance (x, y) such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}_{x,y}[\tau_\delta] \geq \Omega\left(\frac{n + \log(\frac{1}{\delta})}{\epsilon^2 \alpha}\right). \quad (4)$$

This result holds both for General and Zero-Sum Games.

Procedure 1

Require: Number of samples $m \in \mathbb{N}$, $\alpha, \epsilon, \delta \in \mathbb{R}_{\geq 0}$

- 1: **while** $t \in \{1, \dots, m\}$ **do**
- 2: Observe $X_t, Y_t \sim x, y$ and update $\hat{x}(t), \hat{y}(t)$
- 3: **end while**
- 4: Return $\mathcal{G}_\alpha(\hat{x}(m), \hat{y}(m))$ or $\mathcal{Z}_\alpha(\hat{x}(m), \hat{y}(m))$

An important aspect of this result is that the factor α^{-1} multiplies *all* the other terms in the bound. To understand why this happens, we provide some intuition on how we obtained the term $\log(\delta^{-1})/\epsilon^2 \alpha$.⁶ Consider two instances with $n = 2$ such that $y^1 = y^0 = (1, 0)$, $x^0 = (\frac{\alpha}{2}, 1 - \frac{\alpha}{2})$, and $x^1 = (\frac{\alpha+3\alpha\epsilon}{2}, 1 - \frac{\alpha+3\alpha\epsilon}{2})$. Again, let $\Delta_A = a_{21} - a_{11}$. One can verify that, for α and ϵ sufficiently small, it holds

$$\begin{aligned} A \in \mathcal{G}_\alpha^x(x^0, y^0) &\implies \Delta_A \geq -2; \\ A \in \mathcal{G}_\alpha^x(x^1, y^1) &\implies \Delta_A \geq -\frac{2\alpha}{\alpha+3\alpha\epsilon} > -2. \end{aligned}$$

These sets are depicted in Figure 1 (right). With some algebraic manipulations, one can show that the distance between these sets is at least 2ϵ and, therefore, it is important for any algorithm to distinguish (x^0, y^0) and (x^1, y^1) to be (ϵ, δ) -correct. However, for ϵ and α small, the two instances are very similar as the KL divergence between the two is at most $\epsilon^2 \alpha$. Therefore, using change-of-measure arguments, we can obtain the term $\log(\delta^{-1})/\epsilon^2 \alpha$. This shows that, while α provides a form of protection against the adverse geometry encountered when learning from exact equilibria, its effect is limited since all the other parameter the defines the instance are badly impacted by a small α .

4. Algorithm

The algorithm we developed and analyzed to attain minimax optimality is simple, and its pseudocode can be found in Procedure 1. It takes as input the parameter of the problem, i.e., α, ϵ and δ , and an additional parameter $m \in \mathbb{N}$, that denotes

⁶The term $n/(\epsilon^2 \alpha)$ follows from a related but more involved argument, which we defer to Appendix A.3.

the number of samples to collect. Then it simply gathers m observations for the two players x and y , and it stops by returning all the matrices that are an (α) -Nash equilibrium for the empirical strategy profiles. In other words, the recommendation rule is $\mathcal{G}_\alpha(\hat{x}(m), \hat{y}(m))$ for general-sum games and $\mathcal{Z}_\alpha(\hat{x}(m), \hat{y}(m))$ for zero sum games, where $\hat{x}(m)$ and $\hat{y}(m)$ are the maximum likelihood estimators of x and y after m samples (*i.e.*, empirical means). In the following, we drop the dependency on m in $\hat{x}(m)$ and $\hat{y}(m)$, and we simply write $\hat{x}$ and $\hat{y}$.

In the rest of this section, we show that, by properly choosing the parameter m , Procedure 1 is (ϵ, δ) -correct and, furthermore, this choice of m attains the same rate that we presented in the lower bound section. Specifically, in Section 4.1 we discuss our results for the exact equilibrium case and in Section 4.2 for the approximate one.

Representation of $\mathcal{G}_\alpha(\hat{x}, \hat{y})$ and $\mathcal{Z}_\alpha(\hat{x}, \hat{y})$ In the introductory section, we noted that the recommendation rule is in one-to-one correspondence with a set of matrices. Here, we note that the sets produced by our procedure can be represented as linear feasibility problems. Indeed, given $\hat{x}$ and $\hat{y}$, the set $\mathcal{G}_\alpha(\hat{x}, \hat{y})$ and $\mathcal{Z}_\alpha(\hat{x}, \hat{y})$ can be described by collections of linear inequalities.

4.1. Optimal Rates for Exact Equilibria

As discussed in Section 3.1, when $\alpha = 0$ the problem of learning the payoff sets is in general ill-posed, as the sample complexity may be infinite. Therefore, we adopt the same assumption of Theorem 3.2 and restrict our attention to instances in which $\pi_{\min}(x, y) \geq \pi_{\min} > 0$. In this setting, we can show the following.

Theorem 4.1. *Consider $\alpha = 0$ and let*

$$m \in \tilde{\mathcal{O}} \left(\frac{\log \left(\frac{n}{\delta} \right)}{\log \left(\frac{1}{1 - \pi_{\min}} \right)} + \frac{n + \log \left(\frac{1}{\delta} \right)}{\epsilon^2} \right).$$

Then, Procedure 1 is (ϵ, δ) -correct and its sample complexity τ_δ is given by m , both for General and Zero-Sum Games.⁷

Theorem 4.1 shows the same rate of the lower bound of Theorem 3.2, thus showing minimax optimality. As discussed in the proof sketch below, the first term can be interpreted as the number of samples required to learn the support of (x, y) , while the second one, is the number of samples needed for the Hausdorff distance between $\mathcal{G}(x, y)$ and $\mathcal{G}(\hat{x}, \hat{y})$ to be bounded by ϵ , once the support has been identified.

We now present a proof sketch of Theorem 4.1 for the general-sum game problem. Then, we will discuss how to extend this reasoning to the zero-sum setting.

⁷Here, $\tilde{\mathcal{O}}$ hides multiplicative constants and logarithmic factors.

Proof sketch of Theorem 4.1 (General-Sum Games)

The proof works as follows. We fix a generic $m \in \mathbb{N}$, and we derive a high-probability upper bound on the Hausdorff distance between $\mathcal{G}(x, y)$ and $\mathcal{G}(\hat{x}, \hat{y})$. Then, m is chosen so that this upper bound is below ϵ . The line that we follow to derive this upper bound is inspired by the lower bound of Section 3. In particular, we know that the support of (x, y) and $(\hat{x}, \hat{y})$ should match in order to avoid the degenerate behavior that we identified. Hence, we start with the assumption that (x, y) and $(\hat{x}, \hat{y})$ are such that $\text{supp}(x) = \text{supp}(\hat{x})$ and $\text{supp}(y) = \text{supp}(\hat{y})$, and we later discuss how to guarantee that this holds in high probability once m is large enough. Under the assumption that $\text{supp}(x) = \text{supp}(\hat{x})$ and $\text{supp}(y) = \text{supp}(\hat{y})$, we upper bound $H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y}))$ by showing that for any $(A, B) \in \mathcal{G}(x, y)$ there exists $(\hat{A}, \hat{B}) \in \mathcal{G}(\hat{x}, \hat{y})$ such that:⁸

$$\|A - \hat{A}\|_\infty \leq 4\|y - \hat{y}\|_1, \|B - \hat{B}\|_\infty \leq 4\|x - \hat{x}\|_1. \quad (5)$$

Note that this directly gives us an upper bound on the Hausdorff distance of $4(\|x - \hat{x}\|_1 + \|y - \hat{y}\|_1)$. In other words, the error scales linearly with the ℓ_1 -distance between the true strategy profile and the empirical one.

Now, we discuss how to obtain Equation (5) (we do that for A , and the construction for B follows from a similar reasoning). Starting from $A \in \mathcal{G}^x(x, y)$ (and assuming equal support), we know that if $\hat{A} \in \mathcal{G}^x(\hat{x}, \hat{y})$, then for all $i \in \text{supp}(x)$, $\hat{x}^\top \hat{A} \hat{y} = e_i^\top \hat{A} \hat{y}$. Then, since we want to construct $\hat{A}$ close to A , our goal is to "redistribute" the payoffs of A according to the shift of $\hat{x}$ and $\hat{y}$ so that, in all the rows in $\text{supp}(\hat{x})$, the value obtained by the x -player is left unchanged. To this end, for any $i \in \text{supp}(\hat{x})$, we first define the following quantity:

$$\Delta_i = \max_{k \in \text{supp}(\hat{x})} \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{j \in [n]} \hat{y}_j A_{ij}. \quad (6)$$

Here, $\Delta_i \in [0, 2\|y - \hat{y}\|_1]$ represents the maximum payoff difference (for the x -player) in playing the pure strategy i w.r.t. playing any other strategy $k \in \text{supp}(\hat{x})$. Intuitively, this is the shift, for actions in $\text{supp}(\hat{x})$, that $\hat{A}$ should meet so that $\hat{A} \in \mathcal{G}^x(\hat{x}, \hat{y})$. Instead, for all the rows that do not belong to $\text{supp}(\hat{x})$, we only need to ensure that their value remains greater than those of actions in $\text{supp}(\hat{x})$. Given these intuitions, we define $\hat{A}$ as follows:

$$\hat{A}_{ij} := \begin{cases} \frac{A_{ij}}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in S_x^*, \\ \frac{A_{ij} + \Delta_i^x}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in \text{supp}(\hat{x}) \setminus S_x^*, \\ \frac{A_{ij} + 2\|y - \hat{y}\|_1}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \notin \text{supp}(\hat{x}), \end{cases} \quad (7)$$

⁸We also show that the symmetric claim is true, since we need both directions to upper bound the Hausdorff distance.

where $S_{\hat{x}}^* = \operatorname{argmax}_{k \in \operatorname{supp}(\hat{x})} \sum_{j \in [n]} \hat{y}_j A_{kj}$. Ignoring the denominator, which is simply a rescaling to guarantee that $\hat{A}$ remains in $[-1, 1]^{n \times n}$, we observe that the construction does not modify values in $S_{\hat{x}}^*$, but it increases the values of other actions played in $\hat{x}$ so that any action in $\operatorname{supp}(\hat{x})$ yields the same value. Moreover, this construction increases the value of all the actions which are not played by $\hat{x}$, so that they do not obtain values which are better than actions that are actually played by $\hat{x}$. It is then possible to verify that $\hat{A}$ belongs to $\mathcal{G}^x(\hat{x}, \hat{y})$, and that $\|A - \hat{A}\|_\infty \leq 4\|y - \hat{y}\|_1$.

Finally, to conclude the proof, it is sufficient to upper bound with high-probability $\|x - \hat{x}\|_1$ and $\|y - \hat{y}\|_1$, and to guarantee that the support of (x, y) aligns with that of $(\hat{x}, \hat{y})$. Standard concentration tools yield $\mathcal{O}(\sqrt{(n + \log(\delta^{-1}))/m})$ for the former and $\tilde{\mathcal{O}}(\log(\delta^{-1})/\log((1 - \pi_{\min})^{-1}))$ for the latter. Putting everything together this gives Theorem 4.1.

What changes in Zero-Sum Games The proof for the zero-sum game problem follows from a similar argument; however, there is an important difference in how to upper bound the Hausdorff distance once the support of (x, y) and $(\hat{x}, \hat{y})$ matches. Specifically, constructing $\hat{A} \in \mathcal{Z}(\hat{x}, \hat{y})$ is more challenging as now $\hat{A}$ should satisfy the constraints for both players simultaneously. In particular, we show in Section B.2 that a matrix $\hat{A}$ constructed as in Equation (7) fails to meet the constraints for the y -player. To solve this issue (*i.e.*, building a matrix that accounts at the same time for changes in x and y), we show that we can upper bound $H(\mathcal{Z}(x, y), \mathcal{Z}(\hat{x}, \hat{y}))$ as:

$$H(\mathcal{Z}(x, y), \mathcal{Z}(\hat{x}, y)) + H(\mathcal{Z}(\hat{x}, y), \mathcal{Z}(\hat{x}, \hat{y})).$$

This error decomposition allows us to apply the technique for general-sum games even in the zero-sum setting.

4.2. Optimal Rates for Approximate Equilibria

Next, we present the results for the case in which the two players are playing an approximate Nash equilibrium. The following result summarizes our findings.

Theorem 4.2. *Consider $\alpha > 0$ and let*

$$m \in \mathcal{O}\left(\frac{n + \log\left(\frac{1}{\delta}\right)}{\epsilon^2 \alpha} + \frac{\sqrt{n} \log\left(\frac{1}{\delta}\right)}{\epsilon \alpha}\right).$$

Then Procedure 1 is (ϵ, δ) -correct and its sample complexity τ_δ is given by m , both for General and Zero-Sum Games.

For ϵ sufficiently small, *i.e.*, $\epsilon \lesssim n^{-1/2}$, Theorem 4.2 matches the rate of Theorem 3.3. Hence, Procedure 1 is minimax optimal in the relevant regime of small ϵ .

Proof sketch of Theorem 4.2 (General-Sum Games) As for Theorem 4.1, we fix a generic $m \in \mathbb{N}$ and derive a high

probability upper bound on the Hausdorff distance between the two sets as a function of m . Then, we choose m so that this upper bound is below ϵ . We start by employing the same decomposition that we used for zero-sum games and $\alpha = 0$ in order to upper bound $H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y}))$ with

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y)) + H(\mathcal{G}_\alpha(\hat{x}, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})).$$

This allows us to study only the error components that arise from one of the two strategy profiles, x or y . For clarity of exposition, we explain how to upper bound $H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y))$, and in particular we focus on the component corresponding to the first player, namely the sets $\mathcal{G}_\alpha^x(\cdot, \cdot)$ (as for $\mathcal{G}_\alpha^y(\cdot, \cdot)$ we have specular arguments).

Thus, we start by considering any $A \in \mathcal{G}_\alpha^x(x, y)$ and construct $\hat{A} \in \mathcal{G}_\alpha^x(\hat{x}, y)$ which is close to A . To this end, we follow a reasoning that is similar to the above, *i.e.*, we first quantify A 's constraint violation of $\mathcal{G}_\alpha^x(\hat{x}, y)$ and we try to correct that error with an appropriate construction. Specifically, let us denote by g_A the following quantity

$$g_A := \max_{i \in [n]} (\hat{x}^\top A y - e_i^\top A y).$$

If $g_A \leq \alpha$, then $A \in \mathcal{G}_\alpha^x(\hat{x}, y)$ as α is absorbing the error that arises from $\hat{x} \neq x$. If $g_A \geq \alpha$, instead, we construct $\hat{A}$ as $\hat{A} = \frac{\alpha}{g_A} A$. This multiplicative construction ensures that $\hat{A} \in \mathcal{G}_\alpha^x(\hat{x}, y)$ since, for all $i \in [n]$, we have that:

$$\hat{x}^\top \hat{A} y - e_i^\top \hat{A} y = \frac{\alpha}{g_A} (\hat{x}^\top A y - e_i^\top A y) \leq \alpha.$$

The difference between A and $\hat{A}$ is then given by:

$$\|A - \hat{A}\|_\infty = \mathcal{O}\left(\frac{g_A - \alpha}{g_A}\right) \leq \mathcal{O}\left(\frac{(\hat{x} - x)^\top A y}{\alpha}\right),$$

where the second step follows from $g_A \geq \alpha$ and the fact that $A \in \mathcal{G}_\alpha^x(x, y)$.⁹ Hence, to control the Hausdorff distance related to $\mathcal{G}_\alpha^x(x, y)$ and $\mathcal{G}_\alpha^x(\hat{x}, y)$, we need to control:

$$\frac{1}{\alpha} \sup_{A \in \mathcal{G}_\alpha^x(x, y)} (\hat{x} - x)^\top A y. \quad (8)$$

At first, one might be tempted to bound $(\hat{x} - x)^\top A y$ with $2\|x - \hat{x}\|_1$. However, this naive argument yields sub-optimal rates. Since with high probability $\|x - \hat{x}\|_1 \leq \sqrt{(n + \log(\delta^{-1}))/m}$, the number of samples to ensure that Equation (8) is below ϵ would roughly be $(n + \log(\delta^{-1}))/\epsilon^2 \alpha^2$, which does not match the lower bound of Theorem 3.3. The difficulty is that the bound $(\hat{x} - x)^\top A y \leq 2\|x - \hat{x}\|_1$ fails to exploit the fact that $A \in \mathcal{G}_\alpha^x(x, y)$ and does not leverage its geometric structure.

In the following, we show how to obtain better rates by exploiting the structure of $\mathcal{G}_\alpha^x(x, y)$. Indeed, the properties

⁹We add and subtract $x^\top A y$. Then we use $x^\top A y - e_i^\top A y \leq \alpha$.

of $\mathcal{G}_\alpha^x(x, y)$ will ensure that the variance of $(\hat{x} - x)^\top Ay$ is small, thus leading to faster high-probability concentration. To this end, we need to manipulate appropriately Equation (8). First of all, we bound Equation (8) as follows

$$\sup_{A \in \mathcal{G}_\alpha^x(x, y)} \frac{\sum_{i: x_i > 0} (\hat{x}_i - x_i)(Ay)_i}{\alpha} + \frac{2 \sum_{i: x_i = 0} \hat{x}_i}{\alpha}, \quad (9)$$

to get rid of the components where $x_i = 0$ in the optimization problem. Let us ignore the second term for now, and focus on the remaining optimization over A (we return to the second term later in the proof sketch). We will show that this remaining optimization problem presents a hidden fractional knapsack structure. First, let us introduce some notation. For any $A \in \mathcal{G}_\alpha^x(x, y)$, let $\star \in [n]$ be any row that belongs to $\arg\min_{i \in [n]} e_i^\top Ay$. Then, let $c_i = x_i[(Ay)_i - (Ay)_\star]$. Then, we can rewrite that optimization problem as:

$$\frac{1}{\alpha} \sup_{c: c_i \in [0, 2x_i]} \sum_{i: x_i > 0} \frac{\hat{x}_i - x_i}{x_i} c_i \quad \text{s.t.} \quad \sum_{i \in [n]} c_i \leq \alpha. \quad (10)$$

This problem is well-known to be solved by a greedy algorithm that sorts the indexes i 's according to their value (i.e., $\frac{\hat{x}_i - x_i}{x_i}$) and allocates the maximum value of c_i (i.e., $c_i = 2x_i$), according to this order until the budget is fully depleted, i.e., it stops once $\sum_{i: x_i > 0} c_i > \alpha$. Equation (10) can be rewritten as (exact steps in Section C.1):

$$\frac{1}{\alpha} \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i),$$

where $\mathcal{S}_\alpha(x)$ is given by

$$\mathcal{S}_\alpha(x) := \{S \subseteq [n] : x_i > 0 \forall i \in S \wedge \sum_{i \in S} x_i \leq \alpha/2\}.$$

Now, the main point is that each set $S \in \mathcal{S}_\alpha(x)$ is such that $\sum_{i \in S} x_i \leq \alpha/2$. Hence, we can treat $\sum_{i \in S} (\hat{x}_i - x_i)$ as a Bernoulli random variable with mean bounded by $\alpha/2$ (and variance bounded by $\alpha/2$). Using Bernstein's inequality, we have that $\sum_{i \in S} (\hat{x}_i - x_i)$ concentrates as $\mathcal{O}(\sqrt{\alpha \log(\delta^{-1})/m})$. Taking an union bound over all the possible subsets $S \in \mathcal{S}_\alpha(x)$, we obtain a rate of $\mathcal{O}(\sqrt{\alpha(n + \log(\delta^{-1})/m)})$. Plugging this result in Equation (9), we have that only $\mathcal{O}(\frac{n + \log(\delta^{-1})}{\epsilon^2 \alpha})$ samples are needed to have the first error term below ϵ .

Now, we still need to control the second term of Equation (9). It is easy to see that this term is 0. Indeed, given that $\hat{x}_i$ is the empirical mean of x_i , $x_i = 0 \implies \hat{x}_i = 0$. However, this only holds for one direction of the argument, i.e., when we are finding matrices $\hat{A} \in \mathcal{G}_\alpha^x(\hat{x}, y)$ which are close to $A \in \mathcal{G}_\alpha^x(x, y)$. In the opposite case, we obtain $2 \sum_{i: \hat{x}_i = 0} x_i$, which is commonly referred to as *missing mass* (McAllester and Ortiz, 2003) and is not equal to 0.¹⁰

¹⁰There is another asymmetry in bounding that direction of the

However, Rajaraman et al. (2020) has shown that this term concentrates around its expectation fast w.r.t. to the number of samples, i.e., it decays as $\frac{\sqrt{n \log(\delta^{-1})}}{m}$ and hence only $\sqrt{n \log(\delta^{-1})}/\epsilon \alpha$ samples are needed to have that term below ϵ .¹¹ In this way, we concluded the proof of Theorem 4.2.

What changes in Zero-Sum Games For the zero-sum game problem, we follow an analogous argument. The main difference is that we define g_A as:

$$g_A := \max \left\{ \max_{i \in [n]} \hat{x}^\top Ay - e_i^\top Ay, \max_{j \in [n]} \hat{x}^\top Ae_j - \hat{x}^\top Ay \right\}.$$

This is due to the fact that the matrix $\hat{A}$ now needs to satisfy a larger set of constraints. We show that the errors introduced by this modification are bounded as in the general-sum case.

The challenges for $\alpha > 0$ We conclude by highlighting the challenges introduced by the case $\alpha > 0$. The main difficulty lies in the fact that when the players are playing an approximate equilibrium, we cannot rely on strong characterizations of their behavior, i.e., Equation (2), as we did for the $\alpha = 0$ case. Furthermore, Theorem 3.3 shows that the dependence in α is complex, as it multiplies all the parameters of the problem due to geometric arguments that impact how much the Hausdorff distance changes when the strategy profiles have values around α .

5. Conclusions

In this paper, we study the problem of learning from demonstrations the payoff sets that are compatible with an unknown, possibly approximate, equilibrium. We derived tight minimax rates by providing matching lower and upper bounds. As we highlighted, these results required careful considerations on the geometry of the set of feasible payoffs. Our work paves the way for several promising directions for future research. For instance, it would be interesting to extend these results beyond the notion of approximate Nash equilibrium to broader solution concepts, such as correlated equilibria. Another line of possible research, instead, could aim to extend these results beyond bimatrix games with a finite number of actions. Indeed, in several games the set of actions for each player is an infinite set. Furthermore, in this work, we assumed that the strategy profiles (x, y) are fixed. Future research should focus on the case where strategy profiles are evolving, and players are learning the game through regret minimization. Finally, we note that our analysis is instance-independent. In the future, it would

Hausdorff distance. Specifically, when constructing $A \in \mathcal{G}_\alpha(x, y)$ close to some $\hat{A} \in \mathcal{G}_\alpha(\hat{x}, y)$, the set $\mathcal{S}_\alpha(\hat{x})$ is stochastic. In the proof, we show that we can reduce it to $\mathcal{S}_{2\alpha}(x)$ by only paying a minor order term in the sample complexity.

¹¹Furthermore, the expectation decays also as $\mathcal{O}(m^{-1})$.

be exciting to develop instance-dependent lower and upper bounds that directly depend on the unknown strategy profiles (x, y) . Here, one could take inspiration from the optimal instance-dependent results available in the bandit literature (e.g., Garivier and Kaufmann, 2016; Poiani et al., 2026). In particular, in Poiani et al. (2026), the authors deals with optimal instance-dependent rates for problems that admits an infinite set of (euclidean) answers. In the future, it would be interesting to extend these results to the inverse learning problem that we presented in this work.

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Impact Statement

This paper presents work aimed at advancing the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

Pieter Abbeel and Andrew Y Ng. Apprenticeship learning via inverse reinforcement learning. In *Proceedings of the twenty-first international conference on Machine learning*, page 1, 2004.

Sydney N Afriat. The construction of utility functions from expenditure data. *International economic review*, 8(1): 67–77, 1967.

Maria-Florina Balcan, Amit Daniely, Ruta Mehta, Ruth Urner, and Vijay V Vazirani. Learning economic parameters from revealed preferences. In *International Conference on Web and Internet Economics*, pages 338–353. Springer, 2014.

Eyal Beigman and Rakesh Vohra. Learning from revealed preference. In *Proceedings of the 7th ACM Conference on Electronic Commerce*, pages 36–42, 2006.

Stéphane Boucheron, Gábor Lugosi, and Olivier Bousquet. Concentration inequalities. In *Summer school on machine learning*, pages 208–240. Springer, 2003.

Donald J Brown and Rosa L Matzkin. Testable restrictions

on the equilibrium manifold. *Econometrica: Journal of the Econometric Society*, pages 1249–1262, 1996.

Lyve Cui, Haoran Yu, Pierre Pinson, and Dario Paccagnan. Inverse game theory: an incenter-based approach. In *34th International Joint Conference on Artificial Intelligence, IJCAI 2025*, pages 3805–3813. International Joint Conferences on Artificial Intelligence, 2025.

Till Freihaut and Giorgia Ramponi. On feasible rewards in multi-agent inverse reinforcement learning. *arXiv preprint arXiv:2411.15046*, 2024.

Justin Fu, Andrea Tacchetti, Julien Perolat, and Yoram Bachrach. Evaluating strategic structures in multi-agent inverse reinforcement learning. *Journal of Artificial Intelligence Research*, 71:925–951, 2021.

Aurélien Garivier and Emilie Kaufmann. Optimal best arm identification with fixed confidence. In *Conference on Learning Theory*, pages 998–1027. PMLR, 2016.

Herbert Gintis. *Game theory evolving: A problem-centered introduction to modeling strategic behavior*. Princeton university press, 2000.

Denizalp Goktas, Amy Greenwald, Sadie Zhao, Alec Koppel, and Sumitra Ganesh. Efficient inverse multiagent learning. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Representation Learning*, volume 2024, pages 2469–2482, 2024.

Nika Haghtalab, Fei Fang, Thanh H. Nguyen, Arunesh Sinha, Ariel D. Procaccia, and Milind Tambe. Three strategies to success: learning adversary models in security games. In *Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence, IJCAI’16*, page 308–314, 2016.

Chi Jin, Akshay Krishnamurthy, Max Simchowitz, and Tiancheng Yu. Reward-free exploration for reinforcement learning. In *International Conference on Machine Learning*, pages 4870–4879. PMLR, 2020.

Anders Jonsson, Emilie Kaufmann, Pierre Ménard, Omar Darwiche Domingues, Edouard Leurent, and Michal Valko. Planning in markov decision processes with gap-dependent sample complexity. *Advances in Neural Information Processing Systems*, 33:1253–1263, 2020.

Volodymyr Kuleshov and Okke Schrijvers. Inverse game theory: Learning utilities in succinct games. In *International Conference on Web and Internet Economics*, pages 413–427. Springer, 2015.

- David Lindner, Andreas Krause, and Giorgia Ramponi. Active exploration for inverse reinforcement learning. *Advances in Neural Information Processing Systems*, 35: 5843–5853, 2022.
- Chun Kai Ling, Fei Fang, and J. Zico Kolter. What game are we playing? end-to-end learning in normal and extensive form games. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*, pages 396–402. International Joint Conferences on Artificial Intelligence Organization, 7 2018.
- Charles F Manski. *Partial identification of probability distributions*. Springer, 2003.
- Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. *arXiv preprint arXiv:0907.3740*, 2009.
- David McAllester and Luis Ortiz. Concentration inequalities for the missing mass and for histogram rule error. *Journal of Machine Learning Research*, 4(Oct):895–911, 2003.
- Alberto Maria Metelli, Matteo Pirotta, and Marcello Restelli. Compatible reward inverse reinforcement learning. *Advances in neural information processing systems*, 30, 2017.
- Alberto Maria Metelli, Giorgia Ramponi, Alessandro Concetti, and Marcello Restelli. Provably efficient learning of transferable rewards. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021*, volume 139 of *Proceedings of Machine Learning Research*, pages 7665–7676. PMLR, 2021.
- Alberto Maria Metelli, Filippo Lazzati, and Marcello Restelli. Towards theoretical understanding of inverse reinforcement learning. In *International Conference on Machine Learning*, pages 24555–24591. PMLR, 2023.
- Sriraam Natarajan, Gautam Kunapuli, Kshitij Judah, Prasad Tadepalli, Kristian Kersting, and Jude Shavlik. Multi-agent inverse reinforcement learning. In *2010 ninth international conference on machine learning and applications*, pages 395–400. IEEE, 2010.
- Denis Nekipelov, Vasilis Syrgkanis, and Eva Tardos. Econometrics for learning agents. In *Proceedings of the sixteenth acm conference on economics and computation*, pages 1–18, 2015.
- Andrew Y. Ng and Stuart Russell. Algorithms for inverse reinforcement learning. In Pat Langley, editor, *Proceedings of the Seventeenth International Conference on Machine Learning (ICML 2000)*, pages 663–670, 2000.
- Riccardo Poiani, Curti Gabriele, Alberto Maria Metelli, and Marcello Restelli. Sub-optimal experts mitigate ambiguity in inverse reinforcement learning. *Advances in Neural Information Processing Systems*, 37:85778–85823, 2024.
- Riccardo Poiani, Martino Bernasconi, and Andrea Celli. Pure exploration with infinite answers. In *The 29th International Conference on Artificial Intelligence and Statistics*, 2026.
- Nived Rajaraman, Lin Yang, Jiantao Jiao, and Kannan Ramchandran. Toward the fundamental limits of imitation learning. *Advances in Neural Information Processing Systems*, 33:2914–2924, 2020.
- Nathan D Ratliff, J Andrew Bagnell, and Martin A Zinkevich. Maximum margin planning. In *Proceedings of the 23rd international conference on Machine learning*, pages 729–736, 2006.
- Tummalapalli Sudhamsh Reddy, Vamsikrishna Gopikrishna, Gergely Zaruba, and Manfred Huber. Inverse reinforcement learning for decentralized non-cooperative multiagent systems. In *2012 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 1930–1935. IEEE, 2012.
- R Tyrrell Rockafellar and Roger JB Wets. Stochastic variational inequalities: single-stage to multistage. *Mathematical Programming*, 165(1):331–360, 2017.
- Paul A Samuelson. Consumption theory in terms of revealed preference. *Economica*, 15(60):243–253, 1948.
- Arunesh Sinha, Debarun Kar, and Milind Tambe. Learning adversary behavior in security games: A pac model perspective. In *Proceedings of the 2016 International Conference on Autonomous Agents & Multiagent Systems, AAMAS ’16*, page 214–222. International Foundation for Autonomous Agents and Multiagent Systems, 2016.
- Elie Tamer. Partial identification in econometrics. *Annu. Rev. Econ.*, 2(1):167–195, 2010.
- Jingwu Tang, Gokul Swamy, Fei Fang, and Zhiwei S Wu. Multi-agent imitation learning: Value is easy, regret is hard. *Advances in Neural Information Processing Systems*, 37:27790–27816, 2024.
- Hal R Varian. The nonparametric approach to demand analysis. *Econometrica: Journal of the Econometric Society*, pages 945–973, 1982.
- Yevgeniy Vorobeychik, Michael P Wellman, and Satinder Singh. Learning payoff functions in infinite games. *Machine Learning*, 67(1):145–168, 2007.

- Kevin Waugh, Brian D Ziebart, and J Andrew Bagnell. Computational rationalization: the inverse equilibrium problem. In *Proceedings of the 28th International Conference on International Conference on Machine Learning*, pages 1169–1176, 2011.
- Jibang Wu, Weiran Shen, Fei Fang, and Haifeng Xu. Inverse game theory for stackelberg games: the blessing of bounded rationality. *Advances in Neural Information Processing Systems*, 35:32186–32198, 2022.
- Lantao Yu, Jiaming Song, and Stefano Ermon. Multi-agent adversarial inverse reinforcement learning. In *International Conference on Machine Learning*, pages 7194–7201. PMLR, 2019.
- Yue Yu, Jonathan Salfity, David Fridovich-Keil, and Ufuk Topcu. Inverse matrix games with unique quantal response equilibrium. *IEEE Control Systems Letters*, 7: 643–648, 2022.
- Morteza Zadimoghaddam and Aaron Roth. Efficiently learning from revealed preference. In *International Workshop on Internet and Network Economics*, pages 114–127. Springer, 2012.
- Siliang Zeng, Chenliang Li, Alfredo Garcia, and Mingyi Hong. Maximum-likelihood inverse reinforcement learning with finite-time guarantees. *Advances in Neural Information Processing Systems*, 35:10122–10135, 2022.
- Xiangyuan Zhang, Kaiqing Zhang, Erik Miehling, and Tamer Basar. Non-cooperative inverse reinforcement learning. *Advances in neural information processing systems*, 32, 2019.

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Table 1. Table of Symbols

Symbol	Description
n	Number of actions per player
$[n]$	Set $\{1, \dots, n\}$
Δ_n	Probability simplex over $[n]$
e_i	i -th canonical basis vector
$\text{supp}(x)$	Support of $x \in \Delta_n$
A, B	Payoff matrices of a general-sum bimatrix game
x, y	Mixed strategies of the two players
α	Approximation parameter for α -Nash equilibrium
$\mathcal{G}_\alpha(x, y)$	Set of payoff matrices (A, B) for which (x, y) is an α -Nash equilibrium
$\mathcal{G}_\alpha^x(x, y)$	Player-1 feasible payoff matrices
$\mathcal{G}_\alpha^y(x, y)$	Player-2 feasible payoff matrices
$\mathcal{Z}_\alpha(x, y)$	Zero-sum payoff matrices for which (x, y) is an α -Nash equilibrium
$\hat{\mathcal{G}}_\alpha, \hat{\mathcal{Z}}_\alpha$	Learner's estimate of the feasible payoff set
(X_t, Y_t)	Actions observed at round t , sampled from $x \times y$
$\hat{x}, \hat{y}$	Empirical estimates of strategies
m	Number of observed samples
τ_δ	Stopping time of the learning algorithm
$H(\cdot, \cdot)$	Hausdorff distance (w.r.t. ℓ_∞ norm)
$\ \cdot\ _1, \ \cdot\ _\infty$	ℓ_1 and ℓ_∞ norms
ϵ	Target Hausdorff accuracy
δ	Failure probability
$\pi_{\min}(x, y)$	Minimum positive probability in supports of x and y
$\pi_{\min}$	Known lower bound on $\pi_{\min}(x, y)$

A. Proof of the lower bounds

A.1. Proof of Theorem 3.1

Proof. The proof is split into several steps.

Step 1: Instance Construction To prove the result, we consider the case where $n = 2$. Then, we consider two different instances where the unknown strategy profiles (x^0, y^0) and (x^1, y^1) are defined as follows:

$$\begin{aligned} x^0 &= (0, 1), y^0 = (1, 0) \\ x^1 &= (\pi, 1 - \pi), y^1 = (1, 0) \end{aligned}$$

where π is any number in $(0, 1)$.

Step 2: Hausdorff Distance Next, we analyze the distances $H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1))$ and $H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1))$. In particular, we will provide lower bounds on these quantities.

We start from $H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1))$. First of all, we note that:

$$H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1)) \geq \max_{A \in \mathcal{G}^x(x^0, y^0)} \min_{A' \in \mathcal{G}^x(x^1, y^1)} \|A - A'\|_\infty. \quad (11)$$

Next, we analyze this lower bound in our simplified case. For any matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in [-1, 1]^{2 \times 2}$, using the definition of $\mathcal{G}^x(x, y)$, we have that:

- (i) $A \in \mathcal{G}^x(x^0, y^0) \iff a_{21} \leq a_{11} \text{ and } a_{21} \leq a_{21} \iff a_{21} \leq a_{11};$
- (ii) $A \in \mathcal{G}^x(x^1, y^1) \iff \pi a_{11} + (1 - \pi)a_{21} \leq \min\{a_{11}, a_{21}\} \iff a_{21} \leq a_{11} \text{ and } a_{11} \leq a_{21} \iff a_{21} = a_{11}.$

We now use this result to further lower bound the r.h.s. of Equation (11). In particular, we have that:

$$\begin{aligned} \max_{A \in \mathcal{G}^x(x^0, y^0)} \min_{A' \in \mathcal{G}^x(x^1, y^1)} \|A - A'\|_\infty &= \max_{a_{11}, a_{21} \in [-1, 1]^2: a_{11} \geq a_{21}} \min_{t \in [-1, 1]} \max\{|a_{11} - t|, |a_{21} - t|\} \\ &\geq \min_{t \in [-1, 1]} \max\{|1 - t|, |1 + t|\} \\ &= 1, \end{aligned}$$

where the first equality is justified by points (i) and (ii) above, together with the fact that for any $A \in \mathcal{G}(x^0, y^0)$ we can pick $A' \in \mathcal{G}(x^1, y^1)$ such that $a_{22} = a'_{22}$ and $a_{12} = a'_{12}$, and the inequality follows by setting $a_{11} = 1$ and $a_{21} = -1$.

Similarly we can analyze $H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1))$. In this case, we have:

$$\begin{aligned} \text{(i) } A \in \mathcal{Z}(x^0, y^0) &\iff a_{22} \leq a_{21} \leq a_{11}; \\ \text{(ii) } A \in \mathcal{Z}(x^1, y^1) &\iff \pi a_{21} + (1 - \pi)a_{22} \leq \pi a_{11} + (1 - \pi)a_{21} \leq \min\{a_{11}, a_{21}\}. \end{aligned}$$

Hence, we have that:

$$\begin{aligned} H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1)) &\geq \max_{A \in \mathcal{Z}(x^0, y^0)} \min_{A' \in \mathcal{Z}(x^1, y^1)} \|A - A'\|_\infty \\ &\geq \min_{A' \in \mathcal{Z}(x^1, y^1)} \max\{|1 - a'_{11}|, |-1 - a'_{21}|\} \quad (\text{Pick } A \in \mathcal{Z}(x^0, y^0) : a_{11} = 1, a_{21} = -1) \\ &\geq \min_{t \in [-1, 1]} \max\{|1 - t|, |1 + t|\} \quad (A' \in \mathcal{Z}(x^1, y^1) \implies a'_{11} = a'_{22}) \\ &= 1. \end{aligned}$$

Therefore, the following holds:

$$H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1)) \geq 1 \quad (12)$$

$$H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1)) \geq 1. \quad (13)$$

Step 3: Change of measure We first set some notation. For any $x, y \in \Delta_n$, we have that $\mathbb{P}_{x,y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, where $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$. Then, since Equations (12) and (13) hold (and since $\epsilon < 1$), we can apply Lemma D.6. Thus for any (ϵ, δ) -correct algorithm, for both GSGs and ZSGs we obtain

$$\delta \geq \frac{1}{4} \exp(-\text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1})).$$

We proceed by analyzing the r.h.s. of this equation. In the case we are considering we have set $n = 2$. Hence, x, y are Bernoulli distributions, and we obtain that:

$$\begin{aligned} \text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1}) &= \mathbb{E}_{x^0, y^0} \left[\sum_{t=1}^{\tau_\delta} \text{KL}(p_{x^0}(X_t), p_{x^1}(X_t)) \right] && (\text{Since } y^0 = y^1) \\ &= \mathbb{E}_{x^0, y^0} [\tau_\delta] \text{KL}(p_{x^0}, p_{x^1}) && (\text{Wald's identity}) \\ &= \mathbb{E}_{x^0, y^0} [\tau_\delta] \log \left(\frac{1}{1 - \pi} \right). && (\text{By definition of } x^0, x^1) \end{aligned}$$

Then, we obtained that:

$$\delta \geq \frac{1}{4} \exp \left(-\mathbb{E}_{x^0, y^0} [\tau_\delta] \log \left(\frac{1}{1 - \pi} \right) \right),$$

which yields:

$$\mathbb{E}_{x^0, y^0} [\tau_\delta] \geq \frac{\log \left(\frac{1}{4\delta} \right)}{\log \left(\frac{1}{1 - \pi} \right)},$$

thus concluding the proof. $\square$

A.2. Proof Theorem 3.2

Proof outline Theorem 3.2 is obtained as a combination of three distinct lower bounds, *i.e.*, Theorems A.1 to A.3. In these theorems, we prove that there exists instances where the sample complexity of any (ϵ, δ) -correct algorithm is lower bounded by:

$$(i) \frac{\log(\frac{1}{\delta})}{\log\left(\frac{1}{1-\pi_{\min}}\right)}, \quad (ii) \frac{\log\left(\frac{1}{\delta}\right)}{\epsilon^2}, \quad \text{and} \quad (iii) \frac{n}{\epsilon^2},$$

respectively (here, we neglected the exact multiplicative constants, which are listed below). Importantly, these results hold both for General-Sum and Zero-Sum Games. Then, Theorem 3.2 follows by the following argument.

Proof of Theorem 3.2. It is sufficient to consider the instances presented in Theorems A.1 to A.3. The results then follows from applying $\max\{a, b, c\} \geq \frac{1}{3}(a + b + c)$. $\square$

Now, we proceed by proving Theorems A.1 to A.3. We start from Theorem A.1, which is a simple corollary of Theorem 3.1.

Theorem A.1. *Let $\alpha = 0$ and $\epsilon < 1$. Then, there exists a problem instance such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}[\tau_\delta] \geq \frac{\log\left(\frac{1}{4\delta}\right)}{\log\left(\frac{1}{1-\pi_{\min}}\right)}.$$

This holds for both Zero-Sum and General-Sum Games.

Proof. The results follow from applying Theorem 3.1 with $\pi = \pi_{\min}$. $\square$

We now continue with Theorem A.2.

Theorem A.2. *Let $\alpha = 0$ and $\epsilon < \frac{1}{\sqrt{32}}$. Then, there exists a problem instance such that, for any (ϵ, δ) -correct algorithm, it holds that*

$$\mathbb{E}[\tau_\delta] \geq \frac{\log\left(\frac{1}{4\delta}\right)}{16\epsilon^2}.$$

This holds both for Zero-Sum and General-Sum Games.

Proof. The proof is split into three steps

Step 1: Instance Construction We consider the setting with $n = 2$. Then, we consider two different instances where the unknown strategy profiles (x^0, y^0) and (x^1, y^1) are defined as follows:

$$\begin{aligned} x^0 &= \left(\frac{1}{2}, \frac{1}{2}\right), \quad y^0 = \left(\frac{1}{2}, \frac{1}{2}\right) \\ x^1 &= \left(\frac{1}{2} + \gamma, \frac{1}{2} - \gamma\right), \quad y^1 = \left(\frac{1}{2}, \frac{1}{2}\right), \end{aligned}$$

where $\gamma \in (0, 1/2)$ is a real number that will be specified later in the proof.

Step 2: Hausdorff Distance Next, we analyze the distances $H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1))$ and $H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1))$. In particular, we will provide lower bounds on these quantities.

We start from $H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1))$. We have that:

$$\begin{aligned} H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1)) &\geq \max_{B \in \mathcal{G}^y(x^0, y^0)} \min_{B' \in \mathcal{G}^y(x^1, y^1)} \|B - B'\|_\infty \\ &\geq \min_{B' \in \mathcal{G}^y(x^1, y^1)} \max\{|1 - b'_{11}|, |b'_{12}|, |b'_{21}|, |1 - b'_{22}|\}, \end{aligned}$$

where, in the second step, we have set $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathcal{G}^y(x^0, y^0)$ and $B' = \begin{bmatrix} b'_{11} & b'_{12} \\ b'_{21} & b'_{22} \end{bmatrix}$.¹² Now, using the values of (x^1, y^1) , we can rewrite the last optimization problem as follows:

$$\begin{aligned} & \min_{B'} \max\{|1 - b'_{11}|, |b'_{12}|, |b'_{21}|, |1 - b'_{22}|\} \\ \text{s.t. } & b'_{ij} \in [-1, 1] \quad \forall i, j \in [n] \\ & \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (b'_{11} + b'_{12}) + \left(\frac{1}{2} - \gamma \right) (b'_{21} + b'_{22}) \right) \leq b'_{11} \left(\frac{1}{2} + \gamma \right) + b'_{21} \left(\frac{1}{2} - \gamma \right), \\ & \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (b'_{11} + b'_{12}) + \left(\frac{1}{2} - \gamma \right) (b'_{21} + b'_{22}) \right) \leq b'_{12} \left(\frac{1}{2} + \gamma \right) + b'_{22} \left(\frac{1}{2} - \gamma \right). \end{aligned}$$

Introducing an additional variable $t := \|B - B'\|_\infty \geq 0$ and simplifying the constraints, we can rewrite this optimization problem as follows:

$$\begin{aligned} & \min_{B' \in [-1, 1]^{2 \times 2}, t \geq 0} t \\ \text{s.t. } & |1 - b'_{11}| \leq t, \quad |b'_{12}| \leq t, \quad |b'_{21}| \leq t, \quad |1 - b'_{22}| \leq t \\ & (b'_{11} - b'_{12}) \left(\frac{1}{2} + \gamma \right) + (b'_{21} - b'_{22}) \left(\frac{1}{2} - \gamma \right) = 0. \end{aligned} \tag{14}$$

Let $t(B')$ be the optimal value of Problem (14) for any feasible matrix B' . In the following, we will prove that $t(B') \geq \gamma$ for any feasible matrix B' . To this end, let us introduce the following vector $g^\top = [(\frac{1}{2} + \gamma), (-\frac{1}{2} - \gamma), (\frac{1}{2} - \gamma), (-\frac{1}{2} + \gamma)]$. Then, the last linear constraints in (14) can be rewritten as $g^\top z(B') = 0$, where $z(B')$ denotes the flattened version of matrix B' , i.e., $z(B') = (b'_{11}, b'_{12}, b'_{21}, b'_{22})$. Next, let $z_0 = (1, 0, 0, 1)$. Note that $g^\top z_0 = 2\gamma$. Thus, for any feasible B' , $g^\top (z(B') - z_0) = -2\gamma$. Therefore,

$$\begin{aligned} 2\gamma &= |g^\top (z(B') - z_0)| \\ &\leq \|g\|_1 \|z(B') - z_0\|_\infty && \text{(Hölder's inequality)} \\ &= \|g\|_1 t(B') && \text{(Since } z_0 = z(B) \text{ and } t(B') = \|B' - B\|_\infty) \\ &= 2t(B') && \text{(Definition of } g \text{ for } \gamma \in (0, \frac{1}{2})) \end{aligned}$$

Thus, we obtained that, for any feasible matrix B' , $t(B') \geq \gamma$. Choosing $\gamma = 2\epsilon$,¹³ we obtained that

$$H(\mathcal{G}(x^0, y^0), \mathcal{G}(x^1, y^1)) \geq 2\epsilon. \tag{15}$$

Next, we analyze $H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1))$. We have that:

$$\begin{aligned} H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1)) &\geq \max_{A \in \mathcal{Z}(x^0, y^0)} \min_{A' \in \mathcal{Z}(x^1, y^1)} \|A - A'\|_\infty \\ &\geq \min_{A' \in \mathcal{Z}(x^1, y^1)} \max\{|1 - a'_{11}|, |a'_{12}|, |a'_{21}|, |1 - a'_{22}|\}, \end{aligned}$$

where, in the second step, we have set

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathcal{Z}(x^0, y^0).^{14}$$

Now, using the values of (x^1, y^1) , we can rewrite the last optimization problem as follows:

$$\begin{aligned} & \min_{A'} \max\{|1 - a'_{11}|, |a'_{12}|, |a'_{21}|, |1 - a'_{22}|\} \\ \text{s.t. } & a'_{ij} \in [-1, 1] \quad \forall i, j \in \{1, 2\} \\ & \left(\frac{1}{2} + \gamma \right) a'_{11} + \left(\frac{1}{2} - \gamma \right) a'_{21} \leq \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (a'_{11} + a'_{12}) + \left(\frac{1}{2} - \gamma \right) (a'_{21} + a'_{22}) \right) \\ & \left(\frac{1}{2} + \gamma \right) a'_{12} + \left(\frac{1}{2} - \gamma \right) a'_{22} \leq \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (a'_{11} + a'_{12}) + \left(\frac{1}{2} - \gamma \right) (a'_{21} + a'_{22}) \right) \\ & \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (a'_{11} + a'_{12}) + \left(\frac{1}{2} - \gamma \right) (a'_{21} + a'_{22}) \right) \leq \frac{1}{2} (a'_{11} + a'_{12}) \\ & \frac{1}{2} \left(\left(\frac{1}{2} + \gamma \right) (a'_{11} + a'_{12}) + \left(\frac{1}{2} - \gamma \right) (a'_{21} + a'_{22}) \right) \leq \frac{1}{2} (a'_{21} + a'_{22}). \end{aligned}$$

¹²Indeed, $(x^0)^\top B y^0 = \frac{1}{2}$ and $(x^0)^\top B e_j = \frac{1}{2}$ for all $j \in [n]$. Hence, all the constraints of $\mathcal{G}^y(x^0, y^0)$ are satisfied for such B .

¹³This choice is valid since $\gamma \in (0, \frac{1}{2})$ and $\epsilon < 1/4$ by assumption.

¹⁴Indeed, $(x^0)^\top A y^0 = \frac{1}{2}$, $e_j^\top A y^0 = \frac{1}{2}$ and $(x^0)^\top A e_j = \frac{1}{2}$ for all $j \in [n]$. Hence, all the constraints of $\mathcal{Z}(x^0, y^0)$ are satisfied.

Introducing the additional variable $t := \|A - A'\|_\infty \geq 0$ and simplifying the constraints, we can rewrite the above minimization as the following optimization problem:

$$\begin{aligned}
 & \min_{A' \in [-1,1]^{2 \times 2}, t \geq 0} t \\
 \text{s.t. } & |1 - a'_{11}| \leq t, \quad |a'_{12}| \leq t, \quad |a'_{21}| \leq t, \quad |1 - a'_{22}| \leq t, \\
 & (\tfrac{1}{2} + \gamma)(a'_{11} - a'_{12}) + (\tfrac{1}{2} - \gamma)(a'_{21} - a'_{22}) = 0, \\
 & \tfrac{1}{2} \left((\tfrac{1}{2} + \gamma)(a'_{11} + a'_{12}) + (\tfrac{1}{2} - \gamma)(a'_{21} + a'_{22}) \right) \leq \tfrac{1}{2}(a'_{11} + a'_{12}) \\
 & \tfrac{1}{2} \left((\tfrac{1}{2} + \gamma)(a'_{11} + a'_{12}) + (\tfrac{1}{2} - \gamma)(a'_{21} + a'_{22}) \right) \leq \tfrac{1}{2}(a'_{21} + a'_{22}).
 \end{aligned} \tag{16}$$

Now, as we did for General-Sum Games, let $t(A')$ be the optimal value of Problem (16) for any feasible matrix A' . In the following, we will prove that $t(A') \geq \gamma$ for any feasible matrix A' . To this end, let us introduce the following vector $g^\top = [(\frac{1}{2} + \gamma), (-\frac{1}{2} - \gamma), (\frac{1}{2} - \gamma), (-\frac{1}{2} + \gamma)]$. Then, the second constraint in (16) can be rewritten as $g^\top z(A') = 0$, where $z(A')$ denotes the flattened version of matrix A' , i.e., $z(A') = (a'_{11}, a'_{12}, a'_{21}, a'_{22})$. Next, let $z_0 = (1, 0, 0, 1)$. Note that $g^\top z_0 = 2\gamma$. Thus, for any feasible A' , $g^\top (z(A') - z_0) = -2\gamma$. Therefore,

$$\begin{aligned}
 2\gamma &= |g^\top (z(A') - z_0)| \\
 &\leq \|g\|_1 \|z(A') - z_0\|_\infty && \text{(Hölder's inequality)} \\
 &= \|g\|_1 t(A') && \text{(Since } z_0 = z(A) \text{ and } t(A') = \|A' - A\|_\infty) \\
 &= 2t(A'). && \text{(Definition of } g \text{ for } \gamma \in (0, \tfrac{1}{2}))
 \end{aligned}$$

Thus, we obtained that, for any feasible matrix A' , $t(A') \geq \gamma$. Choosing $\gamma = 2\epsilon$,¹⁵ we obtained that

$$H(\mathcal{Z}(x^0, y^0), \mathcal{Z}(x^1, y^1)) \geq 2\epsilon. \tag{17}$$

Step 3: Change of measure For any $x, y \in \Delta_n$, let $\mathbb{P}_{x,y} = \prod_{t=1}^{\tau_\delta} p_x(X_t)p_y(Y_t)$, where $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$. Then, since Equations (15) and (17) holds, we can apply Lemma D.6. Thus, both for GSGs and ZSGs, we have that:

$$\delta \geq \frac{1}{4} \exp(-\text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1})).$$

We proceed by analyzing the r.h.s. of this equation. In the case we are considering we have set $n = 2$. Hence, x, y are Bernoulli distributions, and we obtain that:

$$\begin{aligned}
 \text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1}) &= \mathbb{E}_{x^0, y^0} \left[\sum_{t=1}^{\tau_\delta} \text{KL}(p_{x^0}(X_t), p_{x^1}(X_t)) \right] && \text{(Since } y^0 = y^1) \\
 &= \mathbb{E}_{x^0, y^0} [\tau_\delta] \text{KL}(p_{x^0}, p_{x^1}) && \text{(Wald's identity)} \\
 &= \frac{1}{2} \mathbb{E}_{x^0, y^0} [\tau_\delta] \log \left(\frac{1}{1 - 16\epsilon^2} \right) && \text{(By def. of } x^0, x^1 \text{ and setting } \gamma = 2\epsilon) \\
 &\leq \mathbb{E}_{x^0, y^0} [\tau_\delta] \frac{8\epsilon^2}{1 - 16\epsilon^2} && (-\log(1 - x) \leq \frac{x}{1-x} \text{ for } x \leq 1) \\
 &\leq 16\mathbb{E}_{x^0, y^0} [\tau_\delta] \epsilon^2 && (\epsilon \leq \sqrt{\frac{1}{32}} \implies (1 - 16\epsilon^2) \geq \tfrac{1}{2})
 \end{aligned}$$

Hence, we obtained that:

$$\delta \geq \frac{1}{4} \exp(-16\mathbb{E}_{x^0, y^0} [\tau_\delta] \epsilon^2)$$

Rearranging the terms yields:

$$\mathbb{E}_{x^0, y^0} [\tau_\delta] \geq \frac{\log(\frac{1}{4\delta})}{16\epsilon^2},$$

which concludes the proof. $\square$

¹⁵This choice is valid since $\gamma \in (0, \frac{1}{2})$ and $\epsilon < 1/4$ by assumption.

Finally, we prove Theorem A.3.

Theorem A.3. *Let $\alpha = 0$, $n \geq 20$, $n \bmod 16 = 0$ and $\epsilon \leq \frac{1}{384}$. Then, there exists a problem instance such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}[\tau_\delta] \geq \frac{\log(2)n}{40(192)^2\epsilon^2}.$$

This holds both for Zero-Sum and General-Sum Games.

Proof. As before, the proof is split into three main steps.

Step 1: Instance Construction In the following, taking inspiration from the work of Jin et al. (2020), we construct a set of different n -dimensional instances. We assume that $n \geq 16$ and that $n \bmod 16 = 0$. These instances are parameterized with an n -dimensional vector $v \in \{-1, 1\}^n$ such that $\sum_{k=1}^n v_k = 0$ and are constructed as follows. We denote by $\mathcal{V}$ the set of vectors that satisfies these two requirements. Now, let $\gamma \in (0, \frac{1}{2})$; we define (x^v, y^v) as follows:

$$x^v = (1, 0, \dots, 0) \text{ and } y^v = \left(\frac{1 + \gamma v_1}{n}, \frac{1 + \gamma v_2}{n}, \dots, \frac{1 + \gamma v_n}{n} \right).$$

Observe that $n \geq 2$, $\gamma \in (0, \frac{1}{2})$, and $v \in \{-1, 1\}^n$. Therefore, by construction, we have that $y^v \in \Delta_n$.

In the rest of the proof, we restrict our attention to vectors v belonging to a subset $\bar{\mathcal{V}}$ of $\mathcal{V}$. In particular, consider a pair $v, w \in \bar{\mathcal{V}}$ and let

$$D_{v,w}^+ = \{k \in [n] : v_k = 1 \text{ and } w_k = -1\} \quad D_{v,w}^- = \{k \in [n] : v_k = -1 \text{ and } w_k = 1\},$$

where we will drop the explicit dependence on v, w when clear from context. Then, we require that for any pair $v, w \in \bar{\mathcal{V}}$ it holds

$$|D_{v,w}^+| = |D_{v,w}^-| \geq \frac{n}{64}. \quad (18)$$

It is easy to see that $|D_{v,w}^+| = |D_{v,w}^-|$ is granted for any pair $v, w \in \mathcal{V}$. The requirement that, e.g., $|D_{v,w}^+| \geq \frac{n}{64}$, is instead equivalent to impose that $\sum_{k=1}^n |v_k - w_k| \geq \frac{n}{16}$. Thanks to Lemma D.12, we know that:

$$\exists \bar{\mathcal{V}} \subseteq \mathcal{V} \text{ s.t. } \forall v, w \in \bar{\mathcal{V}}, \quad \sum_{k=1}^n |v_k - w_k| \geq \frac{n}{16} \text{ and } |\bar{\mathcal{V}}| > 2^{\frac{n}{5}}. \quad (19)$$

In the following step, we will consider two generic profiles (x^v, y^v) and (x^w, y^w) such that $v, w \in \bar{\mathcal{V}}$.

Step 2: Hausdorff Distance We analyze the distances $H(\mathcal{G}(x^v, y^v), \mathcal{G}(x^w, y^w))$ and $H(\mathcal{Z}(x^v, y^v), \mathcal{Z}(x^w, y^w))$. In particular, we will provide lower bounds on these quantities.

We start from $H(\mathcal{G}(x^v, y^v), \mathcal{G}(x^w, y^w))$. We have that:

$$\begin{aligned} H(\mathcal{G}(x^v, y^v), \mathcal{G}(x^w, y^w)) &\geq \max_{A \in \mathcal{G}^x(x^v, y^v)} \min_{A' \in \mathcal{G}^x(x^w, y^w)} \|A - A'\|_\infty \\ &\geq \min_{A' \in \mathcal{G}^x(x^w, y^w)} \|A - A'\|_\infty, \end{aligned}$$

where we define the matrix $A \in \mathcal{G}(x^v, y^v)$ as

$$A_{ij} := \begin{cases} 0, & i = 1, j \in [n] \\ \frac{1}{3}, & i > 1, j \in D^+, \\ -\frac{1}{3}\kappa, & i > 1, j \in D^-, \\ 0, & i > 1, j \in [n] \setminus (D^+ \cup D^-), \end{cases} \quad \text{where} \quad \kappa := \frac{\sum_{j \in D^+} y_j^v}{\sum_{j \in D^-} y_j^v}. \quad (20)$$

Before continuing, we prove that A constructed as in Equation (20) satisfies $A \in \mathcal{G}(x^v, y^v)$. To this end, we first verify that $A \in [-1, 1]^{n \times n}$. It suffices to check that $\kappa \leq 3$. With some simple manipulations, we have that:

$$\kappa = \frac{\sum_{j \in D^+} y_j^v}{\sum_{j \in D^-} y_j^v} = \frac{\frac{|D^+|(1+\gamma)}{n}}{\frac{|D^-|(1-\gamma)}{n}} = \frac{1+\gamma}{1-\gamma} \leq 3,$$

where the second equality is due to Equation (18) and the last step is due to $\gamma \in (0, \frac{1}{2})$. Next, we verify that A satisfies the Nash constraints. Since $x^v = e_1$, those constraints are equivalent to $(e_1 - e_i)^\top A y^v \leq 0$ for all $i \in [n]$. Using the definition of A and y^v we have that:

$$\begin{aligned} (e_1 - e_i)^\top A y^v &= \sum_{j=1}^n y_j^v (A_{1j} - A_{ij}) \\ &= - \sum_{j=1}^n y_j^v A_{ij} && (A_{1j} = 0 \text{ for all } j \in [n]) \\ &= -\frac{1}{3} \left(\sum_{j \in D^+} y_j^v - \kappa \sum_{j \in D^-} y_j^v \right) && (\text{Definition of } A) \\ &= 0. && (\text{Definition of } \kappa) \end{aligned}$$

Hence, all constraints are tight and $A \in \mathcal{G}^x(x^v, y^v)$.

We can now continue by lower bounding $\min_{A' \in \mathcal{G}^x(x^w, y^w)} \|A - A'\|_\infty$. In the following, we will prove that:

$$\min_{A' \in \mathcal{G}^x(x^w, y^w)} \|A - A'\|_\infty \geq \frac{2|D^+|\gamma}{3n(1-\gamma)}. \quad (21)$$

To this end, consider any matrix $A' \in \mathcal{G}^x(x^w, y^w)$. Let $\Delta(A') = A' - A$ so that $A' = \Delta(A') + A$. Since $x^w = e_1$, then A' satisfies $(e_1 - e_i)^\top A' y^w \leq 0$ for all $i \in [n]$. Moreover, for all $A' \in \mathcal{G}^x(x^w, y^w)$ it holds that:

$$\begin{aligned} (e_1 - e_i)^\top A' y^w &= (e_1 - e_i)^\top (\Delta(A') + A) y^w \\ &= (e_1 - e_i)^\top \Delta(A') y^w + (e_1 - e_i)^\top A y^w \\ &= (e_1 - e_i)^\top \Delta(A') y^w - \sum_{j=1}^n y_j^w A_{ij} && (A_{1j} = 0 \text{ for all } j \in [n]) \\ &= (e_1 - e_i)^\top \Delta(A') y^w - \frac{1}{3} \left(\frac{|D^+|(1-\gamma)}{n} - \kappa \frac{|D^-|(1+\gamma)}{n} \right) && (\text{Def. of } A, y^w, D^+ \text{ and } D^-.) \\ &= (e_1 - e_i)^\top \Delta(A') y^w - \frac{1}{3} \left(\frac{|D^+|(1-\gamma)}{n} - \frac{|D^+|(1+\gamma)^2}{n(1-\gamma)} \right) && (\text{Def. of } \kappa) \\ &= (e_1 - e_i)^\top \Delta(A') y^w + \frac{4|D^+|\gamma}{3n(1-\gamma)} \\ &\leq 0 && (A' \in \mathcal{G}^x(x^w, y^w)) \end{aligned}$$

Then, it follows that any feasible $A' \in \mathcal{G}^x(x^w, y^w)$ satisfies $(e_i - e_1)^\top \Delta(A') y^w \geq \frac{4|D^+|\gamma}{3n(1-\gamma)}$. This yields the following:

$$\begin{aligned} \frac{4|D^+|\gamma}{3n(1-\gamma)} &\leq (e_i - e_1)^\top \Delta(A') y^w \\ &\leq \sum_{j=1}^n y_j^w (|\Delta(A')_{ij}| + |\Delta(A')_{1j}|) \\ &\leq 2 \sum_{j=1}^n y_j^w \|\Delta(A')\|_\infty \\ &= 2\|A - A'\|_\infty, \end{aligned}$$

which proves Equation (21).

Combining this result with Equation (18) gives $H(\mathcal{G}(x^v, y^v), \mathcal{G}(x^w, y^w)) \geq \frac{\gamma}{96}$. Picking $\gamma = 192\epsilon$,¹⁶ we obtained

$$H(\mathcal{G}(x^v, y^v), \mathcal{G}(x^w, y^w)) \geq 2\epsilon. \quad (22)$$

Next we analyze $H(\mathcal{Z}(x^v, y^v), \mathcal{Z}(x^w, y^w))$. We have that:

$$\begin{aligned} H(\mathcal{Z}(x^v, y^v), \mathcal{Z}(x^w, y^w)) &\geq \max_{A \in \mathcal{Z}(x^v, y^v)} \min_{A' \in \mathcal{Z}(x^w, y^w)} \|A - A'\|_\infty \\ &\geq \min_{A' \in \mathcal{Z}(x^w, y^w)} \|A - A'\|_\infty, \end{aligned}$$

where the matrix $A \in \mathcal{Z}(x^v, y^v)$ is defined as in Equation (20). We already proved above that $A \in [-1, 1]^{n \times n}$. It remains to prove that it satisfies the Nash constraints for zero-sum games, *i.e.*, that A satisfies:

$$e_1^\top A e_j \leq e_1^\top A y^v \leq e_i^\top A y^v \quad \forall i, j \in [n].$$

We already proved above that $e_1^\top A y^v \leq e_i^\top A y^v$ holds for all $i \in [n]$. We only need to check that $e_1^\top A e_j \leq e_1^\top A y^v$ holds for all $j \in [n]$. Since $A_{1j} = 0$ for all $j \in [n]$, we have that:

$$e_1^\top A e_j = A_{1j} = 0 = \sum_{j \in [n]} y_j^v A_{1j} = e_1^\top A y^v,$$

thus showing that $A \in \mathcal{Z}(x^v, y^v)$.

We now continue by lower bounding $\min_{A' \in \mathcal{Z}(x^w, y^w)} \|A - A'\|_\infty$. We can follow the same reasoning that we used for General-Sum Games to prove that:

$$H(\mathcal{Z}(x^v, y^v), \mathcal{Z}(x^w, y^w)) \geq 2\epsilon. \quad (23)$$

Step 3: Change of measure For any $x, y \in \Delta_n$, let $\mathbb{P}_{x,y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, where $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$. Then, since Equations (22) and (23) holds, we can apply Lemma D.7 using the set of instances $\{(x^v, y^v)\}_{v \in \bar{\mathcal{V}}}$ and $(x^0, y^0) = (e_1, (\frac{1}{n}, \dots, \frac{1}{n}))$ as base instance. Thus, for both GSGs and ZSGs we have that for any (ϵ, δ) -correct algorithm it holds

$$\delta \geq 1 - \frac{1}{\log(|\bar{\mathcal{V}}|)} \left(\frac{1}{|\bar{\mathcal{V}}|} \sum_{v \in \bar{\mathcal{V}}} \text{KL}(\mathbb{P}_{x^v, y^v}, \mathbb{P}_{x^0, y^0}) + \log 2 \right).$$

We proceed by analyzing the kl-divergence term in the r.h.s. of this equation. Specifically, we have that:

$$\begin{aligned} \text{KL}(\mathbb{P}_{x^v, y^v}, \mathbb{P}_{x^0, y^0}) &= \mathbb{E}_{(x^v, y^v)} \left[\sum_{t=1}^{\tau_\delta} \text{KL}(p_{y^v}(Y_t), p_{y^0}(Y_t)) \right] && \text{(Since } x^v = x^0) \\ &= \mathbb{E}_{(x^v, y^v)} [\tau_\delta] \text{KL}(p_{y^v}, p_{y^0}) && \text{(Wald's identity)} \\ &\leq 2(192^2) \epsilon^2 \mathbb{E}_{(x^v, y^v)} [\tau_\delta]. && \text{(Lemma D.4)} \end{aligned}$$

Plugging this result in the previous equation and rearranging the terms, we obtain:

$$\frac{1}{|\bar{\mathcal{V}}|} \sum_{v \in \bar{\mathcal{V}}} \mathbb{E}_{(x^v, y^v)} [\tau_\delta] \geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(192)^2 \epsilon^2}.$$

Since $\mathbb{E}_{(x^v, y^v)} [\tau_\delta] \leq \max_{w \in \bar{\mathcal{V}}} \mathbb{E}_{(x^w, y^w)} [\tau_\delta]$ for all $v \in \bar{\mathcal{V}}$, the last equation implies that there exists $\bar{w} \in \bar{\mathcal{V}}$ such that:

$$\mathbb{E}_{(x^{\bar{w}}, y^{\bar{w}})} [\tau_\delta] \geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(192)^2 \epsilon^2}$$

¹⁶This choice is valid. Indeed, $\epsilon \leq \frac{1}{384}$ by assumption and we imposed $\gamma < \frac{1}{2}$.

Finally, we have

$$\begin{aligned}
 \mathbb{E}_{(x^w, y^w)} [\tau_\delta] &\geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(192)^2 \epsilon^2} \\
 &\geq \frac{\log(2)((1 - \delta)\frac{n}{5} + 1)}{2(192)^2 \epsilon^2} \quad (\text{Equation (19)}) \\
 &\geq \frac{\log(2)n}{40(192)^2 \epsilon^2}, \quad (\delta \leq \frac{1}{2} \text{ and } n \geq 20)
 \end{aligned}$$

which concludes the proof. $\square$

A.3. Proof of Theorem 3.3

Similarly to Theorem 3.2, the proof of Theorem 3.3 is obtained by combining two distinct lower bounds, *i.e.*, Theorems A.4 and A.5. In these theorems, we prove that there exist instances where the sample complexity of any (ϵ, δ) -correct algorithm is lower bounded by:

$$(i) \frac{\log(\frac{1}{\delta})}{\epsilon^2 \alpha}, \quad \text{and} \quad (ii) \frac{n}{\epsilon^2 \alpha},$$

here, we only dropped the multiplicative constants. These results hold both for General-Sum Games and Zero-Sum Games. Theorem 3.3 follows by the following argument.

Proof of Theorem 3.3. It is sufficient to consider the case where the agent observes the instances presented in Theorems A.4 and A.5 with probability $(\frac{1}{2}, \frac{1}{2})$. The result is then straightforward. $\square$

We now proceed by proving Theorems A.4 and A.5.

Theorem A.4. *Let $\alpha \in (0, \frac{1}{4})$ and $\epsilon \leq \frac{1}{8}$. Then, there exists a problem instance such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}[\tau_\delta] \geq \frac{\log(\frac{1}{4\delta})}{18\epsilon^2 \alpha}.$$

This holds for both Zero-Sum and General-Sum Games.

Proof. The proof is divided into three steps.

Step 1: Instance Construction To prove the result, we consider the case in which $n = 2$, and we have two different instances where the unknown strategy profiles (x^0, y^0) and (x^1, y^1) are defined as follows:

$$\begin{aligned}
 x^0 &= \left(\frac{\alpha}{2}, 1 - \frac{\alpha}{2}\right), \quad y^0 = (1, 0) \\
 x^1 &= \left(\frac{\alpha}{2} + \frac{\beta}{2}, 1 - \frac{\alpha}{2} - \frac{\beta}{2}\right), \quad y^1 = (1, 0),
 \end{aligned}$$

where $\beta \in (0, \frac{1}{4})$ is a parameter that will be tuned later on. Since $\alpha \in (0, \frac{1}{4})$ and $\beta \in (0, \frac{1}{4})$, x^0 and x^1 both belong to the simplex.

Step 2: Hausdorff Distance Next, we lower bound $H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1))$ and $H(\mathcal{Z}_\alpha(x^0, y^0), \mathcal{Z}_\alpha(x^1, y^1))$.

We start from $H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1))$. We have that:

$$\begin{aligned}
 H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1)) &\geq \max_{A \in \mathcal{G}_\alpha^x(x^0, y^0)} \min_{A' \in \mathcal{G}_\alpha^x(x^1, y^1)} \|A - A'\|_\infty \\
 &\geq \min_{A' \in \mathcal{G}_\alpha(x^1, y^1)} \max\{|1 - a'_{11}|, |1 + a'_{12}|, |1 + a'_{21}|, |1 + a'_{22}|\},
 \end{aligned}$$

where, in the second step, we have set $A = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \in \mathcal{G}_\alpha^x(x^0, y^0)$.¹⁷

¹⁷Indeed, $(x^0)^\top A y^0 = -1 + \alpha$, $e_1^\top A y^0 = \frac{\alpha}{2}$ and $e_2^\top A y^0 = -1$. Hence, $(x^0)^\top A y^0 \leq e_i^\top A y^0 + \alpha$ for all $i \in \{1, 2\}$.

Now, using the values of x^1 and y^1 , we are interested in solving the following problem:

$$\begin{aligned} \min_{A' \in [-1,1]^{2 \times 2}} \max \{ & |1 - a'_{11}|, |1 + a'_{12}|, |1 + a'_{21}|, |1 + a'_{22}| \} \\ \text{s.t. } & \left(\frac{\alpha}{2} + \frac{\beta}{2}\right)a'_{11} + \left(1 - \frac{\alpha}{2} - \frac{\beta}{2}\right)a'_{21} \leq a'_{11} + \alpha \\ & \left(\frac{\alpha}{2} + \frac{\beta}{2}\right)a'_{11} + \left(1 - \frac{\alpha}{2} - \frac{\beta}{2}\right)a'_{21} \leq a'_{21} + \alpha. \end{aligned}$$

Simplifying the constraints we can rewrite this optimization problem as follows:

$$\begin{aligned} \min_{A' \in [-1,1]^{2 \times 2}} \max \{ & |1 - a'_{11}|, |1 + a'_{21}| \} \\ \text{s.t. } & a'_{21} - a'_{11} \leq \frac{2\alpha}{2-\alpha-\beta} \\ & a'_{11} - a'_{21} \leq \frac{2\alpha}{\alpha+\beta}. \end{aligned} \quad (24)$$

Now, we note that:

$$\begin{aligned} \max \{ & |1 - a'_{11}|, |1 + a'_{21}| \} \geq \frac{1}{2} (|1 - a'_{11}| + |1 + a'_{21}|) & (\max\{a, b\} \geq \frac{1}{2}(a + b)) \\ & \geq \frac{1}{2} |2 - (a'_{11} - a'_{21})|. & (\text{Triangular inequality}) \end{aligned}$$

Let $\Delta = a'_{11} - a'_{21}$ and observe that $\Delta \in [-2, 2]$. We can then lower bound the value of Problem (24) with the value of the following optimization problem:

$$\begin{aligned} \min_{\Delta \in [-2,2]} & \frac{1}{2} |2 - \Delta| \\ \text{s.t. } & -\frac{2\alpha}{2-\alpha-\beta} \leq \Delta \leq \frac{2\alpha}{\alpha+\beta}. \end{aligned} \quad (25)$$

This can be easily solved in closed form, yielding $\Delta^* = 2\frac{\alpha}{\alpha+\beta}$. Consequently, we have obtained the following lower bound on the Hausdorff distance: $H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1)) \geq \frac{\beta}{\alpha+\beta}$.

Set $\beta = \frac{8}{3}\epsilon\alpha$. We have obtained that:¹⁸

$$H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1)) \geq 2\epsilon. \quad (26)$$

We continue by lower bounding $H(\mathcal{Z}_\alpha(x^0, y^0), \mathcal{Z}_\alpha(x^1, y^1))$. We have that:

$$\begin{aligned} H(\mathcal{Z}_\alpha(x^0, y^0), \mathcal{Z}_\alpha(x^1, y^1)) & \geq \max_{A \in \mathcal{Z}_\alpha(x^0, y^0)} \min_{A' \in \mathcal{Z}_\alpha(x^1, y^1)} \|A - A'\|_\infty \\ & \geq \min_{A' \in \mathcal{Z}_\alpha(x^1, y^1)} \max \{ |1 - a'_{11}|, |1 + a'_{12}|, |1 + a'_{21}|, |1 + a'_{22}| \}, \end{aligned}$$

where, in the second step, we have set $A = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \in \mathcal{Z}_\alpha(x^0, y^0)$. Indeed, we already proved above that A satisfies $-1 + \alpha = (x^0)^\top A y^0 \leq e_i^\top A y^0 + \alpha$ for all $i \in \{1, 2\}$. It is also easy to see that $(x^0)^\top A e_1 = -1 + \alpha$ and $(x^0)^\top A e_2 = -1$ for all $j \in \{1, 2\}$. Hence, we obtained $A \in \mathcal{Z}_\alpha(x^0, y^0)$. Then, one can proceed as above to obtain:¹⁹

$$H(\mathcal{Z}_\alpha(x^0, y^0), \mathcal{Z}_\alpha(x^1, y^1)) \geq 2\epsilon. \quad (27)$$

Step 3: Change of measure For any $x, y \in \Delta_n$, let $\mathbb{P}_{x,y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, where $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$. Then, since Equations (26) and (27) hold, we can apply Lemma D.6. For both GSGs and ZSGs we have

$$\delta \geq \frac{1}{4} \exp \left(-\text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1}) \right).$$

¹⁸In this step, we make use of $\epsilon < \frac{1}{8}$.

¹⁹It is sufficient to drop all the constraints that arise from $(x^1)^\top A' y^1 \geq (x^1)^\top A' e_j - \alpha$ in the optimization problems.

We proceed by analyzing the r.h.s. of this equation. In the case we are considering we have set $n = 2$. Hence, x, y are Bernoulli distributions, and we obtain that:

$$\begin{aligned}
 \text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1}) &= \mathbb{E}_{x^0, y^0} \left[\sum_{t=1}^{\tau_\delta} \text{KL}(p_{x^0}(X_t), p_{x^1}(X_t)) \right] && (\text{Since } y^0 = y^1) \\
 &= \mathbb{E}_{x^0, y^0} [\tau_\delta] \text{KL}(p_{x^0}, p_{x^1}) && (\text{Wald's identity}) \\
 &\leq 2\mathbb{E}_{x^0, y^0} [\tau_\delta] \frac{\text{TV}(p_{x^0}, p_{x^1})^2}{\alpha} && (\text{Reverse Pinsker's Inequality}) \\
 &\leq 2\mathbb{E}_{x^0, y^0} [\tau_\delta] \frac{(\frac{8}{3}\epsilon\alpha)^2}{\alpha} \\
 &\leq 18\mathbb{E}_{x^0, y^0} [\tau_\delta] \alpha\epsilon^2.
 \end{aligned}$$

Hence, we obtained that:

$$\delta \geq \frac{1}{4} \exp(-18\mathbb{E}_{x^0, y^0} [\tau_\delta] \alpha\epsilon^2).$$

Rearranging the terms yields:

$$\mathbb{E}_{x^0, y^0} [\tau_\delta] \geq \frac{\log(\frac{1}{4\delta})}{18\alpha\epsilon^2}.$$

which concludes the proof. $\square$

Finally, it remains to prove the last result that composes Theorem 3.3.

Theorem A.5. *Let $\alpha \in (0, \frac{1}{2})$ and $\epsilon \leq \frac{1}{128}$. Then, there exists a problem instance such that, for any (ϵ, δ) -correct algorithm, the following holds:*

$$\mathbb{E}[\tau_\delta] \geq \frac{\log(2)n}{20(128)^2\alpha\epsilon^2}.$$

This holds for both Zero-Sum and General-Sum Games.

Proof. The proof is split into three main steps.

Step 1: Instance Construction Similar to Theorem A.3, we construct a large set of different n -dimensional instances. We assume that $n \geq 22$ and that $n - 1 \bmod 16 = 0$. These instances are parameterized with an $n - 1$ -dimensional vector $v \in \{-1, 1\}^{n-1}$ such that $\sum_{k=1}^{n-1} v_k = 0$ and are constructed as follows. We denote by $\mathcal{V}$ the set of vectors that satisfies these two requirements. Now, let $\gamma \in (0, \alpha)$; we define (x^v, y^v) as follows:

$$y^v = (1, 0, \dots, 0) \text{ and } x^v = \left(\frac{\alpha + \gamma v_1}{n-1}, \frac{\alpha + \gamma v_2}{n-1}, \dots, \frac{\alpha + \gamma v_{n-1}}{n-1}, 1 - \alpha \right).$$

Observe that $n - 1 \geq 16$, $\gamma \in (0, \alpha)$, $\alpha \in (0, \frac{1}{2})$ and $v \in \{-1, 1\}^{n-1} \implies x^v \in [0, 1]$. Furthermore $\sum_{k=1}^{n-1} v_k = 0 \implies \sum_{j=1}^n x_k^v = 1$. Therefore, we have that $x^v \in \Delta_n$.

In the rest of the proof, we restrict our attention to vectors belonging to a subset $\bar{\mathcal{V}}$ of $\mathcal{V}$. Precisely, consider any pair $v, w \in \bar{\mathcal{V}}$ and let

$$D_{v,w}^+ = \{k \in [n-1] : v_k = 1 \text{ and } w_k = -1\} \quad D_{v,w}^- = \{k \in [n-1] : v_k = -1 \text{ and } w_k = 1\},$$

where we will drop the explicit dependence on v, w when clear from context. Then, we require that for any $v, w \in \bar{\mathcal{V}}$ it holds

$$|D_{v,w}^+| = |D_{v,w}^-| \geq \frac{n-1}{16}. \quad (28)$$

It is easy to see that $|D_{v,w}^+| = |D_{v,w}^-|$ is satisfied for any pair $v, w \in \mathcal{V}$. The requirement that, e.g., $|D^+| \geq \frac{n-1}{16}$, is instead equivalent to imposing that $\sum_{k=1}^{n-1} |v_k - w_k| \geq \frac{n-1}{16}$. Thanks to Lemma D.12, we know that:

$$\exists \bar{\mathcal{V}} \subseteq \mathcal{V} : \forall v, w \in \bar{\mathcal{V}}, \quad \sum_{k=1}^{n-1} |v_k - w_k| \geq \frac{n-1}{16} \text{ and } |\bar{\mathcal{V}}| > 2^{\frac{n-1}{5}}. \quad (29)$$

In the following step, we will consider two generic profiles (x^v, y^v) and (x^w, y^w) such that $v, w \in \bar{\mathcal{V}}$.

Step 2: Hausdorff Distance We analyze the distances $H(\mathcal{G}_\alpha(x^v, y^v), \mathcal{G}_\alpha(x^w, y^w))$ and $H(\mathcal{Z}_\alpha(x^v, y^v), \mathcal{Z}_\alpha(x^w, y^w))$. In particular, we will provide lower bounds on these quantities.

We start from $H(\mathcal{G}_\alpha(x^v, y^v), \mathcal{G}_\alpha(x^w, y^w))$. We have that:

$$\begin{aligned} H(\mathcal{G}_\alpha(x^v, y^v), \mathcal{G}_\alpha(x^w, y^w)) &\geq \max_{A \in \mathcal{G}_\alpha^x(x^v, y^v)} \min_{A' \in \mathcal{G}_\alpha^x(x^w, y^w)} \|A - A'\|_\infty \\ &\geq \min_{A' \in \mathcal{G}_\alpha^x(x^w, y^w)} \|A - A'\|_\infty, \end{aligned}$$

where the matrix $A \in \mathcal{G}_\alpha^x(x^v, y^v)$ that we used in the second inequality is constructed as follows:

$$A_{ij} := \begin{cases} -1 & i \in [n], j > 1, \\ -\kappa & i \in D^+, j = 1, \\ 1 & i \in D^-, j = 1, \\ 0 & i \in [n-1] \setminus (D^+ \cup D^-), j = 1, \\ -1 & i = n, j = 1. \end{cases} \quad \text{where} \quad \kappa := \frac{\sum_{j \in D^-} x_j^v}{\sum_{i \in D^+} x_i^v}. \quad (30)$$

Before continuing, we prove that A constructed as in Equation (30) satisfies $A \in \mathcal{G}_\alpha^x(x^v, y^v)$. To this end, we first verify that $A \in [-1, 1]^{n \times n}$. It suffices to check $|\kappa| \leq 1$. With some simple manipulations, we have that:

$$|\kappa| = \frac{\sum_{j \in D^-} x_j^v}{\sum_{i \in D^+} x_i^v} = \frac{\frac{|D^-|(\alpha - \gamma)}{n-1}}{\frac{|D^+|(\alpha + \gamma)}{n-1}} = \frac{\alpha - \gamma}{\alpha + \gamma} \leq 1,$$

where the second equality is due to Equation (28) and the definition of x^v . Next, we verify that A satisfy the α -Nash constraints. To this end, we need to verify that $(x^v)^\top A y \leq -1 + \alpha$. We note that:

$$\begin{aligned} (x^v)^\top A y &= \sum_{i \in D^+} x_i^v A_{i1} + \sum_{i \in D^-} x_i^v A_{i1} + x_n A_{n1} \\ &= -\kappa \sum_{i \in D^+} x_i^v + \sum_{i \in D^-} x_i^v + (\alpha - 1) \quad (\text{Def. of } A \text{ and } x^v) \\ &= -1 + \alpha. \quad (\text{Def. of } \kappa) \end{aligned}$$

Hence, all constraints are satisfied and $A \in \mathcal{G}_\alpha^x(x^v, y^v)$.

We can now continue by lower bounding $\min_{A' \in \mathcal{G}_\alpha^x(x^w, y^w)} \|A - A'\|_\infty$. Specifically, let us introduce the auxiliary variable $t = \|A - A'\|_\infty$. Then, using the definition of y^w , we can rewrite this optimization problem as follows:

$$\begin{aligned} \min_{A' \in [-1, 1]^{n \times n}, t \geq 0} \quad & t \\ \text{s.t.} \quad & |A'_{ij} - A_{ij}| \leq t, \quad \forall i, j \in [n]^2 \\ & \sum_{i \in [n]} x_i^w A'_{i1} \leq \min_{i \in [n]} A'_{i1} + \alpha. \end{aligned} \quad (31)$$

Consider any feasible matrix A' . Then, it must hold that:

$$\begin{aligned}
 A'_{1n} + \alpha &\geq (x^w)^\top A'y && \text{(Last constraint of Equation (31))} \\
 &= \sum_{i \neq n} x_i^w A'_{i1} + (1 - \alpha)A'_{n1} && \text{(Def. of } x^w) \\
 &= \sum_{i \neq n} x_i^w A_{i1} + \sum_{i \neq n} (x_i^w (A'_{i1} - A_{i1})) + (1 - \alpha)A'_{n1} \\
 &\geq \sum_{i \neq n} x_i^w A_{i1} - t\alpha + (1 - \alpha)A'_{n1} && (|A'_{ij} - A_{ij}| \leq t \text{ and } \sum_{i \neq n} x_i^w = \alpha) \\
 &= \sum_{i \neq n} (x_i^w - x_i^v)A_{i1} - t\alpha + (1 - \alpha)A'_{n1} && (\sum_{i \neq n} x_i^v A_{i1} = 0) \\
 &= \frac{2\kappa|D^+|\gamma}{n-1} + \frac{2|D^-|\gamma}{n-1} - t\alpha + (1 - \alpha)A'_{n1} && \text{(Def. of } x \text{ and } A) \\
 &= \frac{4|D^+|\gamma}{n-1} \left(\frac{\alpha}{\alpha + \gamma} \right) - t\alpha + (1 - \alpha)A'_{n1} && \text{(Def. of } \kappa) \\
 &\geq \frac{\gamma\alpha}{16(\alpha + \gamma)} - t\alpha + (1 - \alpha)A'_{n1}. && \text{(Equation (28))}
 \end{aligned}$$

Hence, rearranging the terms, and manipulating the equations, we obtain that any feasible A' satisfies:

$$\begin{aligned}
 \frac{\gamma\alpha}{16(\alpha + \gamma)} &\leq \alpha(1 + A'_{n1}) + t\alpha \\
 &\leq 2t\alpha && (|A_{n1} - A'_{n1}| = |1 + A'_{n1}| \leq t)
 \end{aligned}$$

Therefore, we arrived at $H(\mathcal{G}_\alpha(x^v, y^v), \mathcal{G}_\alpha(x^w, y^w)) \geq \frac{\gamma}{32(\alpha + \gamma)}$. Picking $\gamma = 128\alpha\epsilon$, we obtained that:²⁰

$$H(\mathcal{G}_\alpha(x^v, y^v), \mathcal{G}_\alpha(x^w, y^w)) \geq 2\epsilon. \quad (32)$$

Next we analyze $H(\mathcal{Z}_\alpha(x^v, y^v), \mathcal{Z}_\alpha(x^w, y^w))$. We have that:

$$\begin{aligned}
 H(\mathcal{Z}_\alpha(x^v, y^v), \mathcal{Z}_\alpha(x^w, y^w)) &\geq \max_{A \in \mathcal{Z}_\alpha(x^v, y^v)} \min_{A' \in \mathcal{Z}_\alpha(x^w, y^w)} \|A - A'\|_\infty \\
 &\geq \min_{A' \in \mathcal{Z}_\alpha(x^w, y^w)} \|A - A'\|_\infty,
 \end{aligned}$$

where the matrix $A \in \mathcal{Z}_\alpha(x^v, y^v)$ is defined as in Equation (30). We already proved above that $A \in [-1, 1]^{n \times n}$. It remains to prove that it satisfies the α -Nash constraints for zero-sum games, i.e., that A satisfies:

$$(x^v)^\top A e_j - \alpha \leq (x^v)^\top A y \leq e_i^\top A y + \alpha, \quad \forall i, j \in [n]^2.$$

Above, we have shown that $-1 + \alpha = (x^v)^\top A y \leq e_i^\top A y + \alpha, \quad \forall i, j \in [n]^2$. Now, it is sufficient to note that $(x^v)^\top A e_1 = (x^v)^\top A y$ and $(x^v)^\top A e_j = -1$ for all $j > 1$. Hence, $A \in \mathcal{Z}_\alpha(x^v, y^v)$.

Then, one can proceed as above to obtain:

$$H(\mathcal{Z}_\alpha(x^v, y^v), \mathcal{Z}_\alpha(x^w, y^w)) \geq 2\epsilon. \quad (33)$$

Step 3: Change of measure For any $x, y \in \Delta_n$, let $\mathbb{P}_{x,y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, where $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$. Then, since Equation (33) holds, we can apply Lemma D.7 using the set of instances $\{(x^v, y^v)\}_{v \in \mathcal{V}}$ and $(x^0, y^0) = ((\frac{\alpha}{n-1}, \dots, \frac{\alpha}{n-1}, 1 - \alpha), e_1)$ as base instance. Thus, both for GSGs and ZSGs, we have that, for any (ϵ, δ) -correct algorithm, it holds that:

$$\delta \geq 1 - \frac{1}{\log(|\mathcal{V}|)} \left(\frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \text{KL}(\mathbb{P}_{x^v, y^v}, \mathbb{P}_{x^0, y^0}) + \log 2 \right).$$

²⁰In this step, we used that $\epsilon \leq \frac{1}{128}$.

We proceed by analyzing the kl-divergence term in the r.h.s. of this equation. Specifically, we have that:

$$\begin{aligned}
 \text{KL}(\mathbb{P}_{x^v, y^v}, \mathbb{P}_{x^0, y^0}) &= \mathbb{E}_{(x^v, y^v)} \left[\sum_{t=1}^{\tau_\delta} \text{KL}(p_{x^v}(X_t), p_{x^0}(X_t)) \right] && (\text{Since } y^v = y^0) \\
 &= \mathbb{E}_{(x^v, y^v)} [\tau_\delta] \text{KL}(p_{x^v}, p_{x^0}) && (\text{Wald's identity}) \\
 &\leq 2(128^2) \alpha \epsilon^2 \mathbb{E}_{(x^v, y^v)} [\tau_\delta]. && (\text{Lemma D.4})
 \end{aligned}$$

Plugging this result in the previous equation and rearranging the terms, we obtain:

$$\frac{1}{|\bar{\mathcal{V}}|} \sum_{v \in \bar{\mathcal{V}}} \mathbb{E}_{(x^v, y^v)} [\tau_\delta] \geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(128)^2 \alpha \epsilon^2}.$$

Since $\mathbb{E}_{(x^v, y^v)} [\tau_\delta] \leq \max_{w \in \bar{\mathcal{V}}} \mathbb{E}_{(x^w, y^w)} [\tau_\delta]$ for all $v \in \bar{\mathcal{V}}$, the last equation implies that there exists $\bar{w} \in \bar{\mathcal{V}}$ such that:

$$\mathbb{E}_{(x^{\bar{w}}, y^{\bar{w}})} [\tau_\delta] \geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(128)^2 \alpha \epsilon^2}$$

With some final algebraic steps, we can now conclude the proof:

$$\begin{aligned}
 \mathbb{E}_{(x^{\bar{w}}, y^{\bar{w}})} [\tau_\delta] &\geq \frac{(1 - \delta) \log(|\bar{\mathcal{V}}|) + \log 2}{2(128)^2 \epsilon^2} \\
 &\geq \frac{\log(2) \left((1 - \delta) \frac{(n-1)}{5} + 1 \right)}{2(128)^2 \alpha \epsilon^2} && (\text{Equation (29)}) \\
 &\geq \frac{\log(2)n}{20(128)^2 \alpha \epsilon^2}, && (\delta \leq \frac{1}{2} \text{ and } n \geq 22)
 \end{aligned}$$

This concludes the proof. $\square$

B. Proof of Theorem 4.1

This section is structured as follows:

- In Section B.1 we prove the result for the inverse GSG problem (Theorem B.4)
- In Section B.2, we prove the result of the inverse ZSG problem (Theorem B.7)

The proof of Theorem 4.1 then follows directly by combining Theorems B.4 and B.7.

B.1. Proof of Theorem B.4 ($\alpha = 0$, General-Sum Games)

Proof Outline First, in Lemma B.1, we show that, for any pair of strategies (x, y) and $(\hat{x}, \hat{y})$ such that $\text{supp}(x) = \text{supp}(\hat{x})$ and $\text{supp}(y) = \text{supp}(\hat{y})$, then $\mathcal{G}(x, y)$ is close to $\mathcal{G}(\hat{x}, \hat{y})$ by construction. Specifically, for any point $(A, B) \in \mathcal{G}(x, y)$, there exist $(\hat{A}, \hat{B}) \in \mathcal{G}(\hat{x}, \hat{y})$ such that $A \approx \hat{A}$ and $B \approx \hat{B}$. Importantly the approximation errors scale linearly with $\|x - \hat{x}\|_1$ and $\|y - \hat{y}\|_1$. Then, in Lemma B.2, we show how Lemma B.1 implies that the Hausdorff distance between the two sets is directly proportional to $\|x - \hat{x}\|_1$ and $\|y - \hat{y}\|_1$. In Lemma B.3 we combine this result together with probabilistic argument. In this way, we are able to provide an high probability error upper bound on $H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y}))$ that is explicitly related to the number of data m collected by Algorithm 1. Finally, to prove Theorem B.4 we only need to select m appropriately to guarantee that the final error of Algorithm 1 is bounded by ϵ with high probability.

We start with Lemma B.1. Note that here, $x, y, \hat{x}, \hat{y}$ are any strategy on the simplex.

Lemma B.1 (Alternative Matrix Construction; $\alpha = 0$ and General-Sum Games). *Let $\alpha = 0$, $x, y, \hat{x}, \hat{y} \in \Delta_n$. $A \in \mathcal{G}^x(x, y)$ and $B \in \mathcal{G}^y(x, y)$ and $\hat{x}, \hat{y} \in \Delta_n$.*

Then if $\text{supp}(x) = \text{supp}(\hat{x})$ then there exists $\hat{A} \in \mathcal{G}^x(\hat{x}, \hat{y})$ such that

$$\|A - \hat{A}\|_\infty \leq 4\|y - \hat{y}\|_1, \tag{34}$$

and similarly, if $\text{supp}(y) = \text{supp}(\hat{y})$ then there exists $\hat{B} \in \mathcal{G}^y(\hat{x}, \hat{y})$ such that

$$\|B - \hat{B}\|_\infty \leq 4\|x - \hat{x}\|_1. \quad (35)$$

Furthermore, $\hat{A}$ and $\hat{B}$ are defined as:

$$\hat{A}_{ij} := \begin{cases} \frac{A_{ij}}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in \arg\max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj}, \\ \frac{A_{ij} + \Delta_i^x}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in \text{supp}(x) \setminus \arg\max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj}, \\ \frac{A_{ij} + 2\|y - \hat{y}\|_1}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \notin \text{supp}(x) \end{cases}$$

$$\hat{B}_{ij} := \begin{cases} \frac{B_{ij}}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \in \arg\max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i B_{ik}, \\ \frac{B_{ij} + \Delta_j^y}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \in \text{supp}(y) \setminus \arg\max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i B_{ik}, \\ \frac{B_{ij} + 2\|x - \hat{x}\|_1}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \notin \text{supp}(y), \end{cases}$$

where, for all $i \in \text{supp}(x)$ and all $j \in \text{supp}(y)$, Δ_i^x and Δ_j^y are defines as follows:

$$\Delta_i^x = \max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{j \in [n]} \hat{y}_j A_{ij}$$

$$\Delta_j^y = \max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i B_{ik} - \sum_{i \in [n]} \hat{x}_i B_{ij}.$$

Proof. We first prove Equation (34) by construction. For brevity, let $S_x := \text{supp}(x) \subseteq [n]$ and define the set of maximizers of $\sum_{j \in [n]} \hat{y}_j A_{ij}$ over the different rows of the support of x as S_x^* , namely $S_x^* := \arg\max_{k \in S_x} \sum_{j \in [n]} \hat{y}_j A_{kj}$. Note that, since $A_{ij} \in [-1, 1]$ and $\hat{y} \in \Delta_n$, we have that $\Delta_i^x \in [0, 2]$. Furthermore, for all $i \in S_x$

$$\begin{aligned} \Delta_i^x &= \max_{k \in S_x} \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{j \in [n]} \hat{y}_j A_{ij} \\ &= \sum_{j \in [n]} \hat{y}_j (A_{kj} - A_{ij}) && \text{(For any } k \in S_x^*) \\ &= \sum_{j \in [n]} y_j (A_{kj} - A_{ij}) + \sum_{j \in [n]} (\hat{y}_j - y_j) (A_{kj} - A_{ij}) \\ &= \sum_{j \in [n]} (\hat{y}_j - y_j) (A_{kj} - A_{ij}) && \text{(Lemma D.8 and } i, k \in S_x) \\ &\leq 2\|y - \hat{y}\|_1 && (-1 \leq A \leq 1) \\ &:= \Delta, \end{aligned}$$

where in the last step we have introduced $\Delta := 2\|y - \hat{y}\|_1$.

Now, we are ready to define a matrix $\hat{A}$ that we will use to prove Equation (34). Specifically, $\hat{A}$ is defined as follows:

$$\hat{A}_{ij} := \begin{cases} \frac{A_{ij}}{1 + \Delta}, & \text{if } i \in S_x^*, \\ \frac{A_{ij} + \Delta_i^x}{1 + \Delta}, & \text{if } i \in S_x \setminus S_x^*, \\ \frac{A_{ij} + \Delta}{1 + \Delta}, & \text{if } i \notin S_x. \end{cases}$$

First of all, note that since $A \in [-1, 1]^{n \times n}$ and $\Delta_i^x \leq \Delta$, we have that $\hat{A} \in [-1, 1]^{n \times n}$.

At this point, we continue by showing that $\hat{A} \in \mathcal{G}^x(\hat{x}, \hat{y})$. To this end, since $\alpha = 0$ and $\text{supp}(x) = \text{supp}(\hat{x}) = S_x$, due to Lemma D.8 we only need to show that

$$e_i^\top \hat{A} \hat{y} = e_j^\top \hat{A} \hat{y} \quad \forall i, j \in S_x, \quad (36a)$$

$$e_i^\top \hat{A} \hat{y} \leq e_j^\top \hat{A} \hat{y} \quad \forall i \in S_x, j \notin S_x. \quad (36b)$$

Condition (36a) To prove that Condition (36a) holds, we first show that for any $i_1, i_2 \in S_x^*$, we have $e_{i_1}^\top \hat{A} \hat{y} = e_{i_2}^\top \hat{A} \hat{y}$. Specifically, let $i_1, i_2 \in S_x^*$. Then,

$$\begin{aligned} e_{i_1}^\top \hat{A} \hat{y} &= \frac{1}{1 + \Delta} \sum_{j \in [n]} A_{i_1 j} \hat{y}_j \\ &= \frac{1}{1 + \Delta} \sum_{j \in [n]} A_{i_2 j} \hat{y}_j && (S_x^* = \text{argmax}_{i \in S_x} \sum_{j \in [n]} \hat{y}_j A_{ij} \text{ and } i_1, i_2 \in S_x^*) \\ &= e_{i_2}^\top \hat{A} \hat{y}. \end{aligned}$$

To conclude the proof of Condition (36a), we need to consider $i_1 \in S_x^*$, and $i_2 \in S_x \setminus S_x^*$ and prove that $e_{i_1}^\top \hat{A} \hat{y} = e_{i_2}^\top \hat{A} \hat{y}$. Using the definition of $\hat{A}$, we have that:

$$e_{i_1}^\top \hat{A} \hat{y} = e_{i_2}^\top \hat{A} \hat{y} \iff \sum_{j \in [n]} \hat{y}_j A_{i_1 j} = \sum_{j \in [n]} \hat{y}_j A_{i_2 j} + \Delta_{i_2}^x.$$

However, by definition of $\Delta_{i_2}^x$, we have that,

$$\Delta_{i_2}^x = \max_{k \in S} \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{k \in [n]} \hat{y}_j A_{i_2 k} = \sum_{j \in [n]} \hat{y}_j A_{i_1 j} - \sum_{k \in [n]} \hat{y}_j A_{i_2 k},$$

hence, $\sum_{j \in [n]} \hat{y}_j A_{i_1 j} = \sum_{j \in [n]} \hat{y}_j A_{i_2 j} + \Delta_{i_2}^x$ holds, and $e_{i_1}^\top \hat{A} \hat{y} = e_{i_2}^\top \hat{A} \hat{y}$ holds as well. Finally, we note that the above argument also covers the case where $i_1, i_2 \in S_x \setminus S_x^*$. Indeed, for any $i \in S_x \setminus S_x^*$, we have

$$e_i^\top \hat{A} \hat{y} = e_{i^*}^\top \hat{A} \hat{y}$$

for any $i^* \in S_x^*$. Therefore, for any $i_1, i_2 \in S_x \setminus S_x^*$,

$$e_{i_1}^\top \hat{A} \hat{y} = e_{i^*}^\top \hat{A} \hat{y} = e_{i_2}^\top \hat{A} \hat{y},$$

which shows that Condition (36a) holds for all pairs $i_1, i_2 \in S_x$.

Condition (36b). To prove that Condition (36b) holds, since we already proven Condition (36a), it is sufficient to show that, for $i_1 \in S_x^*$ and $i_2 \notin S_x$, we have that $e_{i_1}^\top \hat{A} \hat{y} \leq e_{i_2}^\top \hat{A} \hat{y}$. Consider the following inequalities:

$$\begin{aligned}
 e_{i_1}^\top \hat{A} \hat{y} &= \frac{1}{1+\Delta} \left(\sum_{j \in [n]} \hat{y}_j A_{i_1 j} \right) \\
 &= \frac{1}{1+\Delta} \left(\sum_{j \in [n]} y_j A_{i_1 j} + \sum_{j \in [n]} (\hat{y}_j - y_j) A_{i_1 j} \right) \\
 &\leq \frac{1}{1+\Delta} \left(\sum_{j \in [n]} y_j A_{i_2 j} + \sum_{j \in [n]} (\hat{y}_j - y_j) A_{i_1 j} \right) \quad (\text{Due to } A \in \mathcal{G}^x(x, y) \text{ and Lemma D.8}) \\
 &= \frac{1}{1+\Delta} \left(\sum_{j \in [n]} \hat{y}_j A_{i_2 j} + \sum_{j \in [n]} (\hat{y}_j - y_j) (A_{i_1 j} - A_{i_2 j}) \right) \\
 &\leq \frac{1}{1+\Delta} \left(\sum_{j \in [n]} \hat{y}_j A_{i_2 j} + 2\|y - \hat{y}\|_1 \right) \\
 &= e_{i_2}^\top \hat{A} \hat{y}.
 \end{aligned}$$

Hence, we proved that $\hat{A} \in \mathcal{G}^x(\hat{x}, \hat{y})$.

Finally, to conclude the proof of Equation (34), it remains to bound $\|A - \hat{A}\|_\infty$. Fix any pair of $i, j \in [n]$. Then, if $i \in S_x^*$, we have that:

$$|A_{ij} - \hat{A}_{ij}| = \left| A_{ij} - \frac{A_{ij}}{1+\Delta} \right| = |A_{ij}| \frac{\Delta}{1+\Delta} \leq \frac{\Delta}{1+\Delta} \leq 2\Delta.$$

If, instead, $i \in S_x \setminus S_x^*$:

$$|A_{ij} - \hat{A}_{ij}| = \left| A_{ij} - \frac{A_{ij} + \Delta_i^x}{1+\Delta} \right| = \frac{|\Delta A_{ij} - \Delta_i^x|}{1+\Delta} \leq \frac{|A_{ij}| \Delta + \Delta_i^x}{1+\Delta} \leq \frac{2\Delta}{1+\Delta} \leq 2\Delta,$$

Finally, if $i \notin S_x$, we have that:

$$|A_{ij} - \hat{A}_{ij}| = \left| A_{ij} - \frac{A_{ij} + \Delta}{1+\Delta} \right| = \frac{\Delta|1 - A_{ij}|}{1+\Delta} \leq \frac{2\Delta}{1+\Delta} \leq 2\Delta.$$

Hence, $\|A - \hat{A}\|_\infty \leq 2\Delta$, thus concluding the proof of Equation (34).

We continue by proving Equation (35). In particular, the proof of Equation (35) follows from arguments that are symmetrical to the ones that we presented above. More precisely, consider $B \in \mathcal{G}^y(x, y)$. Let $S_y := \text{supp}(y)$ and for all $j \in [n]$ and define:

$$\Delta_j^y = \max_{k \in S_y} \sum_{i \in [n]} \hat{x}_i B_{ik} - \sum_{i \in [n]} \hat{x}_i B_{ij}.$$

Furthermore, let $S_y^* = \text{argmax}_{j \in [n]} \sum_{i \in [n]} \hat{x}_i B_{ij}$. Then, we define $\hat{B}$ as follows:

$$\hat{B}_{ij} := \begin{cases} \frac{B_{ij}}{1+\Delta}, & \text{if } j \in S_y^*, \\ \frac{B_{ij} + \Delta_i^x}{1+\Delta}, & \text{if } j \in S_y \setminus S_y^*, \\ \frac{B_{ij} + \Delta}{1+\Delta}, & \text{if } j \notin S_y. \end{cases}$$

where $\Delta = 2\|x - \hat{x}\|_1$. The proof of the results than follows from arguments that are analogous to the ones that we presented for Equation (34). $\square$

We now leverage Lemma B.1 to provide an upper bound on the Hausdorff distance.

Lemma B.2 (Matrix Construction $\rightarrow$ Hausdorff Distance). *Let $\alpha = 0$, $x, y, \hat{x}, \hat{y} \in \Delta_n$. Suppose that $\text{supp}(x) = \text{supp}(\hat{x})$ and $\text{supp}(y) = \text{supp}(\hat{y})$. Then, it holds that:*

$$H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y})) \leq 4(\|x - \hat{x}\|_1 + \|y - \hat{y}\|_1).$$

Proof. The proof is a direct application of Lemma B.1. Precisely:

$$\begin{aligned} H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y})) &= \max \left\{ \max_{(A, B) \in \mathcal{G}(x, y)} \min_{(\hat{A}, \hat{B}) \in \mathcal{G}(\hat{x}, \hat{y})} \max\{\|A - \hat{A}\|_\infty, \|B - \hat{B}\|_\infty\}, \right. \\ &\quad \left. \max_{(\hat{A}, \hat{B}) \in \mathcal{G}(\hat{x}, \hat{y})} \min_{(A, B) \in \mathcal{G}(x, y)} \max\{\|A - \hat{A}\|_\infty, \|B - \hat{B}\|_\infty\} \right\} \\ &\leq 4 \max\{\|x - \hat{x}\|_1, \|y - \hat{y}\|_1\} \\ &\leq 4(\|x - \hat{x}\|_1 + \|y - \hat{y}\|_1). \end{aligned} \quad (\text{Lemma B.1})$$

which concludes the proof. Here, we simply remark that, in order to apply Lemma B.1 in the first inequality, it is important to recall that $\mathcal{G}(x, y) = \mathcal{G}^x(x, y) \times \mathcal{G}^y(x, y)$. $\square$

At this point, we can combine Lemma B.2 with probabilistic arguments to obtain a high probability error bound on the Hausdorff distance that is directly related to the number of samples observed by our algorithm.

Lemma B.3 (High-Probability Hausdorff Bound). *Let $\alpha = 0$ and $\hat{x}, \hat{y} \in \Delta_n$ be the maximum likelihood estimators of x, y after m samples. Suppose that m satisfies:*

$$m \geq \frac{\log\left(\frac{4n}{\delta}\right)}{\log\left(\frac{1}{1-\pi_{\min}}\right)}, \quad (37)$$

then, with probability at least $1 - \delta$, it holds that:

$$H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y})) \leq 8 \sqrt{\frac{2 \log\left(\frac{4}{\delta}\right) + 2n \log(6m)}{m}}.$$

Proof. We first define the following good event $\mathcal{E}$ and shows that it holds with probability at least $1 - \delta$. We define $\mathcal{E} = \mathcal{E}_1 \cap \mathcal{E}_2$, where

$$\begin{aligned} \mathcal{E}_1 &= \left\{ \|x - \hat{x}\|_1 \leq \sqrt{\frac{2 \log\left(\frac{4}{\delta}\right) + 2n \log(6m)}{m}} \right\} \cap \left\{ \|y - \hat{y}\|_1 \leq \sqrt{\frac{2 \log\left(\frac{4}{\delta}\right) + 2n \log(6m)}{m}} \right\} \\ \mathcal{E}_2 &= \{\text{supp}(x) = \text{supp}(\hat{x})\} \cap \{\text{supp}(y) = \text{supp}(\hat{y})\}. \end{aligned}$$

Then, we have that $\mathbb{P}(\mathcal{E}) = 1 - \mathbb{P}(\mathcal{E}^c) \geq 1 - (\mathbb{P}(\mathcal{E}_1^c) + \mathbb{P}(\mathcal{E}_2^c))$. Hence, we simply upper bound $\mathbb{P}(\mathcal{E}_1^c)$ and $\mathbb{P}(\mathcal{E}_2^c)$.

$$\begin{aligned} \mathbb{P}(\mathcal{E}_1^c) &\leq \mathbb{P}\left(\|x - \hat{x}\|_1 > \sqrt{\frac{2 \log\left(\frac{4}{\delta}\right) + 2n \log(6m)}{m}}\right) + \mathbb{P}\left(\|y - \hat{y}\|_1 > \sqrt{\frac{2 \log\left(\frac{4}{\delta}\right) + 2n \log(6m)}{m}}\right) \\ &\leq \frac{\delta}{2}. \end{aligned} \quad (\text{Lemma D.3})$$

Furthermore,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_2^c) &\leq \mathbb{P}(\text{supp}(x) \neq \text{supp}(\hat{x})) + \mathbb{P}(\text{supp}(y) \neq \text{supp}(\hat{y})) \\ &\leq \frac{\delta}{2}. \end{aligned} \quad (\text{Equation (37) and Lemma D.1})$$

Thus, we obtained that $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$.

Therefore, due to the definition of $\mathcal{E}$ and by using Lemma B.2, we have that, with probability at least $1 - \delta$:

$$\begin{aligned} H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y})) &\leq 4(\|x - \hat{x}\|_1 + \|y - \hat{y}\|_1) \\ &\leq 8\sqrt{\frac{2\log(\frac{4}{\delta}) + 2n\log(6m)}{m}}, \end{aligned}$$

which concludes the proof. $\square$

We are now ready to show that Algorithm 1 is an (ϵ, δ) -correct method for learning $\mathcal{G}(x, y)$ with minimax optimal sample complexity.

Theorem B.4. *Consider $\alpha = 0$ and let*

$$m \in \tilde{\mathcal{O}}\left(\frac{\log(\frac{1}{\delta})}{\log(1/(1 - \pi_{\min}))} + \frac{n + \log(\frac{1}{\delta})}{\epsilon^2}\right).$$

Then Algorithm 1 is (ϵ, δ) -correct for General-Sum Games and its sample complexity τ_δ is given by m .

Proof. First, we give a precise expression of the number of samples m needed by Algorithm 1.

$$m = \max\left\{\frac{\log(\frac{4n}{\delta})}{\log(\frac{1}{1 - \pi_{\min}})}, 2\left(2 + \frac{4\log(4/\delta)}{(\epsilon/\sqrt{128})^2} + \frac{4n}{(\epsilon/\sqrt{128})^2} \log\left(\frac{12n}{(\epsilon/\sqrt{128})^2}\right)\right)\right\} \quad (38)$$

$$\in \tilde{\mathcal{O}}\left(\frac{\log(\frac{1}{\delta})}{\log(\frac{1}{1 - \pi_{\min}})} + \frac{\log(\frac{1}{\delta}) + n}{\epsilon^2}\right). \quad (39)$$

Now, since $m \geq \frac{\log(\frac{4n}{\delta})}{\log(\frac{1}{1 - \pi_{\min}})}$, by Lemma B.3, we have that, with probability at least $1 - \delta$:

$$H(\mathcal{G}(x, y), \mathcal{G}(\hat{x}, \hat{y})) \leq 8\sqrt{\frac{2\log(\frac{4}{\delta}) + 2n\log(6m)}{m}}$$

Thus, we need to show that, for m chosen as in Equation (38), $8\sqrt{\frac{2\log(\frac{4}{\delta}) + 2n\log(6m)}{m}} \leq \epsilon$. That holds if and only if

$$\frac{\log(\frac{4}{\delta}) + n\log(6m)}{m} \leq \left(\frac{\epsilon}{\sqrt{128}}\right)^2.$$

This, however, is granted by the definition of m . To this end, it is sufficient to apply Lemma D.9 with $c_1 = 4$, $c_2 = 6$, $K = (\epsilon/\sqrt{128})$.

Hence, after m samples, Algorithm 1 is (ϵ, δ) -correct for $\alpha = 0$ and General-Sum Games. $\square$

B.2. Proof of Theorem B.7 ($\alpha = 0$, Zero-Sum Games)

Proof outline The proof of Theorem B.7 is similar in spirit to the one that we presented for Theorem B.4. There are, however, a couple of important differences that complicate the analysis. Importantly, these differences arise from the additional constraints that are present in the definition of $\mathcal{Z}(x, y)$, which require a single matrix to verify constraints both for the x and the y players. Consider, indeed, any pair of strategies (x, y) and $(\hat{x}, \hat{y})$ such that $\text{supp}(x) = \text{supp}(\hat{x})$ and

$\text{supp}(y) = \text{supp}(\hat{y})$. Then, for any $A \in \mathcal{Z}(x, y)$, the construction of $\hat{A}$ that we provided in Lemma B.1 for general-sum games fails to satisfy the constraints of $\mathcal{Z}(\hat{x}, \hat{y})$. To this end, consider the following simple numerical example. Let:

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad x = y = (0.5, 0.5), \quad \hat{x} = (0.6, 0.4), \hat{y} = (0.4, 0.6)$$

In this case, Lemma B.1 yields:

$$\hat{A} = \begin{pmatrix} 1 & -0.42857143 \\ -0.71428571 & 0.71428571 \end{pmatrix}.$$

$$\hat{A} \approx \begin{pmatrix} 1 & -0.43 \\ -0.71 & 0.71 \end{pmatrix}.$$

Since $\hat{A} \notin \mathcal{Z}(\hat{x}, \hat{y})$, we cannot rely on the previous argument for this more complicated case. More generally, we found particularly challenging building analytically a matrix $\hat{A}$ that takes into account variations both in x and y strategy profiles. To this end, however, we developed a trick to take into account the single variations of the x and the y vector. Specifically, we start our analysis with Corollary D.11 which states that

$$H(\mathcal{Z}(x, y), \mathcal{Z}(\hat{x}, \hat{y})) \leq H(\mathcal{Z}(x, y), \mathcal{Z}(\hat{x}, y)) + H(\mathcal{Z}(\hat{x}, y), \mathcal{Z}(\hat{x}, \hat{y})).$$

This result allows us to construct matrices to upper bound the Hausdorff distance again by construction. Specifically, we can fix two matrices $A_1 \in \mathcal{Z}(x, y)$ and $A_2 \in \mathcal{Z}(\hat{x}, y)$ and find two matrices $A_3 \in \mathcal{Z}(\hat{x}, y)$ and $A_4 \in \mathcal{Z}(\hat{x}, \hat{y})$ such that $A_1 \approx A_3$ and $A_2 \approx A_4$ (Lemma B.5 and Lemma B.6). Importantly, note that A_1 and A_3 belong to sets that “vary only across the x player’s strategy” (i.e., y is fixed), while A_2 and A_4 belong to sets that “vary only across the y player’s strategy” (i.e., $\hat{x}$ is fixed). Importantly the approximation errors of both matrices scale linearly with $\|x - \hat{x}\|_1$ and $\|y - \hat{y}\|_1$. Then, once this is done, we proceed identically to Theorem B.4.

We are now ready to start the proof of Theorem B.7. As anticipated, given Corollary D.11, we start by fixing the x -vector and, for any matrix $A \in \mathcal{Z}(x, y)$ we build an alternative matrix $\hat{A} \in \mathcal{Z}(x, \hat{y})$. A careful reader might note that the matrix constructed in the following lemma is similar to the one presented in Lemma B.1 for general-sum games. However, here both the statement and the proofs are different. Indeed, (i) to construct $\hat{A}$ we require that the support of y and $\hat{y}$ coincide and (ii) we need to ensure that our construction satisfies a larger number of constraints.

We remark that in the following lemma, $x, y, \hat{y}$ are generic strategies on the simplex.

Lemma B.5 (Alternative Matrix Construction (fix x); $\alpha = 0$; Zero-Sum Games). *Let $\alpha = 0$, and $x, y, \hat{y} \in \Delta_n$. Let $A \in \mathcal{Z}(x, y)$. It holds that:*

$$\text{supp}(y) = \text{supp}(\hat{y}) \implies \exists \hat{A} \in \mathcal{Z}(x, \hat{y}) : \|A - \hat{A}\|_\infty \leq 4\|y - \hat{y}\|_1$$

Furthermore, $\hat{A}$ is defined as follows:

$$\hat{A}_{ij} := \begin{cases} \frac{A_{ij}}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in \arg\max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj}, \\ \frac{A_{ij} + \Delta_i}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \in \text{supp}(x) \setminus \arg\max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj}, \\ \frac{A_{ij} + 2\|y - \hat{y}\|_1}{1 + 2\|y - \hat{y}\|_1}, & \text{if } i \notin \text{supp}(x), \end{cases}$$

where, for all $i \in \text{supp}(x)$, Δ_i is given by

$$\Delta_i = \max_{k \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{j \in [n]} \hat{y}_j A_{ij}$$

Proof. Given the definition of $\hat{A}$, we already know from Lemma B.1 that all the following statements are true:

- (i) $\|A - \hat{A}\|_\infty \leq 4\|y - \hat{y}\|_1$
- (ii) $e_i^\top \hat{A} \hat{y} = e_j^\top \hat{A} \hat{y} \quad \forall i, j \in \text{supp}(x)$
- (iii) $e_i^\top \hat{A} \hat{y} \leq e_j^\top \hat{A} \hat{y} \quad \forall i \in \text{supp}(x), \forall j \notin \text{supp}(x)$.

Hence, from Lemma D.8 it only remains to verify that $\hat{A}$ satisfies:

$$x^\top \hat{A} e_j = x^\top \hat{A} \hat{y} \quad \forall j \in \text{supp}(\hat{y}) \quad (40a)$$

$$x^\top \hat{A} e_j \leq x^\top \hat{A} \hat{y} \quad \forall j \notin \text{supp}(\hat{y}). \quad (40b)$$

Before proving Equations (40a) and (40b) we make some preliminary considerations. First, we recall from Lemma D.8 that, since $A \in \mathcal{Z}(x, y)$, it holds that:

$$x^\top A y = x^\top A e_j = e_i^\top A y, \quad \forall i \in \text{supp}(x), j \in \text{supp}(y), \quad (41)$$

$$x^\top A e_j \leq x^\top A y \leq e_i^\top A y, \quad \forall i \notin \text{supp}(x), \forall j \notin \text{supp}(y). \quad (42)$$

Now, let $\Delta = 2\|y - \hat{y}\|_1$ and let $j \in [n]$. Then, we have that:

$$\begin{aligned} x^\top \hat{A} e_j &= \sum_{i \in [n]} x_i \hat{A}_{ij} \\ &= \frac{1}{1 + \Delta} \left(x^\top A e_j + \sum_{i: x_i > 0} x_i \Delta_i \right). \end{aligned} \quad (\text{Definition of } \hat{A})$$

Now, let us focus in detail on the term $\sum_{i: x_i > 0} x_i \Delta_i$. We have that:

$$\begin{aligned} \sum_{i: x_i > 0} x_i \Delta_i &= \sum_{i: x_i > 0} x_i \sum_{j \in [n]} \hat{y}_j (A_{kj} - A_{ij}) \quad (\text{Definition of } \Delta_i \text{ and } k \in \arg\max_{\star \in \text{supp}(x)} \sum_{j \in [n]} \hat{y}_j A_{\star, j}) \\ &= \sum_{j \in [n]} \hat{y}_j \left(A_{kj} - \sum_{i: x_i > 0} x_i A_{ij} \right) \\ &= \sum_{j \in [n]} \hat{y}_j (A_{kj} - x^\top A e_j) \\ &= \sum_{j \in [n]} \hat{y}_j A_{kj} - \sum_{j: y_j > 0} \hat{y}_j (x^\top A e_j) \quad (\text{supp}(y) = \text{supp}(\hat{y})) \\ &= \sum_{j \in [n]} \hat{y}_j A_{kj} - x^\top A y \quad (\text{Equation (41)}) \end{aligned}$$

Hence, we obtained that, for all $j \in [n]$:

$$x^\top \hat{A} e_j = \frac{1}{1 + \Delta} \left(x^\top A e_j - x^\top A y + \max_{k \in \text{supp}(x)} e_k^\top A \hat{y} \right) \quad (43)$$

We are now ready to conclude the proof. Consider $j_1, j_2 \in [n]$ such that $y_{j_1} > 0$ and $y_{j_2} = 0$. Then, combining Equations (41) and (43), we have that:

$$\begin{aligned} x^\top \hat{A} e_{j_1} &= \frac{\max_{k \in \text{supp}(x)} e_k^\top A \hat{y}}{1 + \Delta} \\ x^\top \hat{A} e_{j_2} &\leq \frac{\max_{k \in \text{supp}(x)} e_k^\top A \hat{y}}{1 + \Delta}, \end{aligned}$$

thus proving Conditions (40a) and (40b). Hence, $\hat{A} \in \mathcal{Z}(x, \hat{y})$. $\square$

We now fix (x, y) and for $A \in \mathcal{Z}(x, y)$, we build $\hat{A} \in \mathcal{Z}(\hat{x}, y)$. The proof of the following lemma is similar to that of Lemma B.5, and the proof is reported mainly for the sake of completeness.

Here, $x, \hat{x}, y$ are again generic strategies on the simplex.

Lemma B.6 (Alternative Matrix Construction (fix y); $\alpha = 0$; Zero-Sum Games). *Let $\alpha = 0$, $A \in \mathcal{Z}(x, y)$. It holds that:*

$$\text{supp}(x) = \text{supp}(\hat{x}) \implies \exists \hat{A} \in \mathcal{Z}(\hat{x}, y) : \|A - \hat{A}\|_\infty \leq 4\|x - \hat{x}\|_1$$

Furthermore, $\hat{A}$ is defined as follows:

$$\hat{A}_{ij} := \begin{cases} \frac{A_{ij}}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \in \arg\max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i A_{ik}, \\ \frac{A_{ij} + \Delta_j}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \in \text{supp}(y) \setminus \arg\max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i A_{ik}, \\ \frac{A_{ij} - 2\|x - \hat{x}\|_1}{1 + 2\|x - \hat{x}\|_1}, & \text{if } j \notin \text{supp}(y), \end{cases}$$

where, for all $j \in \text{supp}(x)$, Δ_j is defined as

$$\Delta_j = \max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i A_{ik} - \sum_{i \in [n]} \hat{x}_i A_{ij}.$$

Proof. First, as in Lemma B.1, one can trivially verify that, since $A \in [-1, 1]^{n \times n}$, $\hat{A} \in [-1, 1]^{n \times n}$ as well. Furthermore, by following analogous arguments, it is also easy to see that $\|A - \hat{A}\|_\infty \leq 4\|x - \hat{x}\|_1$

By applying Lemma D.8 it thus remains to verify that:

$$\hat{x}^\top \hat{A} e_i = \hat{x}^\top \hat{A} e_j, \quad \forall i, j \in \text{supp}(y) \quad (44a)$$

$$\hat{x}^\top \hat{A} e_i \geq \hat{x}^\top \hat{A} e_j, \quad \forall i \in \text{supp}(y), j \notin \text{supp}(y) \quad (44b)$$

$$e_i^\top \hat{A} y = e_i^\top \hat{A} e_j, \quad \forall i, j \in \text{supp}(x) \quad (44c)$$

$$e_i^\top \hat{A} y \leq e_j^\top \hat{A} y, \quad \forall i \in \text{supp}(x), j \notin \text{supp}(x). \quad (44d)$$

Before continuing, we introduce some additional notation for the sake of brevity. Let $\Delta = 2\|x - \hat{x}\|_1$ and define $S = \text{supp}(y)$ and $S_y^* = \arg\max_{k \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i A_{ik}$.

Conditions (44a) and (44b) Given the expression of $\hat{A}$, Equation (44a) is a direct consequence of Lemma B.1. We now prove Equation (44b). Since Equation (44a) holds, we only need to consider $j_1 \in \text{supp}(y)$ and $j_2 \notin \text{supp}(y)$ and show that $\hat{x}^\top \hat{A} e_{j_1} \geq \hat{x}^\top \hat{A} e_{j_2}$. Specifically, we consider $j_1 \in S_y^*$. Then, it holds that:

$$\begin{aligned}
 \hat{x}^\top \hat{A}e_{j_1} &= \frac{1}{1+\Delta} \sum_{i \in [n]} \hat{x}_i A_{ij_1} \\
 &= \frac{1}{1+\Delta} \left(x^\top A e_{j_1} + \sum_{i \in [n]} (\hat{x}_i - x_i) A_{ij_1} \right) \\
 &\geq \frac{1}{1+\Delta} \left(x^\top A e_{j_2} + \sum_{i \in [n]} (\hat{x}_i - x_i) A_{ij_1} \right) \quad (A \in \mathcal{Z}(x, y), \text{ Lemma D.8 and } j_1 \in \text{supp}(y)) \\
 &= \frac{1}{1+\Delta} \left(\sum_{i \in [n]} \hat{x}_i A_{ij_2} + \sum_{i \in [n]} (\hat{x}_i - x_i) (A_{ij_1} - A_{ij_2}) \right) \\
 &\geq \frac{1}{1+\Delta} \left(\sum_{i \in [n]} \hat{x}_i A_{ij_2} - 2\|x - \hat{x}\|_1 \right) \\
 &= \hat{x}^\top \hat{A}e_{j_2},
 \end{aligned}$$

which concludes the proof.

Conditions (44c) and (44d) Here, we proceed in a way that is analogous to Lemma B.5. Specifically, consider any $i \in [n]$. Then, it holds that:

$$\begin{aligned}
 e_i^\top \hat{A}y &= \sum_{j \in [n]} y_j \hat{A}_{ij} \\
 &= \frac{1}{1+\Delta} \left(e_i^\top Ay + \sum_{j: y_j > 0} y_j \Delta_j \right). \quad (\text{Definition of } \hat{A})
 \end{aligned}$$

Let us focus on $\sum_{j: y_j > 0} y_j \Delta_j$. We have that:

$$\begin{aligned}
 \sum_{j: y_j > 0} y_j \Delta_j &= \sum_{j: y_j > 0} y_j \sum_{i \in [n]} \hat{x}_i (A_{ik} - A_{ij}) \quad (\text{Definition of } \Delta_j \text{ and } k \in \arg\max_{\star \in \text{supp}(y)} \sum_{i \in [n]} \hat{x}_i A_{i\star}) \\
 &= \sum_{i \in [n]} \hat{x}_i \left(A_{ik} - \sum_{j: y_j > 0} y_j A_{ij} \right) \\
 &= \sum_{i \in [n]} \hat{x}_i (A_{ik} - e_i^\top Ay) \\
 &= \sum_{i \in [n]} \hat{x}_i A_{ik} - \sum_{i: x_i > 0} \hat{x}_i (e_i^\top Ay) \quad (\text{supp}(x) = \text{supp}(\hat{x})) \\
 &= \sum_{i \in [n]} \hat{x}_i A_{ik} - x^\top Ay \quad (A \in \mathcal{Z}(x, y))
 \end{aligned}$$

Therefore, we obtained that, for all $i \in [n]$:

$$e_i^\top \hat{A}y = \frac{1}{1+\Delta} \left(e_i^\top Ay - x^\top Ay + \max_{k \in \text{supp}(y)} \hat{x}^\top A e_k \right).$$

Now, let $i_1 \in \text{supp}(\hat{x})$ and $i_2 \notin \text{supp}(\hat{x})$. We have that:

$$\begin{aligned}
 e_{i_1}^\top \hat{A}y &= \frac{1}{1+\Delta} \max_{k \in \text{supp}(y)} \hat{x}^\top A e_k \\
 e_{i_2}^\top \hat{A}y &\geq \frac{1}{1+\Delta} \max_{k \in \text{supp}(y)} \hat{x}^\top A e_k
 \end{aligned}$$

which proves (44c) and (44d), thus showing that $\hat{A} \in \mathcal{Z}(\hat{x}, y)$. $\square$

Combining Lemmas B.5 and B.6 with Corollary D.11, we have that, whenever $\text{supp}(x) = \text{supp}(\hat{x})$ and $\text{supp}(y) = \text{supp}(\hat{y})$, then:

$$H(\mathcal{Z}(x, y), \mathcal{Z}(\hat{x}, \hat{y})) \leq 8(\|x - \hat{x}\|_1 + \|y - \hat{y}\|_1) \quad (45)$$

Hence, following arguments that are identical to the ones presented for Theorem B.4, we obtain the following result.

Theorem B.7. Consider $\alpha = 0$ and let

$$m \in \tilde{\mathcal{O}} \left(\frac{\log(\frac{1}{\delta})}{\log(1/(1 - \pi_{\min}))} + \frac{n + \log(\frac{1}{\delta})}{\epsilon^2} \right).$$

Then Algorithm 1 is (ϵ, δ) -correct for Zero-Sum Games and its sample complexity τ_δ is given by m .

Proof. Use Equation (45). Then, the proof follows from the same arguments of Theorem B.4. $\square$

C. Proof of Theorem 4.2

This section is structured as follows:

- In Section C.1 we prove the result for the inverse GSG problem (Theorem C.5)
- In Section C.2, we prove the result of the inverse ZSG problem (Theorem C.8)

The proof of Theorem 4.2 then follows directly by combining Theorems C.5 and C.8.

C.1. Proof of Theorem C.5 ($\alpha > 0$, General-Sum Games)

Proof outline We begin our proof by considering two pair of strategies (x, y) and $(\hat{x}, \hat{y})$ and, for any pair of matrix $(A, B) \in \mathcal{G}_\alpha(x, y)$ we construct:

- $(\hat{A}, \hat{B}) \in \mathcal{G}_\alpha(\hat{x}, y)$ and $\mathcal{G}_\alpha(x, \hat{y})$ such that $\hat{A} \approx A$ and $\hat{B} \approx B$
- $(\hat{A}, \hat{B}) \in \mathcal{G}_\alpha(x, \hat{y})$ and $\mathcal{G}_\alpha(x, y)$ such that $\hat{A} \approx A$ and $\hat{B} \approx B$

This first result is presented in Lemma C.1. In Lemma C.2, we discuss how well these matrices that we constructed approximate the original pair of matrices (A, B) . This lemma is the key result that we discussed in Section 4.2. Specifically, it shows how these errors scales as, e.g.,:

$$\mathcal{O} \left(\frac{1}{\alpha} \sup_{S \in \mathcal{S}(\alpha)} \sum_{i \in S} (\hat{x}_i - x_i) + \frac{1}{\alpha} \sum_{i: \hat{x}_i = 0} x_i \right).$$

Next, using Corollary D.11, we will use this result to formally upper bound, with high probability, the Hausdorff distance between $\mathcal{G}_\alpha(x, y)$ and $\mathcal{G}_\alpha(\hat{x}, \hat{y})$. This is done in Lemmas C.3 and C.4. Finally, Theorem C.5 simply follows by selecting the number of samples in a way such that the high-probability Hausdorff distance upper bound is below ϵ .

We now start with matrix construction argument. Note that in the following lemma $x, y, \hat{x}, \hat{y}$ are generic strategies on the simplex.

Lemma C.1 (Matrix construction, $\alpha > 0$, General-Sum Games). *Let $\alpha > 0$ and $x, y, \hat{x}, \hat{y} \in \Delta_n$. Consider $(A, B) \in \mathcal{G}_\alpha(x, y)$. It holds that:*

$$\exists (\hat{A}, \hat{B}) \in \mathcal{G}_\alpha(\hat{x}, y) : \|A - \hat{A}\|_\infty \leq 2 \left(\frac{(\hat{x} - x)^\top A y}{\alpha} \right)_+ \text{ and } \|B - \hat{B}\|_\infty \leq 2 \left(\frac{\max_{j \in [n]} (\hat{x} - x)^\top B (y - e_j)}{\alpha} \right)_+ \quad (46)$$

$$\exists (\hat{A}, \hat{B}) \in \mathcal{G}_\alpha(x, \hat{y}) : \|A - \hat{A}\|_\infty \leq 2 \left(\frac{\max_{i \in [n]} (x - e_i)^\top A (\hat{y} - y)}{\alpha} \right)_+ \text{ and } \|B - \hat{B}\|_\infty \leq 2 \left(\frac{x^\top B (\hat{y} - y)}{\alpha} \right)_+. \quad (47)$$

Proof. We first prove Equation (46).

Specifically, for $(A, B) \in \mathcal{G}_\alpha(x, y)$ we define $(\hat{A}, \hat{B})$ as $(\hat{A}, \hat{B}) = (\lambda_A A, \lambda_B B)$, where $\lambda_A, \lambda_B \in \mathbb{R}$ are defined as follows

$$\lambda_A = \min \left\{ 1, \frac{\alpha}{\max_{i \in [n]} \hat{x}^\top A y - e_i^\top A y} \right\}, \quad \lambda_B = \min \left\{ 1, \frac{\alpha}{\max_{j \in [n]} \hat{x} B y - \hat{x} B e_j} \right\}.$$

From now on, w.l.o.g., we assume that $\lambda_A, \lambda_B > 0$. Otherwise, we could simply pick $\hat{A} = A$ and $\hat{B} = B$ and all the Nash constraints are verified.

Next, we verify that $(A, B) \in \mathcal{G}_\alpha(\hat{x}, y)$. First, under the assumption that $\lambda_A, \lambda_B > 0$, it follows that $\lambda_A, \lambda_B \in (0, 1]$, and hence $\hat{A}, \hat{B} \in [-1, 1]^{n \times n}$. Secondly, for any $i \in [n]$:

$$\begin{aligned} \hat{x}^\top \hat{A} y - e_i^\top \hat{A} y &= \lambda_A (\hat{x}^\top A y - e_i^\top A y) \\ &= \min \left\{ 1, \frac{\alpha}{\max_{i \in [n]} \hat{x}^\top A y - e_i^\top A y} \right\} (\hat{x}^\top A y - e_i^\top A y) \\ &\leq \alpha, \end{aligned}$$

where, (i) if $\lambda_A < 1$, the last step is trivial, and (ii) if $\lambda_A = 1$, then, $\alpha > \max_{i \in [n]} \hat{x}^\top A y - e_i^\top A y$. Thus, we have proved that $\hat{A} \in \mathcal{G}_\alpha^x(\hat{x}, y)$. The proof for showing that $\hat{B} \in \mathcal{G}_\alpha^y(\hat{x}, y)$ is equivalent.

Next, we proceed by analyzing $\|A - \hat{A}\|_\infty$. We consider the case where $\lambda_A < 1$, otherwise $A = \hat{A}$ and Equation (46) is trivial. We have that:

$$\begin{aligned} \|A - \hat{A}\|_\infty &\leq (1 - \lambda_A) \|A\|_\infty \\ &\leq 2(1 - \lambda_A) \\ &\leq 2 \frac{\max_{i \in [n]} \hat{x}^\top A y - e_i^\top A y - \alpha}{\alpha} \\ &= 2 \frac{(\hat{x} - x)^\top A y + x^\top A y - \min_{i \in [n]} e_i^\top A y - \alpha}{\alpha} \\ &\leq 2 \frac{(\hat{x} - x)^\top A y}{\alpha}, \end{aligned} \quad (A \in \mathcal{G}_\alpha^x(x, y))$$

which proves the first part of Equation (46). One can follow the same steps to prove the second part of Equation (46).

The proof of Equation (47) follows from analogous argument. It is indeed sufficient to define $(\hat{A}, \hat{B}) = (\tilde{\lambda}_A A, \tilde{\lambda}_B B)$ where:

$$\tilde{\lambda}_A = \min \left\{ 1, \frac{\alpha}{\max_{i \in [n]} x^\top A \hat{y} - e_i^\top A \hat{y}} \right\}, \quad \tilde{\lambda}_B = \min \left\{ 1, \frac{\alpha}{\max_{j \in [n]} x^\top B \hat{y} - x^\top B e_j} \right\}.$$

The proof is thus concluded. $\square$

Next, the following lemma shows how to upper bound the terms that appeared in Lemma C.1. We briefly recall that, for any $p \in \Delta^n$, $S_\alpha(p)$ is defined as

$$S_\alpha(p) = \left\{ S \subseteq [n] : \forall i \in S \ p_i > 0 \text{ and } \sum_{i \in S} p_i \leq \frac{\alpha}{2} \right\}.$$

Also in the following lemma, $x, y, \hat{x}, \hat{y}$ are generic strategies on the simplex.

Lemma C.2 (Key Intermediate Step). *Let $\alpha > 0$ and $x, y, \hat{x}, \hat{y} \in \Delta_n$. The following holds:*

$$\sup_{A \in \mathcal{G}_\alpha^x(x, y)} ((\hat{x} - x)^\top A y)_+ \leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i=0} \hat{x}_i \quad (48)$$

$$\sup_{B \in \mathcal{G}_\alpha^y(x, y)} \left(\max_{j \in [n]} (\hat{x} - x)^\top B(y - e_j) \right)_+ \leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i=0} \hat{x}_i \quad (49)$$

$$\sup_{A \in \mathcal{G}_\alpha^x(x, y)} \left(\max_{i \in [n]} (x - e_i)^\top A(\hat{y} - y) \right)_+ \leq 4 \sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + 2 \sum_{j: y_j=0} \hat{y}_j \quad (50)$$

$$\sup_{B \in \mathcal{G}_\alpha^y(x, y)} (x^\top B(\hat{y} - y))_+ \leq 4 \sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + 2 \sum_{j: y_j=0} \hat{y}_j. \quad (51)$$

Proof. First, we observe that, due to symmetry, once Equation (48) and Equation (49) hold, Equation (50) and Equation (51) hold as well.

We now start by proving Equation (48). We assume w.l.o.g. that the supremum is positive. Furthermore, for any $A \in \mathcal{G}_\alpha(x, y)$, denote by $\star \in [n]$ any index that attains the argmin in $\min_{i \in [n]} (Ay)_i$. Then, we have that:

$$\begin{aligned} \sup_{A \in \mathcal{G}_\alpha^x(x, y)} ((\hat{x} - x)^\top A y)_+ &= \sup_{A \in \mathcal{G}_\alpha^x(x, y)} \sum_{i \in [n]} (\hat{x}_i - x_i) [(Ay)_i - (Ay)_\star] \quad (\hat{x}, x \in \Delta_n) \\ &\leq \left(\sup_{A \in \mathcal{G}_\alpha^x(x, y)} \sum_{i: x_i > 0} (\hat{x}_i - x_i) [(Ay)_i - (Ay)_\star] \right) + 2 \sum_{i: x_i=0} \hat{x}_i. \end{aligned}$$

Let us focus on the remaining optimization problem over A . We are interested in solving:²¹

$$\begin{aligned} \max_{A \in [-1, 1]^{n \times n}} \quad & \sum_{i: x_i > 0} (\hat{x}_i - x_i) [(Ay)_i - (Ay)_\star] \\ \text{s.t.} \quad & \sum_{i: x_i > 0} x_i [(Ay)_i - (Ay)_\star] \leq \alpha. \end{aligned} \quad (52)$$

Let us introduce the following auxiliary variable $c_i = x_i [(Ay)_i - (Ay)_\star] \in [0, 2x_i]$. Then, (52) is upper bounded by:

$$\begin{aligned} \max_{(c_1, \dots, c_n)} \quad & \sum_{i: x_i > 0} c_i \frac{(\hat{x}_i - x_i)}{x_i} \\ \text{s.t.} \quad & \sum_{i: x_i > 0} c_i \leq \alpha \text{ and } c_i \in [0, 2x_i]. \end{aligned} \quad (53)$$

This is a fractional knapsack problem. It is well-known that such problems can be solved by a greedy algorithm that sorts the indexes i 's according to their value (i.e., $\frac{\hat{x}_i - x_i}{x_i}$) and allocates the maximum value of c_i according to this order until all the budget (i.e., $\sum_{i: x_i > 0} c_i \leq \alpha$) expires. More formally, any optimal solution $c^\star$ of (53) has the following shape. Let $S \subseteq \{i \in [n] : x_i > 0\}$ be any subset such that

$$\min_{i \in S} \frac{\hat{x}_i - x_i}{x_i} \geq \max_{i \in S^c} \frac{\hat{x}_i - x_i}{x_i},$$

where $S^c = \{i \in [n] : x_i > 0\} \setminus S$. Then $c_i^\star = 2x_i$ for all $i \in S$. Furthermore, there exists at most one $j \in S^c$ such that $c_j^\star \in [0, 2x_j]$. For all the remaining elements, we have that $c_i^\star = 0$. Hence, the optimal value of (53) can be written (and upper bounded) as follows:

$$\begin{aligned} 2 \sum_{i \in S} (\hat{x}_i - x_i) + \frac{\hat{x}_j - x_j}{x_j} c_j^\star &\leq 4 \sum_{i \in S} (\hat{x}_i - x_i) \\ &\leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i). \end{aligned} \quad (\text{Def. of } \mathcal{S}_\alpha(x))$$

²¹The sup is attained since it is an optimization problem over a compact domain of a continuous function.

where the first inequality step follows from the fact that, for $j \in S^c$ it holds that $c_j \frac{\hat{x}_j - x_j}{x_j} \leq c_i \frac{\hat{x}_i - x_i}{x_i}$ for any $i \in S$. This concludes the proof of Equation (48).

Next, we continue by proving Equation (49). Again, we assume w.l.o.g. that the sup is greater or equal than 0. Then, we have that:

$$\begin{aligned}
 \sup_{B \in \mathcal{G}_\alpha^y(x, y)} \left(\max_{j \in [n]} (\hat{x} - x)^\top B(y - e_j) \right)_+ &= \max_{B \in \mathcal{G}_\alpha^y(x, y)} \max_{\star \in [n]} \sum_{i \in [n]} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] \\
 &\leq \max_{B \in \mathcal{G}_\alpha^y(x, y)} \max_{\star \in [n]} \sum_{i: x_i > 0} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] + 2 \sum_{i: x_i = 0} \hat{x}_i \\
 &= \max_{\star \in [n]} \max_{\substack{B \in \mathcal{G}_\alpha(x, y) \\ \star \in \arg\max (\hat{x} - x)^\top B(y - e_\star)}} \sum_{i: x_i > 0} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] + 2 \sum_{i: x_i = 0} \hat{x}_i \\
 &\leq \max_{\star \in [n]} \max_{B \in \mathcal{G}_\alpha(x, y)} \sum_{i: x_i > 0} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] + 2 \sum_{i: x_i = 0} \hat{x}_i.
 \end{aligned}$$

Hence, fix any $\star \in [n]$ in the outer maximization problem, and restrict our attention to the maximization problem over B , that is

$$\begin{aligned}
 &\max_{B \in [-1, 1]^{n \times n}} \sum_{i: x_i > 0} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] \\
 \text{s.t. } &\sum_{i: x_i > 0} x_i \left[\sum_{j \in [n]} y_j B_{ij} - B_{ik} \right] \leq \alpha \quad \forall k \in [n].
 \end{aligned} \tag{54}$$

Since the α -Nash constraint holds for all $k \in [n]$, it holds also for $\star$. Hence, we can upper bound (54) with the following problem:

$$\begin{aligned}
 &\max_{B \in [-1, 1]^{n \times n}} \sum_{i: x_i > 0} (\hat{x}_i - x_i) \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] \\
 \text{s.t. } &\sum_{i: x_i > 0} x_i \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right] \leq \alpha
 \end{aligned} \tag{55}$$

Introducing the variable $c_i = x_i \left[\sum_{j \in [n]} y_j B_{ij} - B_{i\star} \right]$ and rewriting the objective function as $\sum_{i: x_i > 0} c_i \frac{\hat{x}_i - x_i}{x_i}$, we obtain again a fractional knapsack problem that can be solved in closed form, as we presented above. Following the same steps as above, the proof is concluded. $\square$

Here, we show how to exploit the previous results to upper bound the Hausdorff distance. Here, we will use the fact that $\hat{x}$ and $\hat{y}$ are the empirical mean of x and y after m samples.

Lemma C.3 (Matrix Construction $\rightarrow$ Hausdorff Distance). *Let $\alpha > 0$ and $x, y, \hat{x}, \hat{y} \in \Delta_n$. Specifically, $\hat{x}$ and $\hat{y}$ are the maximum likelihood estimators of x, y after m samples. Let $f_\alpha(x, \hat{x})$ and $f_\alpha(y, \hat{y})$ be defined as:*

$$\begin{aligned}
 f_\alpha(x, \hat{x}) &= \frac{16}{\alpha} \left(\sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + \sup_{S \in \mathcal{S}_\alpha(\hat{x})} \sum_{i \in S} (x_i - \hat{x}_i) + \sum_{i: \hat{x}_i = 0} x_i \right) \\
 f_\alpha(y, \hat{y}) &= \frac{16}{\alpha} \left(\sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + \sup_{S \in \mathcal{S}_\alpha(\hat{y})} \sum_{j \in S} (y_j - \hat{y}_j) + \sum_{j: \hat{y}_j = 0} y_j \right).
 \end{aligned}$$

It holds that:

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \leq f_\alpha(x, \hat{x}) + f_\alpha(y, \hat{y}).$$

Proof. From Corollary D.11, we have that:

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \leq H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y)) + H(\mathcal{G}_\alpha(\hat{x}, y), \mathcal{G}_\alpha(\hat{x}, \hat{y}))$$

Let us focus on the first term. From Lemma C.1, we have that:

$$\begin{aligned} H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y)) &\leq 2 \sup_{A \in \mathcal{G}_\alpha^x(x, y)} \left(\frac{(\hat{x} - x)^\top A y}{\alpha} \right)_+ + 2 \sup_{A \in \mathcal{G}_\alpha^x(\hat{x}, y)} \left(\frac{(x - \hat{x})^\top A y}{\alpha} \right)_+ + \\ &\quad + \sup_{A \in \mathcal{G}_\alpha^y(x, y)} 2 \left(\frac{\max_{j \in [n]} (\hat{x} - x)^\top B(y - e_j)}{\alpha} \right)_+ + \\ &\quad + \sup_{A \in \mathcal{G}_\alpha^y(\hat{x}, y)} 2 \left(\frac{\max_{j \in [n]} (x - \hat{x})^\top B(y - e_j)}{\alpha} \right)_+ \end{aligned}$$

This, in turn, can be upper bounded using Lemma C.2. We thus obtain:

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y)) \leq \frac{16}{\alpha} \left(\sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + \sup_{S \in \mathcal{S}_\alpha(\hat{x})} \sum_{i \in S} (x_i - \hat{x}_i) + \sum_{i: \hat{x}_i = 0} x_i \right),$$

where we used the fact that $\sum_{x_i=0} \hat{x}_i = 0$. Similarly, for $H(\mathcal{G}_\alpha(\hat{x}, y), \mathcal{G}_\alpha(\hat{x}, \hat{y}))$, we obtain:

$$H(\mathcal{G}_\alpha(\hat{x}, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \leq \frac{16}{\alpha} \left(\sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + \sup_{S \in \mathcal{S}_\alpha(\hat{y})} \sum_{j \in S} (y_j - \hat{y}_j) + \sum_{j: \hat{y}_j = 0} y_j \right),$$

thus concluding the proof. $\square$

The following lemma shows how to upper bound with probabilistic argument the terms that appeared in the previous lemma.

Lemma C.4 (Haussdorf High-Probability upper bound). *Let m be such that:*

$$m \geq 1 + \frac{4 \left(n + \log \left(\frac{24}{\delta} \right) \right)}{\alpha}. \quad (56)$$

With probability at least $1 - \delta$ it holds that:

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \leq \frac{64}{\alpha} \left(\sqrt{\frac{4\alpha \left(n + \log \left(\frac{6}{\delta} \right) \right)}{m}} + \frac{5 \left(n + \log \left(\frac{6}{\delta} \right) \right)}{m} + \frac{3\sqrt{n} \log \left(\frac{6}{\delta} \right)}{m} \right).$$

Proof. We first fix any $m \geq 1$ and we consider the following good event

$$\mathcal{E} = \mathcal{E}_{\text{miss}} \cap \mathcal{E}_{\text{set}} \cap \mathcal{E}_{\text{err}, x} \cap \mathcal{E}_{\text{err}, y}$$

where these events are defined as:

$$\begin{aligned}\mathcal{E}_{\text{miss}} &= \left\{ \sum_{i:\hat{x}_i=0} x_i - \mathbb{E} \left[\sum_{i:\hat{x}_i=0} x_i \right] \leq \frac{3\sqrt{n} \log(\frac{6}{\delta})}{m} \right\} \cap \left\{ \sum_{j:\hat{y}_j=0} y_j - \mathbb{E} \left[\sum_{j:\hat{y}_j=0} y_j \right] \leq \frac{3\sqrt{n} \log(\frac{6}{\delta})}{m} \right\} \\ \mathcal{E}_{\text{set},x} &= \bigcap_{S \subseteq [n]} \left\{ \left| \sum_{i \in S} (\hat{x}_i - x_i) \right| \leq \sqrt{\frac{2 \sum_{i \in S} \hat{x}_i (n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right\} \\ \mathcal{E}_{\text{set},y} &= \bigcap_{S \subseteq [n]} \left\{ \left| \sum_{j \in S} (\hat{y}_j - y_j) \right| \leq \sqrt{\frac{2 \sum_{j \in S} \hat{y}_j (n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right\} \\ \mathcal{E}_{\text{err},x} &= \bigcap_{S \in \mathcal{S}_{2\alpha}(x)} \left\{ \left| \sum_{i \in S} (\hat{x}_i - x_i) \right| \leq \sqrt{\frac{4\alpha (n + \log(\frac{6}{\delta}))}{m}} + \frac{4(n + \log(\frac{6}{\delta}))}{m} \right\} \\ \mathcal{E}_{\text{err},y} &= \bigcap_{S \in \mathcal{S}_{2\alpha}(y)} \left\{ \left| \sum_{j \in S} (\hat{y}_j - y_j) \right| \leq \sqrt{\frac{4\alpha (n + \log(\frac{6}{\delta}))}{m}} + \frac{4(n + \log(\frac{6}{\delta}))}{m} \right\}\end{aligned}$$

We now continue by showing that $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$. In particular, using a union bound, we obtain that $\mathbb{P}(\mathcal{E}) = 1 - \mathbb{P}(\mathcal{E}^c) \geq 1 - (\mathbb{P}(\mathcal{E}_{\text{miss}}^c) + \mathbb{P}(\mathcal{E}_{\text{set},x}^c) + \mathbb{P}(\mathcal{E}_{\text{set},y}^c) + \mathbb{P}(\mathcal{E}_{\text{err},x}^c) + \mathbb{P}(\mathcal{E}_{\text{err},y}^c))$. Hence, we continue by upper bounding the complement of each event that compose $\mathcal{E}$. We start from $\mathbb{P}(\mathcal{E}_{\text{miss}}^c)$. We have that:

$$\mathbb{P}(\mathcal{E}_{\text{miss}}^c) \leq \mathbb{P} \left(\sum_{i:\hat{x}_i=0} x_i - \mathbb{E} \left[\sum_{i:\hat{x}_i=0} x_i \right] \geq \frac{\sqrt{n} \log(\frac{6}{\delta})}{m} \right) + \mathbb{P} \left(\sum_{i:\hat{y}_i=0} y_i - \mathbb{E} \left[\sum_{i:\hat{y}_i=0} y_i \right] \geq \frac{\sqrt{n} \log(\frac{6}{\delta})}{m} \right) \leq \frac{\delta}{3}.$$

where the last inequality is a direct application of concentration results for the missing mass, *i.e.*, Lemma D.2.

Next, we analyze $\mathbb{P}(\mathcal{E}_{\text{set},x}^c)$ and $\mathbb{P}(\mathcal{E}_{\text{set},y}^c)$. Using Lemma D.5, we have that:

$$\mathbb{P}(\mathcal{E}_{\text{set},x}^c) = \mathbb{P} \left(\exists S \subseteq [n] : \left| \sum_{i \in S} (\hat{x}_i - x_i) \right| > \sqrt{\frac{2 \sum_{i \in S} \hat{x}_i (n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right) \leq \frac{\delta}{6}.$$

Finally, it remains to analyze $\mathbb{P}(\mathcal{E}_{\text{err},x}^c)$ and $\mathbb{P}(\mathcal{E}_{\text{err},y}^c)$. We present our argument for $\mathbb{P}(\mathcal{E}_{\text{err},x}^c)$ and the computations for $\mathbb{P}(\mathcal{E}_{\text{err},y}^c)$ follows from analogous arguments. Specifically, using Lemma D.5, we have that:

$$\mathbb{P}(\mathcal{E}_{\text{err},x}^c) = \mathbb{P} \left(\exists S \in \mathcal{S}(2\alpha) : \left| \sum_{i \in S} (\hat{x}_i - x_i) \right| \geq \sqrt{\frac{4\alpha (n + \log(\frac{6}{\delta}))}{m}} + \frac{4(n + \log(\frac{6}{\delta}))}{m} \right) \leq \frac{\delta}{6}.$$

Hence, we obtained that $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$.

At this point, we proceed by upper bounding the Hausdorff distance under the good event $\mathcal{E}$. We start by applying Lemma C.3 and we analyze $f_\alpha(x, \hat{x})$.²² We recall that:

$$f_\alpha(x, \hat{x}) = \frac{16}{\alpha} \left(\sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + \sup_{S \in \mathcal{S}_\alpha(\hat{x})} \sum_{i \in S} (x_i - \hat{x}_i) + \sum_{i:\hat{x}_i=0} x_i \right)$$

²²The arguments for upper bounding $f_\alpha(y, \hat{y})$ are symmetric.

At this point, let us focus on $\mathcal{S}_\alpha(\hat{x})$. With probability at least $1 - \delta$ it holds that:

$$\begin{aligned}
 \mathcal{S}_\alpha(\hat{x}) &= \{S \subseteq [n] : \hat{x}_i > 0 \forall i \in S \text{ and } \sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2}\} \\
 &\subseteq \{S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2}\} \quad (\hat{x}_i > 0 \implies x_i > 0) \\
 &= \left\{ S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2} \text{ and } \sum_{i \in S} x_i \leq \frac{\alpha}{2} + \left| \sum_{i \in S} x_i - \hat{x}_i \right| \right\} \\
 &\subseteq \left\{ S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2} \text{ and } \sum_{i \in S} x_i \leq \frac{\alpha}{2} + \sqrt{\frac{2 \sum_{i \in S} \hat{x}_i (n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right\} \\
 &= \left\{ S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2} \text{ and } \sum_{i \in S} x_i \leq \frac{\alpha}{2} + \sqrt{\frac{\alpha(n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right\} \\
 &\subseteq \left\{ S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} x_i \leq \frac{\alpha}{2} + \sqrt{\frac{\alpha(n + \log(\frac{24}{\delta}))}{m}} + \frac{4(n + \log(\frac{24}{\delta}))}{m-1} \right\} \\
 &\subseteq \{S \subseteq [n] : x_i > 0 \forall i \in S \text{ and } \sum_{i \in S} x_i \leq \alpha\} \quad \text{(Equation (56))} \\
 &= S_{2\alpha}(x).
 \end{aligned}$$

where (i) the equality steps in the middle of these arguments follows from the fact that, with high probability, $\sum_{i \in S} \hat{x}_i \leq \frac{\alpha}{2}$ implies the additional conditions that are mentioned in the formula, and (ii) the last $\subseteq$ step follows from the fact that, if m satisfies Equation (56), then:

$$\sqrt{\frac{\alpha(n + \log(\frac{24}{\delta}))}{m}} + \frac{4 \log(\frac{24}{\delta})}{m-1} \leq \frac{\alpha}{2}.$$

Plugging this result within $f_\alpha(x, \hat{x})$, we have that, with probability at least $1 - \delta$,

$$\begin{aligned}
 f_\alpha(x, \hat{x}) &\leq \frac{32}{\alpha} \left(\sup_{S \in \mathcal{S}_{2\alpha}(x)} \left| \sum_{i \in S} \hat{x}_i - x_i \right| + \sum_{i: \hat{x}_i=0} x_i \right) \\
 &\leq \frac{32}{\alpha} \left(\sqrt{\frac{4\alpha(n + \log(\frac{6}{\delta}))}{m}} + \frac{4(n + \log(\frac{6}{\delta}))}{m} + \sum_{i: \hat{x}_i=0} x_i \right) \quad \text{(Def. of } \mathcal{E} \text{)} \\
 &\leq \frac{32}{\alpha} \left(\sqrt{\frac{4\alpha(n + \log(\frac{6}{\delta}))}{m}} + \frac{4(n + \log(\frac{6}{\delta}))}{m} + \frac{3\sqrt{n} \log(\frac{6}{\delta})}{m} + \mathbb{E} \left[\sum_{i: \hat{x}_i=0} x_i \right] \right). \quad \text{(Def. of } \mathcal{E} \text{)}
 \end{aligned}$$

It thus remains to control $\mathbb{E}[\sum_{i: \hat{x}_i=0} x_i]$. It holds that:

$$\mathbb{E} \left[\sum_{i: \hat{x}_i=0} x_i \right] = \sum_{i \in [n]} x_i \mathbb{P}[\hat{x}_i = 0] = \sum_{i \in [n]} x_i (1 - x_i)^m \leq n \max_{p \in [0,1]} p(1-p)^m \leq \frac{n}{m}.$$

Following similar steps, we arrived at the following high-probability bound:

$$\begin{aligned}
 H(G_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) &\leq f_\alpha(x, \hat{x}) + f_\alpha(y, \hat{y}) \\
 &\leq \frac{64}{\alpha} \left(\sqrt{\frac{4\alpha(n + \log(\frac{6}{\delta}))}{m}} + \frac{5(n + \log(\frac{6}{\delta}))}{m} + \frac{3\sqrt{n} \log(\frac{6}{\delta})}{m} \right),
 \end{aligned}$$

which concludes the proof. $\square$

We are now ready to prove Theorem C.5.

Theorem C.5. *Let $\alpha > 0$ and let*

$$m \in \mathcal{O} \left(\frac{n + \log(\frac{1}{\delta})}{\alpha \epsilon^2} + \frac{\sqrt{n} \log(\frac{1}{\delta})}{\alpha \epsilon} \right).$$

Then, Algorithm 1 is (ϵ, δ) -correct for General-Sum Games and its sample complexity τ_δ is given by m .

Proof. Let m_1, m_2, m_3, m_4 be defined as follows:

$$\begin{aligned} m_1 &= 1 + \frac{4(n + \log(\frac{24}{\delta}))}{\alpha} \\ m_2 &= \frac{16(192)^2(n + \log(6/\delta))}{\alpha \epsilon^2} \\ m_3 &= \frac{960(n + \log(6/\delta))}{\alpha \epsilon} \\ m_4 &= \frac{576\sqrt{n} \log(6/\delta)}{\epsilon \alpha}. \end{aligned}$$

Finally, let $m = \lceil \max\{m_1, m_2, m_3, m_4\} \rceil$. Since $m \geq m_1$, we can apply Lemma C.4, and using the definition of m , one can verify that, with probability at least $1 - \delta$,²³

$$H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \leq \epsilon.$$

thus concluding the proof. $\square$

C.2. Proof of Theorem C.8 ($\alpha > 0$, Zero-Sum Games)

Proof outline The proof of Theorem C.8 is in the same spirit as that of Theorem C.5. Specifically, first, for any pair of strategies (x, y) and $(\hat{x}, \hat{y})$, we fix a matrix $A \in \mathcal{Z}_\alpha(x, y)$ and we construct a pair of matrices $(\hat{A}_1, \hat{A}_2) \in \mathcal{Z}_\alpha(\hat{x}, y) \times \mathcal{Z}_\alpha(x, \hat{y})$ (Lemma C.6). Remarkably, $\hat{A}_1 \approx A$ and $\hat{A}_2 \approx A$ with error rates that are similar to those that we presented for the General-Sum Setting. Then, in Lemma C.7, we upper bound these errors by heavily exploiting the key intermediate step that we presented for GSG's (i.e., Lemma C.2). Once this result is obtained, one can follow the same exact proof for General-Sum as for Zero-Sum Games.

Lemma C.6 (Matrix construction, $\alpha > 0$, Zero-Sum Games). *Let $\alpha > 0$ and $x, y, \hat{x}, \hat{y} \in \Delta_n$. Consider $A \in \mathcal{Z}_\alpha(x, y)$. It holds that:*

$$\exists \hat{A} \in \mathcal{Z}_\alpha(\hat{x}, y) : \|A - \hat{A}\|_\infty \leq 2 \left(\max \left\{ \frac{(\hat{x} - x)^\top A y}{\alpha}, \frac{\max_{j \in [n]} (\hat{x} - x)^\top A (e_j - y)}{\alpha} \right\} \right)_+ \quad (57)$$

$$\exists \hat{A} \in \mathcal{Z}_\alpha(x, \hat{y}) : \|A - \hat{A}\|_\infty \leq 2 \left(\max \left\{ \frac{\max_{i \in [n]} (x - e_i)^\top A (\hat{y} - y)}{\alpha}, \frac{x^\top A (\hat{y} - y)}{\alpha} \right\} \right)_+. \quad (58)$$

Proof. We first prove Equation (57).

Let $A \in \mathcal{Z}_\alpha(x, y)$. We define $\hat{A} = \lambda A$ where $\lambda \in \mathbb{R}$ is given by:

$$\lambda = \min \left\{ 1, \frac{\alpha}{\max\{\max_{i \in [n]} \hat{x}^\top A y - e_i^\top A y, \max_{j \in [n]} \hat{x}^\top A e_j - \hat{x}^\top A y\}} \right\}.$$

In the following, we assume w.l.o.g., that $\lambda \geq 0$. If $\lambda < 0$, one could pick $\hat{A} = A$ and all the equilibrium constraints would be satisfied.

²³Here, we note that $m \geq m_i$ imposes that the $i + 1$ -th error term in the statement of Lemma C.4 is bounded by $\frac{\epsilon}{3}$.

Now, we verify that $A \in \mathcal{Z}_\alpha(\hat{x}, y)$. First, since $\lambda \in (0, 1]$, we have that $\hat{A} \in [-1, 1]^{n \times n}$. Secondly, for all $i \in [n]$ it holds that:

$$\begin{aligned} \hat{x}^\top \hat{A}y - e_i^\top \hat{A}y &= \lambda(\hat{x}^\top Ay - e_i^\top Ay) \\ &= \min \left\{ 1, \frac{\alpha}{\max\{\max_{i \in [n]} \hat{x}^\top Ay - e_i^\top Ay, \max_{j \in [n]} \hat{x}^\top Ae_j - \hat{x}^\top Ay\}} \right\} (\hat{x}^\top Ay - e_i^\top Ay) \\ &\leq \alpha, \end{aligned}$$

where, (i) if $\lambda = 1$, we have that $\max_{i \in [n]} \hat{x}^\top Ay - e_i^\top Ay \leq \alpha$, and (ii) if $\lambda < 1$, the last step is trivial. Following similar arguments, one can prove that $\hat{x}^\top \hat{A}e_j - \hat{x}^\top \hat{A}y \leq \alpha$ for all $j \in [n]$. Hence, $\hat{A} \in \mathcal{Z}_\alpha(\hat{x}, y)$.

Next, we proceed by analyzing $\|A - \hat{A}\|_\infty$. We consider the case where $\lambda < 1$, otherwise $A = \hat{A}$ and Equation (57) is direct. We have that:

$$\begin{aligned} \|A - \hat{A}\|_\infty &\leq (1 - \lambda)\|A\|_\infty \\ &\leq 2(1 - \lambda) \\ &\leq 2 \frac{\max\{\max_{i \in [n]} \hat{x}^\top Ay - e_i^\top Ay, \max_{j \in [n]} \hat{x}^\top Ae_j - \hat{x}^\top Ay\} - \alpha}{\alpha} \\ &\leq 2 \max \left\{ \frac{\max_{i \in [n]} \hat{x}^\top Ay - e_i^\top Ay - \alpha}{\alpha}, \frac{\max_{j \in [n]} \hat{x}^\top Ae_j - \hat{x}^\top Ay - \alpha}{\alpha} \right\} \\ &\leq 2 \max \left\{ \frac{(\hat{x} - x)^\top Ay}{\alpha}, \frac{\max_{j \in [n]} (\hat{x} - x)^\top A(e_j - y)}{\alpha} \right\}, \quad (A \in \mathcal{Z}_\alpha(x, y)) \end{aligned}$$

thus concluding the proof of Equation (57).

The proof of Equation (58) follows from analogous arguments. One has simply to define $\hat{A} = \tilde{\lambda}A$ where $\tilde{\lambda}$ is given by:

$$\tilde{\lambda} = \min \left\{ 1, \frac{\alpha}{\max\{\max_{i \in [n]} x^\top A\hat{y} - e_i^\top A\hat{y}, \max_{j \in [n]} x^\top Ae_j - x^\top A\hat{y}\}} \right\}.$$

□

The following results extend the key intermediate that we presented for General-Sum Games (*i.e.*, Lemma C.2) to the Zero-Sum Games problem. The proof exploits a connection between the quantities that arise in Lemma C.6 and those that we already upper-bounded in Lemma C.2.

Lemma C.7 (Key Intermediate Step (Zero-Sum Games)). *Let $\alpha > 0$ and $x, y, \hat{x}, \hat{y} \in \Delta_n$. The following holds:*

$$\sup_{A \in \mathcal{Z}_\alpha(x, y)} (\hat{x} - x)^\top Ay \leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i = 0} \hat{x}_i \quad (59)$$

$$\sup_{A \in \mathcal{Z}_\alpha(x, y)} \max_{j \in [n]} (\hat{x} - x)^\top A(e_j - y) \leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i = 0} \hat{x}_i \quad (60)$$

$$\sup_{A \in \mathcal{Z}_\alpha(x, y)} \max_{i \in [n]} (x - e_i)^\top A(\hat{y} - y) \leq 4 \sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + 2 \sum_{j: y_j = 0} \hat{y}_j \quad (61)$$

$$\sup_{A \in \mathcal{Z}_\alpha(x, y)} x^\top A(y - \hat{y}) \leq 4 \sup_{S \in \mathcal{S}_\alpha(y)} \sum_{j \in S} (\hat{y}_j - y_j) + 2 \sum_{j: y_j = 0} \hat{y}_j \quad (62)$$

Proof. We start by proving Equation (59). Specifically, we have that:

$$\begin{aligned} \sup_{A \in \mathcal{Z}_\alpha(x, y)} (\hat{x} - x)^\top Ay &\leq \sup_{A \in \mathcal{G}_\alpha^x(x, y)} (\hat{x} - x)^\top Ay \\ &\leq 4 \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i = 0} \hat{x}_i, \quad (\text{Lemma C.2}) \end{aligned}$$

where in the first inequality we have simply dropped the constraints $x^\top A e_j - x^\top A y \leq \alpha$ for all $j \in [n]$. We can follow the same argument (*i.e.*, simply drop the constraints) also for proving Equation (61).

Next, we continue with Equation (60). We have that:

$$\begin{aligned} \sup_{A \in \mathcal{Z}_\alpha(x, y)} \max_{j \in [n]} (\hat{x} - x)^\top A (e_j - y) &\leq \sup_{B \in \mathcal{G}_\alpha^y(x, y)} \max_{j \in [n]} (\hat{x} - x)^\top B (y - e_j) \\ &\leq \sup_{S \in \mathcal{S}_\alpha(x)} \sum_{i \in S} (\hat{x}_i - x_i) + 2 \sum_{i: x_i = 0} \hat{x}_i, \end{aligned} \quad (\text{Lemma C.2})$$

where, in the first step, we have dropped the constraints $x^\top A y - e_i^\top A y \leq \alpha$ and we applied a change of variable $B = -A$. We can use the same argument (*i.e.*, dropping constraints and change of variable) also for proving Equation (62). $\square$

We can now prove Theorem C.8

Theorem C.8. *Let $\alpha > 0$ and let*

$$m \in \mathcal{O} \left(\frac{n + \log \left(\frac{1}{\delta} \right)}{\alpha \epsilon^2} + \frac{\sqrt{n} \log \left(\frac{1}{\delta} \right)}{\alpha \epsilon} \right).$$

Then, Algorithm 1 is (ϵ, δ) -correct for Zero-Sum Games, and its sample complexity τ_δ is given by m .

Proof. Combining Lemma C.6 and Lemma C.7, we can upper bound the Hausdorff distance $H(\mathcal{Z}_\alpha(x, y), \mathcal{Z}_\alpha(\hat{x}, \hat{y}))$ with the same quantities that we obtained in General Sum Games (*i.e.*, those appearing in Lemma C.3). Hence, we can proceed as in the proof of General-Sum Games, and we obtain the same result. $\square$

D. Auxiliary Lemmas

In this section, we provide statements of auxiliary lemma that are used throughout our analysis.

D.1. Concentrations and Probabilities

Lemma D.1. *Consider $D \in \mathbb{N}_{>0}$ and $p \in \Delta_D$. Let $\hat{p}_m$ be the maximum likelihood estimate of p obtained with $m \geq 1$ samples. Consider a threshold level $\beta > 0$ and let $\mathcal{I}(\beta) = \{i \in [D] : x_i \geq \beta\}$. If $m \geq \frac{\log \left(\frac{D}{\delta} \right)}{\log \left(\frac{1}{1-\beta} \right)}$, then, with probability at least $1 - \delta$, it holds that:*

$$\forall i \in \mathcal{I}(\beta), N_i(m) \geq 1.$$

As a corollary, let $p_{\min} \in \mathbb{R}$ as any real number that satisfies $p_{\min} > 0$ and $p_{\min} \leq \min_{i \in [D]: p_i > 0} p_i$. Then, if $m \geq \frac{\log \left(\frac{D}{\delta} \right)}{\log \left(\frac{1}{1-p_{\min}} \right)}$ it holds that

$$\mathbb{P}(\text{supp}(p) \neq \text{supp}(\hat{p}_m)) \leq \delta.$$

Proof. With probabilistic arguments, we have that:

$$\begin{aligned} \mathbb{P}(\exists i \in \mathcal{I}(\beta) : N_i(m) = 0) &\leq \sum_{i \in \mathcal{I}(\beta)} \mathbb{P}(N_i(m) = 0) && (\text{Boole's inequality}) \\ &\leq \sum_{i: x_i \geq \beta} (1 - x_i)^m \\ &\leq D(1 - \beta)^m \\ &\leq \delta, && (m \geq \frac{\log \left(\frac{D}{\delta} \right)}{\log \left(\frac{1}{1-\beta} \right)}) \end{aligned}$$

where, in the second inequality, we have used the fact that, for any i , $N_i(m) = 0$ if and only if, for all the samples m , the outcome i is never observed.

The corollary is a direct application of this claim by setting $\beta = p_{\min}$. $\square$

Lemma D.2 (Theorem A.2 in [Rajaraman et al. \(2020\)](#)). Consider $D \in \mathbb{N}_{>0}$ and a distribution $p \in \Delta_D$. Consider a sequence $P_1, \dots, P_m$ of i.i.d. samples drawn from p . For all $i \in [D]$, let $N_i(p) = \sum_{t=1}^m \mathbf{1}\{P_t = i\}$. Then, for $\delta \in (0, 1)$, it holds that:

$$\mathbb{P} \left(\sum_{i \in [D]} p_i \mathbf{1}\{N_i(m) = 0\} - \mathbb{E} \left[\sum_{i \in [D]} p_i \mathbf{1}\{N_i(m) = 0\} \right] \geq \frac{3\sqrt{D} \log(\frac{1}{\delta})}{n} \right) \leq \delta.$$

Lemma D.3. Consider $D \in \mathbb{N}_{>0}$ and $p \in \Delta_D$. Let $\hat{p}$ be the maximum likelihood estimator of p after m i.i.d. samples. Then, for any $\delta \in (0, 1]$, it holds that:

$$\mathbb{P} \left(\|p - \hat{p}_m\|_1 > \sqrt{\frac{2 \log(1/\delta) + 2D \log(6m)}{m}} \right) \leq \delta.$$

Proof. By Pinsker's inequality we have:

$$\|\hat{p}_m - p\|_1 \leq \sqrt{2\text{KL}(\hat{p}_m, p)}$$

where $\text{KL}(p_1, p_2)$ denotes the Kullback–Leibler divergence between distributions p_1 and p_2 .

Now, applying Proposition 1 in [Jonsson et al. \(2020\)](#), we obtain

$$\mathbb{P} \left[\forall m \in \mathbb{N}^*, m\text{KL}(\hat{p}_m, p) \leq \log \left(\frac{1}{\delta} \right) + (D-1) \log \left(e \left(1 + \frac{m}{D-1} \right) \right) \right] \geq 1 - \delta.$$

Thus, we obtained that:

$$\mathbb{P} \left[\|\hat{p}_m - p\|_1 \leq \sqrt{\frac{2}{m} \left(\log \left(\frac{1}{\delta} \right) + (D-1) \log \left(e \left(1 + \frac{m}{D-1} \right) \right) \right)} \right] \geq 1 - \delta.$$

At this point, we observe that:

$$\sqrt{\frac{2}{m} \left(\log \left(\frac{1}{\delta} \right) + (n-1) \log \left(e \left(1 + \frac{m}{D-1} \right) \right) \right)} \leq \sqrt{\frac{2 \log(1/\delta) + 2D \log(6m)}{m}}$$

Thus concluding the proof. $\square$

Lemma D.4. Let $n > 1$ and $v \in \{-1, 1\}^{n-1}$ such that $\sum_{k=1}^{n-1} v_k = 0$. Furthermore, let $\frac{1}{2} > \beta > \gamma > 0$, $\mathbb{P} = \left(\frac{\beta + \gamma \beta v_1}{n-1}, \dots, \frac{\beta + \gamma \beta v_{n-1}}{n-1}, 1 - \beta \right)$, and $\mathbb{Q} = \left(\frac{\beta}{n-1}, \dots, \frac{\beta}{n-1}, 1 - \beta \right)$. Then, it holds that:

$$\text{KL}(\mathbb{P}, \mathbb{Q}) \leq 2\beta\gamma^2.$$

Proof. We have that:

$$\begin{aligned} \text{KL}(\mathbb{P}, \mathbb{Q}) &= \sum_{i \in [n-1]: v_i=1} \frac{\beta + \gamma\beta}{n-1} \log(1 + \gamma) + \sum_{i \in [n-1]: v_i=-1} \frac{\beta - \gamma\beta}{n-1} \log(1 - \gamma) \\ &= \frac{1}{2} \beta (1 + \gamma) \log(1 + \gamma) + \frac{1}{2} \beta (1 - \gamma) \log(1 - \gamma) \\ &= \frac{\beta}{2} (\log(1 - \gamma^2) + \gamma \log(1 + \gamma) - \gamma \log(1 - \gamma)) \\ &\leq \frac{\beta}{2} (\gamma \log(1 + \gamma) - \gamma \log(1 - \gamma)) \quad (\log(1 - \gamma^2) \leq 0) \\ &\leq \frac{\beta}{2} \left(\gamma^2 + \gamma \left(\frac{1}{1 - \gamma} - 1 \right) \right) \quad (\log(1 + x) \leq x \text{ and } -\log(1 - x) \leq \frac{1}{1-x} - 1 \text{ for } x \in (0, 1)) \\ &\leq 2\beta\gamma^2. \end{aligned}$$

This concludes the proof. $\square$

Lemma D.5 (Variance Aware Error Control). *Let $D \in \mathbb{N}_{>0}$ and $p \in \Delta_D$. Let $\hat{p}$ be the maximum likelihood estimator of p after m i.i.d. samples where $m > 2$. Let $\beta \in (0, 1)$ and let $\mathcal{S}(\beta) = \{S \subseteq [D] : \forall i \in S, p_i > 0 \text{ and } \sum_{i \in S} p_i \leq \beta\}$.*

Then, for $\delta \in (0, 1)$ it holds that:

$$\mathbb{P} \left(\exists S \in \mathcal{S}(\beta) : \left| \sum_{i \in S} (p_i - \hat{p}_i) \right| > \sqrt{\frac{4\beta(D + \log(\frac{1}{\delta}))}{m}} + \frac{4(D + \log(\frac{1}{\delta}))}{m} \right) \leq \delta. \quad (63)$$

Furthermore, for $\delta \in (0, 1)$ it holds that:

$$\mathbb{P} \left(\exists S \subseteq [n] : \left| \sum_{i \in S} \hat{p}_i \right| > \sqrt{\frac{2 \sum_{i \in S} \hat{p}_i (D + \log(\frac{4}{\delta}))}{m}} + \frac{4(D + \log(\frac{4}{\delta}))}{m-1} \right) \leq \delta. \quad (64)$$

Proof. We start by proving Equation (63). Fix a set $S \in \mathcal{S}(\beta)$, then $\sum_{i \in S} \hat{p}_i$ is the empirical mean of a Bernoulli r.v. with mean $\mu = \sum_{i \in S} p_i$. Hence, for $\delta' \in (0, 1)$, a standard application of Bernstein's inequality (Boucheron et al., 2003) yields

$$\mathbb{P} \left(\left| \sum_{i \in S} (p_i - \hat{p}_i) \right| > \sqrt{\frac{4\beta \log(\frac{1}{\delta'})}{m}} + \frac{4 \log(\frac{1}{\delta'})}{m} \right) \leq \delta',$$

where we have used that $\text{Var}(\sum_{i \in S} p_i) \leq \sum_{i \in S} p_i \leq \beta$.

Now, for any value of β , the cardinality of $\mathcal{S}(\beta)$ is bounded by 2^D . Hence, for any $\delta' \in (0, 1)$ we have that:

$$\mathbb{P} \left(\exists S \in \mathcal{S}(\beta) : \left| \sum_{i \in S} (p_i - \hat{p}_i) \right| > \sqrt{\frac{4\beta \log(\frac{1}{\delta'})}{m}} + \frac{4 \log(\frac{1}{\delta'})}{m} \right) \leq 2^D \delta'.$$

Picking $\delta' = \frac{\delta}{2^D}$ concludes the proof.

Equation (64) follows from similar arguments, but we replaced Bernstein's inequality with its empirical version (Maurer and Pontil, 2009). In particular, the empirical Bernstein's inequality, replace the variance in the Bernstein inequality with the sample variance. In the following, we show how the sample variance of a Bernoulli r.v. with mean q can be upper bounded with $2\hat{q}$. Let Denote by $Z_1, \dots, Z_m \sim q$ the m independent samples from q . Then:

$$\begin{aligned} \hat{\text{Var}}(\hat{q}) &= \frac{1}{m(m-1)} \sum_{i < j} (Z_i - Z_j)^2 \\ &= \frac{1}{m-1} \sum_{i=1}^m (Z_i - p)^2 \\ &= \frac{m}{m-1} \hat{q}(1 - \hat{q}) \\ &\leq 2\hat{q} \end{aligned} \quad (\text{For } m \geq 2)$$

From this follows the result of Equation (64). $\square$

The following change of measure arguments follows from the same arguments of the ones used in Metelli et al. (2023).

Lemma D.6 (Change of measure argument). *Let $\alpha \geq 0$ and consider any (ϵ, δ) -correct algorithm. Let (x^0, y^0) and (x^1, y^1) be the strategy profiles of two different problem instances. Then, the following holds:*

$$H(\mathcal{G}_\alpha(x^0, y^0), \mathcal{G}_\alpha(x^1, y^1)) \geq 2\epsilon \implies \delta \geq \frac{1}{4} \exp(-\text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1})) \quad (65)$$

$$H(\mathcal{Z}_\alpha(x^0, y^0), \mathcal{Z}_\alpha(x^1, y^1)) \geq 2\epsilon \implies \delta \geq \frac{1}{4} \exp(-\text{KL}(\mathbb{P}_{x^0, y^0}, \mathbb{P}_{x^1, y^1})) \quad (66)$$

where $\mathbb{P}_{x, y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, with (X_t, Y_t) denoting the joint action sampled at time t and $p_q(\cdot)$ denoting the density function of $q \in \Delta_n$.

Lemma D.7 (Change of measure with several instances). *Let $\alpha \geq 0$ and $\{(x^k, y^k)\}_{k=1}^K$ be K distinct instances. Let (x^0, y^0) be a reference instance. It holds that:*

$$\begin{aligned} \forall i, j \in [K], i \neq j, H(\mathcal{G}_\alpha(x^i, y^i), \mathcal{G}_\alpha(x^j, y^j)) \geq 2\epsilon &\implies \delta \geq 1 - \frac{1}{\log K} \left(\frac{1}{K} \sum_{k \in [K]} \text{KL}(\mathbb{P}_{x^i, y^i}, \mathbb{P}_{x^0, y^0}) + \log 2 \right) \\ \forall i, j \in [K], i \neq j, H(\mathcal{Z}_\alpha(x^i, y^i), \mathcal{Z}_\alpha(x^j, y^j)) \geq 2\epsilon &\implies \delta \geq 1 - \frac{1}{\log K} \left(\frac{1}{K} \sum_{k \in [K]} \text{KL}(\mathbb{P}_{x^i, y^i}, \mathbb{P}_{x^0, y^0}) + \log 2 \right), \end{aligned}$$

where $\mathbb{P}_{x, y} = \prod_{t=1}^{\tau_\delta} p_x(X_t) p_y(Y_t)$, and $p_q(\cdot)$ denotes the density function of $q \in \Delta_n$.

D.2. Others

Lemma D.8. *Let $A \in [-1, 1]^{n \times n}$, $y \in \Delta_n$, $x \in \Delta_n$. Then it holds that:*

$$(i) \ x^\top A y \leq e_i^\top A y \ \forall i \in [n] \iff (ii) \ \begin{cases} e_i^\top A y = e_j^\top A y & \forall i, j \in \text{supp}(x), \\ e_i^\top A y \leq e_j^\top A y & \forall i \in \text{supp}(x), \forall j \notin \text{supp}(x). \end{cases} \quad (67)$$

$$(i) \ x^\top A e_j \leq x^\top A y \leq e_i^\top A y \ \forall i, j \in [n] \iff (ii) \ \begin{cases} x^\top A y = x^\top A e_j = e_i^\top A y & \forall i \in \text{supp}(x), j \in \text{supp}(y), \\ x^\top A e_j \leq x^\top A y \leq e_i^\top A y & \forall i \notin \text{supp}(x), \forall j \notin \text{supp}(y). \end{cases} \quad (68)$$

Proof. This is a well-known result (e.g. [Gintis \(2000, Section 3.6\)](#)). For completeness, we prove Equation (67). The proof of Equation (68) follows analogous arguments. We first show that (i) $\Rightarrow$ (ii). Assume $x^\top A y \leq e_i^\top A y$ for all $i \in [n]$. Rewrite $x^\top A y$ as follows:

$$x^\top A y = \sum_{k \in [n]} x_k \sum_{j \in [n]} y_j A_{kj} = \sum_{k \in \text{supp}(x)} x_k \sum_{j \in [n]} y_j A_{kj} = \sum_{k \in \text{supp}(x)} x_k (e_k^\top A y)$$

since $x_k = 0$ for $k \notin \text{supp}(x)$. Note that $\sum_{k \in S} x_k = 1$. Then, since $x_k > 0$ for $k \in \text{supp}(x)$ and each $e_k^\top A y \geq x^\top A y$ (by assumption), it must follow that $e_k^\top A y = x^\top A y$ for all $k \in \text{supp}(x)$. Hence, we have proved the first condition of (ii) and it remains to prove that for all $j \notin \text{supp}(x)$ and all $i \in \text{supp}(x)$, we have that $e_i^\top A y \leq e_j^\top A y$.

Fix any $j \notin \text{supp}(x)$. By assumption (i), we have $x^\top A y \leq e_j^\top A y$. Combining this with the equality $e_i^\top A y = x^\top A y$ for all $i \in \text{supp}(x)$, we obtain

$$e_i^\top A y \leq e_j^\top A y \quad \forall i \in \text{supp}(x), \forall j \notin \text{supp}(x),$$

which concludes the proof of (i) $\Rightarrow$ (ii).

We conclude the proof by showing that (ii) $\Rightarrow$ (i). Assume (ii) holds. Let c denote the common value $e_i^\top A y$ for $i \in \text{supp}(x)$. Then:

$$x^\top A y = \sum_{i \in \text{supp}(x)} x_i (e_i^\top A y) = \sum_{i \in S} x_i c = c.$$

Hence, $x^\top A y \leq e_i^\top A y$ for all $i \in \text{supp}(x)$.

For any $j \notin \text{supp}(x)$, (ii) gives $e_j^\top A y \geq c = x^\top A y$, which concludes the proof. $\square$

Lemma D.9. *Let $c_1, c_2, K \in \mathbb{R}_{>0}$ and consider*

$$\bar{T} = \inf \left\{ t \in \mathbb{N} : \frac{\log(c_1/\delta) + n \log(c_2 t)}{t} \leq K^2 \right\}.$$

Then:

$$\bar{T} \leq 2 \left(2 + \frac{4 \log(c_1/\delta)}{K^2} + \frac{4n}{K^2} \log \left(\frac{2nc_2}{K^2} \right) \right).$$

Proof. First, it is easy to see that:

$$\begin{aligned}
 \bar{T} &\leq \max \left\{ \inf \left\{ t \in \mathbb{N} : \frac{\log(c_1/\delta)}{t} \leq \frac{K^2}{2} \right\}, \inf \left\{ t \in \mathbb{N} : \frac{n \log(c_2 t)}{t} \leq \frac{K^2}{2} \right\} \right\} \\
 &\leq 2 \left(\inf \left\{ t \in \mathbb{N} : \frac{\log(c_1/\delta)}{t} \leq \frac{K^2}{2} \right\} + \inf \left\{ t \in \mathbb{N} : \frac{n \log(c_2 t)}{t} \leq \frac{K^2}{2} \right\} \right) \\
 &\leq 2 \left(1 + \frac{4 \log(c_1/\delta)}{K^2} + \inf \left\{ t \in \mathbb{N} : \frac{n \log(c_2 t)}{t} \leq \frac{K^2}{2} \right\} \right) \\
 &:= 2 \left(1 + \frac{4 \log(c_1/\delta)}{K^2} + \tilde{T} \right).
 \end{aligned}$$

It remains to analyze the term:

$$\tilde{T} = \inf \left\{ t \in \mathbb{N} : \frac{n \log(c_2 t)}{t} \leq \frac{K^2}{2} \right\} = \left\{ t \in \mathbb{N} : t \geq \frac{2n \log(c_2 t)}{K^2} \right\}.$$

By definition, for $\tilde{T} - 1$, we have that:

$$\tilde{T} - 1 < \frac{2n \log(c_2(\tilde{T} - 1))}{K^2}. \quad (69)$$

Since $\log(x) \leq \sqrt{x}$, we have that:

$$\sqrt{\tilde{T} - 1} < \frac{2n\sqrt{c_2}}{K^2},$$

which leads to:

$$\tilde{T} - 1 < \left(\frac{2n\sqrt{c_2}}{K^2} \right)^2.$$

Plugging this result within Equation (69), we obtain:

$$\tilde{T} < \frac{4n}{K^2} \log \left(\frac{2nc_2}{K^2} \right) + 1,$$

which leads to the desired result. $\square$

Lemma D.10 (General Decomposition Error). *Let $(\mathcal{Y}, d)$ be a metric space and let $\mathcal{X}$ be any set. Consider a compact-valued correspondence $F : \mathcal{X} \times \mathcal{X} \rightrightarrows \mathcal{Y}$, and let $x_1, x_2, \hat{x}_1, \hat{x}_2 \in \mathcal{X}$. It holds that:*

$$H_d(F(x_1, x_2), F(\hat{x}_1, \hat{x}_2)) \leq H_d(F(x_1, x_2), F(\hat{x}_1, x_2)) + H_d(F(\hat{x}_1, x_2), F(\hat{x}_1, \hat{x}_2)).$$

Proof. For any $y \in \mathcal{Y}$, let $c(y) \in \operatorname{argmin}_{c \in F(\hat{x}_1, x_2)} d(y, c)$. Note that, by Heine-Cantor, this minimum is attained since $d(y, \cdot)$ is a continuous function and F is compact-valued. Then, recall that:

$$H_d(F(x_1, x_2), F(\hat{x}_1, \hat{x}_2)) = \max \left\{ \sup_{y \in F(x_1, x_2)} \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} d(y, \hat{y}), \sup_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} \inf_{y \in F(x_1, x_2)} d(y, \hat{y}) \right\} \quad (70)$$

Now, focus on the first term. We have that:

$$\begin{aligned}
 \sup_{y \in F(x_1, x_2)} \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} d(y, \hat{y}) &\leq \sup_{y \in F(x_1, x_2)} \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} (d(y, c(y)) + d(c(y), \hat{y})) && \text{(Triangular inequality)} \\
 &= \sup_{y \in F(x_1, x_2)} \left(d(y, c(y)) + \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} d(c(y), \hat{y}) \right) \\
 &\leq \sup_{y \in F(x_1, x_2)} d(y, c(y)) + \sup_{y \in F(x_1, x_2)} \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} d(c(y), \hat{y}) \\
 &\leq H_d(F(x_1, x_2), F(\hat{x}_1, x_2)) + \sup_{y \in F(x_1, x_2)} \inf_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} d(c(y), \hat{y}) && \text{(Def. of } c(y)) \\
 &\leq H_d(F(x_1, x_2), F(\hat{x}_1, x_2)) + H_d(F(\hat{x}_1, x_2), F(\hat{x}_1, \hat{x}_2)) && (c(y) \in F(\hat{x}_1, x_2))
 \end{aligned}$$

Similarly, by using the triangular inequality with $c(\hat{y})$, one has that:

$$\sup_{\hat{y} \in F(\hat{x}_1, \hat{x}_2)} \inf_{y \in F(x_1, x_2)} d(y, \hat{y}) \leq H_d(F(x_1, x_2), F(\hat{x}_1, x_2)) + H_d(F(\hat{x}_1, x_2), F(\hat{x}_1, \hat{x}_2)).$$

Plugging these results within Equation (70), concludes the proof. $\square$

Corollary D.11 (Decomposition Error in Inverse Game Theory). *Let $x, y, \hat{x}, \hat{y} \in \Delta_n$. Then, for $\alpha \geq 0$, it holds that:*

$$\begin{aligned} H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) &\leq H(\mathcal{G}_\alpha(x, y), \mathcal{G}_\alpha(\hat{x}, y)) + H(\mathcal{G}_\alpha(\hat{x}, y), \mathcal{G}_\alpha(\hat{x}, \hat{y})) \\ H(\mathcal{Z}_\alpha(x, y), \mathcal{Z}_\alpha(\hat{x}, \hat{y})) &\leq H(\mathcal{Z}_\alpha(x, y), \mathcal{Z}_\alpha(\hat{x}, y)) + H(\mathcal{Z}_\alpha(\hat{x}, y), \mathcal{Z}_\alpha(\hat{x}, \hat{y})) \end{aligned}$$

Proof. This is a direct application of Lemma D.10. $\square$

Lemma D.12 (Packing argument (Metelli et al., 2023), Lemma E.6). *Let $\mathcal{V} = \left\{v \in \{-1, 1\}^D : \sum_{j=1}^D v_j = 0\right\}$. Then the $\frac{D}{16}$ -packing number of $\mathcal{V}$ with respect to the metric $d(v, v') = \sum_{j=1}^D |v_j - v'_j|$ is lower bounded by $2^{D/5}$.*